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Enzymatic method for the production of L-glufosinate P-esters

Abstract

A first enzymatic method for the production of an L-glufosinate P-ester includes reacting an L-glufosinate P-ester carbamoylate to give the corresponding L-glufosinate P-ester in the presence of a carbamoylase as a catalyzer. A second enzymatic method includes the enantioselective production of an L-glufosinate P-ester from a mixture M.sub.IIIA of L- and D-glufosinate P-ester hydantoins. The L-glufosinate P-ester obtained in the two methods may be saponified to give L-glufosinate.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is a National Stage entry under § 371 of International Application No. PCT/EP2022/056539, filed on Mar. 14, 2022. The content of this application is hereby incorporated by reference in its entirety.

REFERENCE TO A SEQUENCE LISTING

(2) The present application is accompanied by an XML file as a computer readable form containing the sequence listing entitled, “006128USPCT-SeqList.txt”, created on Sep. 11, 2024, with a file size of 86,979 bytes, the content of which is hereby incorporated by reference in its entirety.

BACKGROUND OF THE INVENTION

Field of the Invention

(3) The present invention relates in a first aspect to an enzymatic method for the production of an L-glufosinate P-ester. This method is characterized by a step (c) in which an L-glufosinate P-ester carbamoylate is reacted to give the corresponding L-glufosinate P-ester. This step (c) is catalyzed by a carbamoylase, preferably by an L-enantioselective carbamoylase.

(4) The L-glufosinate P-ester carbamoylate employed in step (c) may be obtained by reaction from the corresponding L-glufosinate hydantoin P-ester. This optional, antecedent step (b) is catalyzed by a hydantoinase, preferably an L-enantioselective hydantoinase. Even more preferred, the L-glufosinate hydantoin P-ester employed in step (b) may be obtained by reaction from the corresponding D-glufosinate hydantoin P-ester. This optional, antecedent step (a) is preferably catalyzed by a hydantoin racemase.

(5) In a second aspect, the present invention relates to an enzymatic method for enantioselective production of an L-glufosinate P-ester from a mixture M.sub.IIIA of L- and D-glufosinate P-ester hydantoins. In the method according to the second aspect of the invention, the mixture M.sub.IIIA is provided in a step (i-A) and employed in steps (ii) and (iii). Steps (ii) and (iii) correspond to steps (b) and (c) described above, respectively, wherein in step (c), an L-carbamoylase is employed.

(6) In an optional, preferred step (i-B) of the method according to the second aspect of the invention, corresponding to preferred step (a) described above, a hydantoin racemase is preferably used to convert at least a part of the D-glufosinate P-ester hydantoin in the mixture M.sub.IIIA into its corresponding L-isomer, giving a mixture M.sub.IIIB, which is then employed in steps (ii) and (iii).

(7) The L-glufosinate P-ester obtained in the methods according to the first or second aspect of the invention may be saponified to give L-glufosinate.

Description of Related Art

(8) Organic phosphorous compounds, i.e. chemical agents comprising a carbon-phosphor bond, are widely applied as herbicides in the area of plant protection. Agents such as the herbicides glyphosate (Roundup®, Touchdown®) and glufosinate (Basta®, Liberty®) as well as the growth

regulator glyphosine (Polaris®) are used for this purpose (as described for example by G. Hörlein, Rev. Environ. Contam. Toxicol. 1994, 138, 73-145).

(9) The esters of P-methyl phosphinic acid (for example, P-methyl phosphinic acid butyl ester; "MPBE"; CAS-No: 6172-80-1) have a key role as synthetic building blocks in the synthesis of the non-selective herbicide glufosinate. These esters are accessible via two fundamental synthetic pathways (summarized in FIGS. 3 a and 3 b, page 130, of the article of K. Haack, Chem. Unserer Zeit 2003, 37, 128-138): a. Reacting diethyl chlorophosphite [CIP(OC.sub.2H.sub.5).sub.2] with CH.sub.3MgCl provides methyl diethoxy phosphine [H.sub.3CP(OC.sub.2H.sub.5).sub.2; "DEMP"; CAS-No. 15715-41-0], which is partially hydrolyzed to give the corresponding methylphosphinic acid ethyl ester (MPEE; CAS-Nr: 16391-07-4). b. Alternatively, methane can be reacted with phosphor trichloride at 500° C. to give methyl dichloro phosphane H.sub.3CPCl.sub.2. The latter can then be solvolized in alcohols to give the corresponding methyl phosphinic acid esters.

(10) The esters of P-methyl phosphinic acid add to carbon-carbon double bonds regioselectively. This property is used in the synthesis of glufosinate for the formation of the second phosphor-carbon bond. For example, H.sub.3CPH(O)OR (R=Alkyl) reacts with 1-cyano allyl acetate in an addition reaction to provide an intermediate. Subsequent exchange of the acetate substituent with ammonia and hydrolysis of the cyano group and the ester group of the phosphinic acid moiety give glufosinate.

(11) Acrylic acid ester is a cheaper alternative starting material. It can react with the ester of P-methyl phosphinic acid to 3-[alkoxy(methyl)phosphinyl]propionic acid alkyl ester. Claisen reaction of this diester with diethyl oxalate, hydrolysis and decarboxylation provide the corresponding α -keto acid, which can be reductively aminated to give glufosinate.

(12) These and further synthetic routes towards glufosinate are also described in the art, e.g. in WO 1999/009039 A1, EP 0 508 296 A1.

(13) A general disadvantage of all synthetic routes to glufosinate is that the obtained glufosinate is a racemic mixture. However, as there is no herbicidal activity of the D-enantiomer, L-glufosinate [hereinafter "LGA"; CAS-Nr. 35597-44-5; other names "(S)-glufosinate", "(-)-glufosinate"] is the enantiomer of economical interest.

(14) ##STR00001##

(15) For example, CN 111662325 A discloses a synthetic pathway in which L-homoserine is reacted to the respective hydantoin, followed by addition of methane phosphor dichloride, which results in a methane phosphane which is disubstituted with L-homoserine hydantoin. After a final Arbuzov reaction and hydrolysis, LGA is obtained. Although the disclosure describes a high enantiomeric excess (= "ee"), this ee could not be reproduced by the inventors of the present invention. A reason could be that during this synthesis pathway, strong acidic (HCl for saponification of the hydantoin phosphor bond) and alkaline conditions (NaOH, 100° C. for ring opening) are applied. Such conditions usually lead to racemization, as described by M. Bovarnick & H. T. Clarke, Journal of the American Chemical Society 1938, 60, 2426-2430, by R. A. Lazarus, J. Org. Chem. 1990, 55, 4755-4757, and by A. S. Bommarius, M. Kottenhahn, H. Klenk, K. Drauz: "A direct route from hydantoins to D-amino acids employing a resting cell biocatalyst with D-hydantoinase and D-carbamoylase activity" on page 164 and 167 in "Microbial Reagents in Organic Synthesis" Series C: Mathematical and Physical Sciences-Vol. 381, S. Servi (Ed.), 1992, Springer Science+Business Media, B. V., Dordrecht.

(16) For enantioselective syntheses of LGA, chemical and enzymatic pathways are described in the art.

(17) WO 2020/145513 A1 and WO 2020/145514 A1 describe a chemical route to LGA. In this route, an L-homoserine derivative such as O-acetyl L-homoserine or O-succinyl L-homoserine is used as starting material and L-glufosinate is obtained by a sequence of reactions including lactonization and halogenation.

(18) WO 2020/145627 A1 describes a similar route, wherein during halogenation, a bromo derivative is obtained.

(19) The route disclosed by CN 106083922 A resembles these synthetic pathways but starts off from L-methionine.

(20) CN 108516991 A describes another synthetic pathway to LGA, starting from the azeotropic dehydration of L-homoserine to give L-3,6-bis(2-haloethyl)-2,5-diketopiperazine, followed by introduction of a methylphosphonate diester group, and hydrolysis.

(21) WO 2017/151573 A1 discloses a two-step enzymatic synthesis of LGA from D-glufosinate. In the first step, D-glufosinate is oxidatively deaminated to give 2-oxo-4-[hydroxy(methyl)phosphinoyl]-butyric acid ("PPO"), followed by the specific amination of PPO to LGA as the second step. The first step is carried out by catalysis of a D-amino acid oxidase, the second step is catalyzed by a transaminase.

(22) WO 2020/051188 A1 discloses a similar method of converting racemic glufosinate to the L-enantiomer. In addition, it discloses a step in which the α -ketoacid or ketone byproduct formed during amination of PPO with an amine donor is converted by ketoglutarate decarboxylase to further shift the equilibrium to LGA.

(23) WO 2019/018406 A1 discloses a method of purifying LGA from a mixture comprising LGA and glutamate. Glutamate is converted to pyroglutamate enzymatically by glutaminy-peptidyl cyclotransferase, and LGA is then purified from the resulting mixture with an ion-exchange resin.

(24) WO 2013/072486 A1 disclose hydantoinase mutants which have a greater activity towards D-amino acids.

(25) WO 00/58449 A1 disclose hydantoinase mutants which have a greater activity towards L-amino acids.

(26) The object of the present invention is to provide a further enzymatic process for producing glufosinate or glufosinate P-esters. This process should provide products with a high excess of the L-enantiomer over the D-enantiomer. Moreover, this process should allow to use new substrates which heretofore were not used in the enzymatic synthesis of glufosinate or glufosinate esters. Moreover, there is a need in the art for an enantioselective method for the production of L-glufosinate or L-glufosinate esters from starting materials comprising D- and L-enantiomers, such as racemic mixtures, in a minimal number of synthetic steps.

SUMMARY OF THE INVENTION

(27) Namely, it was surprisingly found that the desired L-glufosinate P-esters can be produced by an enzymatically catalyzed reaction from the corresponding carbamoylates of formula L-(II):

(28) ##STR00002##

(29) wherein R^{sup.2L} is a radical selected from the following formulae (VIII), (IX), (X), preferably selected from the following formulae (IX), (X), more preferably R^{sup.2L} has the following formula (IX):

(30) ##STR00003##

(31) When R^{sup.2L} is a radical of the formula (VIII), the radical R^{sup.2L} may be selected from the formulae L-(VIII) and D-(VIII), and preferably has a formula according to L-(VIII):

(32) ##STR00004##

(33) When R^{sup.2L} is a radical of the formula (IX), the radical R^{sup.2L} may be selected from the formulae L-(IX) and D-(IX), and preferably has a formula according to L-(IX):

(34) ##STR00005##

(35) When R^{sup.2L} is a radical of the formula (X), the radical R^{sup.2L} may be selected from the formulae L-(X) and D-(X), and preferably has a formula according to L-(X):

(36) ##STR00006##

(37) Such carbamoylates were not used in the enzymatically catalyzed production of L-glufosinate or its P-esters before. This finding was even more surprising, as it turned out that the presence of a radical selected from the following formulae (VIII), (IX), (X) is mandatory, as there is no

corresponding reaction of compounds according to formula L-(II) in which R.sup.2L=H.

(38) α .1) The present invention hence solves the problems mentioned above by providing in a first aspect a method for the production of a glufosinate P-ester according to formula L-(I), comprising a step (c) in which a compound according to the above-mentioned formula L-(II) is reacted to give a compound according to formula L-(I):

(39) ##STR00007##

(40) wherein R.sup.1L is a radical selected from the above-mentioned formulae (VIII), (IX), (X), preferably selected from the above-mentioned formulae (IX), (X), more preferably R.sup.1L has formula (X).

(41) When R.sup.1L is a radical of the formula (VIII), the radical R.sup.1L may be selected from the above-mentioned formulae L-(VIII) and D-(VIII), preferably L-(VIII).

(42) When R.sup.1L is a radical of the formula (IX), the radical R.sup.1L may be selected from the above-mentioned formulae L-(IX) and D-(IX), preferably L-(IX).

(43) When R.sup.1L is a radical of the formula (X), the radical R.sup.1L may be selected from the above-mentioned formulae L-(X) and D-(X), preferably L-(X).

(44) The reaction according step (c) is catalyzed by a carbamoylase E.sub.1, preferably by an L-enantioselective carbamoylase E.sub.1 ("L-enantioselective carbamoylase"="L-carbamoylase").

(45) The L-carbamoylase E.sub.1 is preferably categorized in the EC class 3.5.1.87.

(46) The polypeptide sequence of the L-carbamoylase E.sub.1 is preferably selected from SEQ ID NO: 1 and variants thereof, SEQ ID NO: 2 and variants thereof, SEQ ID NO: 3 and variants thereof, SEQ ID NO: 4 and variants thereof, SEQ ID NO: 5 and variants thereof, SEQ ID NO: 6 and variants thereof, SEQ ID NO: 7 and variants thereof, SEQ ID NO: 8 and variants thereof, SEQ ID NO: 9 and variants thereof.

(47) α .2) In a preferred embodiment of the method according to the first aspect of the invention, the compound according to formula L-(II) that is employed in step (c) is obtained by a step (b) in which a compound according to formula L-(III) is reacted to give the compound according to formula L-(II):

(48) ##STR00008##

(49) wherein the reaction according to step (b) is catalyzed by a hydantoinase E.sub.2, preferably an L-enantioselective hydantoinase E.sub.2 ("L-enantioselective hydantoinase"="L-hydantoinase").

(50) R.sup.3L is a radical selected from the above-mentioned formulae (VIII), (IX), (X), preferably selected from the formulae (VIII), (IX), more preferably R.sup.3L has formula (VIII).

(51) When R.sup.3L is a radical of the formula (VIII), the radical R.sup.3L may be selected from the above-mentioned formulae L-(VIII) and D-(VIII), preferably L-(VIII).

(52) When R.sup.3L is a radical of the formula (IX), the radical R.sup.3L may be selected from the above-mentioned formulae L-(IX) and D-(IX), preferably L-(IX).

(53) When R.sup.3L is a radical of the formula (X), the radical R.sup.3L may be selected from the above-mentioned formulae L-(X) and D-(X), preferably L-(X).

(54) The hydantoinase E.sub.2, in particular the L-hydantoinase E.sub.2, is preferably categorized in the EC class 3.5.2.2.

(55) The polypeptide sequence of the hydantoinase E.sub.2, in particular the L-hydantoinase E.sub.2, is preferably selected from SEQ ID NO: 10 and variants thereof, SEQ ID NO: 11 and variants thereof, SEQ ID NO: 12 and variants thereof, SEQ ID NO: 13 and variants thereof, SEQ ID NO: 14 and variants thereof, SEQ ID NO: 15 and variants thereof, SEQ ID NO: 16 and variants thereof.

(56) α .3) In an even more preferred embodiment of the method according to the first aspect of the invention, the compound according to formula L-(III) that is employed in step (b) is obtained by a step (a) in which a compound according to formula D-(III) is reacted to give the compound according to formula L-(III):

(57) ##STR00009##

(58) wherein the reaction according to step (a) is preferably catalyzed by a hydantoin racemase E.sub.3.

(59) R.sup.3D is a radical selected from the above-mentioned formulae (VIII), (IX), (X), preferably selected from the formulae (VIII), (IX), more preferably R.sup.3D has formula (VIII).

(60) When R.sup.3D is a radical of the formula (VIII), the radical R.sup.3D may be selected from the above-mentioned formulae L-(VIII) and D-(VIII), preferably L-(VIII).

(61) When R.sup.3D is a radical of the formula (IX), the radical R.sup.3D may be selected from the above-mentioned formulae L-(IX) and D-(IX), preferably L-(IX).

(62) When R.sup.3D is a radical of the formula (X), the radical R.sup.3D may be selected from the above-mentioned formulae L-(X) and D-(X), preferably L-(X).

(63) The hydantoin racemase E.sub.3 is preferably categorized in the EC class 5.1.99.5.

(64) The polypeptide sequence of the hydantoin racemase E.sub.3 in step (a) is preferably selected from SEQ ID NO: 17 and variants thereof, SEQ ID NO: 18 and variants thereof, SEQ ID NO: 19 and variants thereof, SEQ ID NO: 20 and variants thereof, SEQ ID NO: 21 and variants thereof, SEQ ID NO: 22 and variants thereof, SEQ ID NO: 23 and variants thereof, SEQ ID NO: 24 and variants thereof, SEQ ID NO: 25 and variants thereof, SEQ ID NO: 26 and variants thereof.

(65) It was further surprisingly found that the above-mentioned finding can be advantageously used for a method for enantioselective production of an L-glufosinate P-ester, in particular from mixtures of stereoisomers, preferably from mixtures of enantiomers, such as racemic mixtures.

(66) β .1) Hence, the present invention relates in a second aspect to a method for the production of an L-glufosinate P-ester according to formula L-(I):

(67) ##STR00010##

(68) wherein R.sup.1L is a radical selected from the above-mentioned formulae (VIII), (IX), (X), preferably selected from the above-mentioned formulae (IX), (X), more preferably R.sup.1L has formula (X).

(69) When R.sup.1L is a radical of the formula (VIII), the radical R.sup.1L may be selected from the above-mentioned formulae L-(VIII) and D-(VIII), preferably L-(VIII).

(70) When R.sup.1L is a radical of the formula (IX), the radical R.sup.1L may be selected from the above-mentioned formulae L-(IX) and D-(IX), preferably L-(IX).

(71) When R.sup.1L is a radical of the formula (X), the radical R.sup.1L may be selected from the above-mentioned formulae L-(X) and D-(X), preferably L-(X).

(72) In this method, a mixture M.sub.IIIA comprising at least one compound L-(III) and at least one compound D-(III) is provided [step (i)-A]:

(73) ##STR00011##

(74) R.sup.3L is a radical selected from the above-mentioned formulae (VIII), (IX), (X), preferably selected from the formulae (VIII), (IX), more preferably R.sup.3L has formula (VIII).

(75) When R.sup.3L is a radical of the formula (VIII), the radical R.sup.3L may be selected from the above-mentioned formulae L-(VIII) and D-(VIII), preferably L-(VIII).

(76) When R.sup.3L is a radical of the formula (IX), the radical R.sup.3L may be selected from the above-mentioned formulae L-(IX) and D-(IX), preferably L-(IX).

(77) When R.sup.3L is a radical of the formula (X), the radical R.sup.3L may be selected from the above-mentioned formulae L-(X) and D-(X), preferably L-(X).

(78) R.sup.3D is a radical selected from the above-mentioned formulae (VIII), (IX), (X), preferably selected from the formulae (VIII), (IX), more preferably R.sup.3D has formula (VIII).

(79) When R.sup.3D is a radical of the formula (VIII), the radical R.sup.3D may be selected from the above-mentioned formulae L-(VIII) and D-(VIII), preferably L-(VIII).

(80) When R.sup.3D is a radical of the formula (IX), the radical R.sup.3D may be selected from the above-mentioned formulae L-(IX) and D-(IX), preferably L-(IX).

(81) When R.sup.3D is a radical of the formula (X), the radical R.sup.3D may be selected from the

above-mentioned formulae L-(X) and D-(X), preferably L-(X).

(82) β .2) In a step (ii), the mixture M.sub.IIIA is subjected to a reaction with a hydantoinase E.sub.2, which is more preferably as defined under item α .2), giving a composition M.sub.II comprising at least one compound according to formula L-(II) and optionally at least one compound according to formula D-(II), wherein L-(II) and D-(II) have the following formulae:

(83) ##STR00012##

(84) R.sup.2L is a radical selected from the above-mentioned formulae (VIII), (IX), (X), preferably selected from the formulae (IX), (X), more preferably R.sup.2L has formula (IX).

(85) R.sup.2D is a radical selected from the above-mentioned formulae (VIII), (IX), (X), preferably selected from the formulae (IX), (X), more preferably R.sup.2D has formula (IX).

(86) When R.sup.2L is a radical of the formula (VIII), the radical R.sup.2L may be selected from the above-mentioned formulae L-(VIII) and D-(VIII), preferably L-(VIII).

(87) When R.sup.2D is a radical of the formula (VIII), the radical R.sup.2D may be selected from the above-mentioned formulae L-(VIII) and D-(VIII), preferably L-(VIII).

(88) When R.sup.2L is a radical of the formula (IX), the radical R.sup.2L may be selected from the above-mentioned formulae L-(IX) and D-(IX), preferably L-(IX).

(89) When R.sup.2D is a radical of the formula (IX), the radical R.sup.2D may be selected from the above-mentioned formulae L-(IX) and D-(IX), preferably L-(IX).

(90) When R.sup.2L is a radical of the formula (X), the radical R.sup.2L may be selected from the above-mentioned formulae L-(X) and D-(X), preferably L-(X).

(91) When R.sup.2D is a radical of the formula (X), the radical R.sup.2D may be selected from the above-mentioned formulae L-(X) and D-(X), preferably L-(X).

(92) β .3) Then, in a further step (iii), M.sub.II is subjected to a reaction, which is catalyzed by an L-carbamoylase E.sub.1, which is preferably as defined under item α .1), giving a M.sub.I comprising at least one compound according to formula L-(I) and optionally at least one compound according to formula D-(I), wherein D-(I) has the following formula:

(93) ##STR00013##

(94) wherein R.sup.1D is a radical selected from the above-mentioned formulae (VIII), (IX), (X), preferably selected from the above-mentioned formulae (IX), (X), more preferably R.sup.1D has formula (X).

(95) When R.sup.1D is a radical of the formula (VIII), the radical R.sup.1D may be selected from the above-mentioned formulae L-(VIII) and D-(VIII), preferably L-(VIII).

(96) When R.sup.1D is a radical of the formula (IX), the radical R.sup.1D may be selected from the above-mentioned formulae L-(IX) and D-(IX), preferably L-(IX).

(97) When R.sup.1D is a radical of the formula (X), the radical R.sup.1D may be selected from the above-mentioned formulae L-(X) and D-(X), preferably L-(X).

(98) The method according to the second aspect of the invention is enantioselective, because M.sub.I comprises either only L-(I) or, in case M.sub.I comprises both stereoisomers L-(I) and D-(I), the molar ratio of L-(I) to D-(I) in M.sub.I is greater than the molar ratio of L-(III) to D-(III) in M.sub.IIIA.

(99) β .4) Furthermore, it was surprisingly found that the method for enantioselective production according to the second aspect of the invention can be further improved. Namely, in an optional step (i-B) which is carried out after step (i-A) [and which may be carried out before or concomitantly with steps (ii) and (iii), preferably concomitantly with steps (ii) and (iii)], at least a part of the compounds D-(III) comprised by the mixture M.sub.IIIA are reacted, preferably enzymatically catalyzed by a hydantoin racemase E.sub.3, into L-(III), which is the isomer of D-(III), giving a composition M.sub.IIIB comprising L-(III) and optionally its isomer D-(III).

(100) In case the reaction according to step (i-B) is enzymatically catalyzed, it is preferably catalyzed by a hydantoin racemase E.sub.3.

(101) The hydantoin racemase E.sub.3 is preferably as defined under item α .3).

Description

3. BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 shows the pOM21c plasmid map.

(2) FIG. 2 shows the pOM22c plasmid map.

4. DETAILED DESCRIPTION OF THE INVENTION

4.1 Definitions

(3) Any of the enzymes used according to any aspect of the present invention, may be an isolated enzyme. In particular, the enzymes used according to any aspect of the present invention may be used in an active state and in the presence of all cofactors, substrates, auxiliary and/or activating polypeptides or factors essential for its activity. Such factors may be metal ions such as Mn.sup.2+ or Co.sup.2+.

(4) In particular, an enzyme according to the present application may be a carbamoylase E.sub.1, a hydantoinase E.sub.2, or a hydantoin racemase E.sub.3.

(5) "Enzymatically catalyzed" means that the respective reaction is catalyzed by an enzyme, which may be a carbamoylase E.sub.1, a hydantoinase E.sub.2, or a hydantoin racemase E.sub.3.

(6) The enzyme used according to any aspect of the present invention may be recombinant. The term "recombinant" as used herein, refers to a molecule or is encoded by such a molecule, particularly a polypeptide or nucleic acid that, as such, does not occur naturally but is the result of genetic engineering or refers to a cell that comprises a recombinant molecule. For example, a nucleic acid molecule is recombinant if it comprises a promoter functionally linked to a sequence encoding a catalytically active polypeptide and the promoter has been engineered such that the catalytically active polypeptide is overexpressed relative to the level of the polypeptide in the corresponding wild type cell that comprises the original unaltered nucleic acid molecule.

(7) A "polypeptide" (one or more peptides) is a chain of chemical building blocks called amino acids that are linked together by chemical bonds called peptide bonds. A protein or polypeptide, including an enzyme, may be "native" or "wild-type", meaning that it occurs in nature or has the amino acid sequence of a native protein, respectively. These terms are sometimes used interchangeably. A polypeptide may or may not be glycosylated.

(8) The term "overexpressed", as used herein, means that the respective polypeptide encoded or expressed is expressed at a level higher or at higher activity than would normally be found in the cell under identical conditions in the absence of genetic modifications carried out to increase the expression, for example in the respective wild type cell.

4.2 Methods to Obtain Enzymes

(9) The enzymes that can be used in the method according to the present invention can be synthesized by methods that are known to the skilled person.

(10) One approach, which is a preferred approach according to the invention, is to express the enzyme(s) in microorganism(s) such as *Escherichia coli* (= "*E. coli*"), *Saccharomyces cerevisiae*, *Pichia pastoris*, and others, and to add the whole cells to the reactions as whole cell biocatalysts. Another approach is to express the enzyme(s), lyse the microorganisms, and add the cell lysate. Yet another approach is to purify, or partially purify, the enzyme(s) from a lysate and add pure or partially pure enzyme(s) to the reaction. If multiple enzymes are required for a reaction, the enzymes can be expressed in one or several microorganisms, including expressing all enzymes within a single microorganism.

(11) For example, the skilled person can obtain the enzymes according to the invention by expression, in particular, overexpression, [hereinafter, "expression, in particular overexpression" is abbreviated as (over)expression", and "express, in particular overexpress" is abbreviated as (over)express"] of these enzymes in a cell and subsequent isolation thereof, e.g. as described in DE 100 31 999 A1. Episomal plasmids, for example, are employed for increasing the expression of the

respective genes. In such plasmids, the nucleic acid molecule to be (over) expressed or encoding the polypeptide or enzyme to be (over) expressed may be placed under the control of a strong inducible promoter such as the lac promoter, located upstream of the gene. A promoter is a DNA sequence consisting of about 40 to 50 base pairs which constitutes the binding site for an RNA polymerase holoenzyme and the transcriptional start point (M. Pátek, J. Holátko, T. Busche, J. Kalinowski, J. Nešvera, Microbial Biotechnology 2013, 6, 103-117), whereby the strength of expression of the controlled polynucleotide or gene can be influenced. A “functional linkage” is obtained by the sequential arrangement of a promoter with a gene, which leads to a transcription of the gene.

(12) Suitable strong promoters or methods of producing such promoters for increasing expression are known from the literature (e.g. S. Lisser & H. Margalit, Nucleic Acid Research 1993, 21, 1507-1516; M. Pátek and J. Nesvera in H. Yukawa and M Inui (eds.), *Corynebacterium glutamicum*, Microbiology Monographs 23, Springer Verlag Berlin Heidelberg 2013, 51-88; B. J. Eikmanns, E. Kleinertz, W. Liebl, H. Sahm, Gene 1991, 102, 93-98). For instance, native promoters may be optimized by altering the promoter sequence in the direction of known consensus sequences with respect to increasing the expression of the genes functionally linked to these promoters (M. Pátek, B. J. Eikmanns, J. Pátek, H. Sahm, Microbiology 1996, 142, 1297-1309; M. Pátek, J. Holátko, T. Busche, J. Kalinowski, J. Nešvera, Microbial Biotechnology 2013, 6, 103-117).

(13) Constitutive promoters are also suitable for the (over)expression, in which the gene encoding the enzyme activity is expressed continuously under the control of the promoter such as, for example, the glucose dependent deo promoter. Chemically induced promoters are also suitable, such as tac, lac, rha or trp. The most widespread system for the induction of promoters is the lac operon of *E. coli*. In this case, either lactose or isopropyl β -D-thiogalactopyranoside (IPTG) is used as inducer. Also, systems using arabinose (e.g. the pBAD system) or rhamnose (e.g. *E. coli* KRX) are common as inducers. A system for physical induction is, for example, the temperature-induced cold shock promoter system based on the *E. coli* cspA promoter from Takara or Lambda PL and also osmotically inducible promoters, for example, osmB (e.g. WO 95/25785 A1).

(14) Suitable plasmids or vectors are in principle all embodiments available for this purpose to the person skilled in the art. The state of the art describes standard plasmids that may be used for this purpose, for example the pET system of vectors exemplified by pET-3a or pET-26b (+) (commercially available from Novagen). Further plasmids and vectors can be taken, for example, pOM21 described in WO 2004/111227 A2, pOM22 described in WO 00/058449 A1 or pOM18 described in WO 2013/072486 A1 or from the brochures of the companies Novagen, Promega, New England Biolabs, Clontech or Gibco BRL. Further preferred plasmids and vectors can be found in: Glover, D. M. (1985) DNA cloning: a practical approach, Vol. I-III, IRL Press Ltd., Oxford; Rodriguez, R. L. and Denhardt, D. T (eds) (1988) Vectors: a survey of molecular cloning vectors and their uses, 179-204, Butterworth, Stoneham; Goeddel, D. V. (1990) Systems for heterologous gene expression, Methods Enzymol. 185, 3-7; Sambrook, J.; Fritsch, E. F. and Maniatis, T. (1989), Molecular cloning: a laboratory manual, 2nd ed., Cold Spring Harbor Laboratory Press, New York. Of these plasmids, pOM21 and pOM22 are preferred.

(15) The plasmid vector, which contains the gene to be amplified, is then converted to the desired strain, e.g. by conjugation or transformation. The method of conjugation is described, for example, by A. Schäfer, J. Kalinowski, A. Pühler, Applied and Environmental Microbiology 1994, 60, 756-759. Methods for transformation are described, for example, in G. Thierbach, A. Schwarzer, A. Pühler, Applied Microbiology and Biotechnology 1988, 29, 356-362, L. K. Dunican & E. Shivnan, Bio/Technology 1989, 7, 1067-1070 and A. Tauch, O. Kirchner, L. Wehmeier, J. Kalinowski, A. Pühler, FEMS Microbiology Letters 1994, 123, 343-347. After homologous recombination by means of a “cross-over” event, the resulting strain contains at least two copies of the gene concerned.

(16) The desired enzyme can be isolated by disrupting cells which contain the desired activity in a

manner known to the person skilled in the art, for example with the aid of a ball mill, a French press or of an ultrasonic disintegrator and subsequently separating off cells, cell debris and disruption aids, such as, for example, glass beads, by centrifugation for 10 minutes at 13,000 rpm and 4° C. Using the resulting cell-free crude extract, enzyme assays with subsequent LC-ESI-MS detection of the products can then be carried out. Alternatively, the enzyme can be enriched in the manner known to the person skilled in the art by chromatographic methods (such as nickel-nitrilotriacetic acid affinity chromatography, streptavidin affinity chromatography, gel filtration chromatography or ion-exchange chromatography) or else purified to homogeneity. Quantification of the enzyme can be performed by methods known to the person skilled in the art, for example by determination of the concentration of the respective polypeptide of the enzyme (e.g. carbamoylase, hydantoinase and racemase) in the obtained solution by SDS page and analysis of the respective bands via the software GelQuant® (BiochemLabSolutions).

(17) Moreover, whether or not a nucleic acid or polypeptide is (over) expressed, may be determined by way of quantitative PCR reaction in the case of a nucleic acid molecule, SDS polyacrylamide electrophoreses, Western blotting or comparative activity assays in the case of a polypeptide. Genetic modifications may be directed to transcriptional, translational, and/or post-translational modifications that result in a change of enzyme activity and/or selectivity under selected and/or identified culture conditions.

4.3 Variants

(18) In the context of the present invention, the term “variant” with respect to polypeptide sequences refers to a polypeptide sequence with a degree of identity to the reference sequence (“sequence identity”) of at least 60%, preferably at least 70%, more preferably at least 71%, more preferably at least 72%, more preferably at least 73%, more preferably at least 74%, more preferably at least 75%, more preferably at least 76%, more preferably at least 77%, more preferably at least 78%, more preferably at least 79%, more preferably at least 80%, more preferably at least 81%, more preferably at least 82%, more preferably at least 83%, more preferably at least 84%, more preferably at least 85%, more preferably at least 86%, more preferably at least 87%, more preferably at least 88%, more preferably at least 89%, more preferably at least 90%, more preferably at least 91%, more preferably at least 92%, more preferably at least 93%, more preferably at least 94%, more preferably at least 95%, more preferably at least 96%, more preferably at least 97%, more preferably at least 98%, more preferably at least 99%, more preferably at least 99.9%. In still further particular embodiments, the degree of identity is at least 98.0%, more preferably at least 98.2%, more preferably at least 98.4%, more preferably at least 98.6%, more preferably at least 98.8%, more preferably at least 99.0%, more preferably at least 99.1%, more preferably at least 99.2%, more preferably at least 99.3%, more preferably at least 99.4%, more preferably at least 99.5%, more preferably at least 99.6%, more preferably at least 99.7%, more preferably at least 99.8%, or at least more preferably at least 99.9%.

(19) It goes without saying that a “variant” of a certain polypeptide sequence is not identical to the polypeptide sequence.

(20) Such variants may be prepared by introducing deletions, insertions, substitutions, or combinations thereof, in particular in amino acid sequences, as well as fusions comprising such macromolecules or variants thereof.

(21) Modifications of amino acid residues of a given polypeptide sequence which lead to no significant modifications of the properties and function of the given polypeptide are known to those skilled in the art. Thus for example many amino acids can often be exchanged for one another without problems; examples of such suitable amino acid substitutions are: Ala by Ser; Arg by Lys; Asn by Gln or His; Asp by Glu; Cys by Ser; Gln by Asn; Glu by Asp; Gly by Pro; His by Asn or Gln; Ile by Leu or Val; Leu by Met or Val; Lys by Arg or Gln or Glu; Met by Leu or Ile; Phe by Met or Leu or Tyr; Ser by Thr; Thr by Ser; Trp by Tyr; Tyr by Trp or Phe; Val by Ile or Leu. It is

also known that modifications, particularly at the N- or C-terminus of a polypeptide in the form of for example amino acid insertions or deletions, often exert no significant influence on the function of the polypeptide.

(22) In line with this, preferable variants according to the invention of any of SEQ ID NO: 1, SEQ ID NO: 2, SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 9, SEQ ID NO: 10, SEQ ID NO: 11, SEQ ID NO: 12, SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15, SEQ ID NO: 16, SEQ ID NO: 17, SEQ ID NO: 18, SEQ ID NO: 19, SEQ ID NO: 20, SEQ ID NO: 21, SEQ ID NO: 22, SEQ ID NO: 23, SEQ ID NO: 24, SEQ ID NO: 25, SEQ ID NO: 26, respectively, have a polypeptide sequence that comprises the complete polypeptide sequence of SEQ ID NO: 1, SEQ ID NO: 2, SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 9, SEQ ID NO: 10, SEQ ID NO: 11, SEQ ID NO: 12, SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15, SEQ ID NO: 16, SEQ ID NO: 17, SEQ ID NO: 18, SEQ ID NO: 19, SEQ ID NO: 20, SEQ ID NO: 21, SEQ ID NO: 22, SEQ ID NO: 23, SEQ ID NO: 24, SEQ ID NO: 25, SEQ ID NO: 26, respectively, or at least the amino acids of the respective sequence SEQ ID NO: 1, SEQ ID NO: 2, SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 9, SEQ ID NO: 10, SEQ ID NO: 11, SEQ ID NO: 12, SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15, SEQ ID NO: 16, SEQ ID NO: 17, SEQ ID NO: 18, SEQ ID NO: 19, SEQ ID NO: 20, SEQ ID NO: 21, SEQ ID NO: 22, SEQ ID NO: 23, SEQ ID NO: 24, SEQ ID NO: 25, SEQ ID NO: 26 that are essential for the function, for example the catalytic activity of a protein, or the fold or structure of the protein. The other amino acids may be deleted, substituted or replaced by insertions or essential amino acids are replaced in a conservative manner to the effect that the activity of the enzyme, in particular the L-carbamoylase (SEQ ID NO: 1, SEQ ID NO: 2, SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 9), hydantoinase (SEQ ID NO: 10, SEQ ID NO: 11, SEQ ID NO: 12, SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15, SEQ ID NO: 16), hydantoin racemase (SEQ ID NO: 17, SEQ ID NO: 18, SEQ ID NO: 19, SEQ ID NO: 20, SEQ ID NO: 21, SEQ ID NO: 22, SEQ ID NO: 23, SEQ ID NO: 24, SEQ ID NO: 25, SEQ ID NO: 26), is preserved.

3.4 Sequence Identity

(23) The person skilled in the art is aware that various computer programs are available for the calculation of similarity or identity between two nucleotide or polypeptide sequences.

(24) Preferred methods for determining the sequence identity initially generate the greatest alignment between the sequences to be compared. Computer programs for determining the sequence identity include, but are not limited to, the GCG program package including GAP [J. Deveroy et al., Nucleic Acid Research 1984, 12, page 387, Genetics Computer Group University of Wisconsin, Medicine (WI)], and BLASTP, BLASTN and FASTA (S. Altschul et al., Journal of Molecular Biology 1990, 215, 403-410). The BLAST program can be obtained from the National Center for Biotechnology Information (NCBI) and from other sources (BLAST Handbook, S. Altschul et al., NCBI NLM NIH Bethesda ND 22894; S. Altschul et al., above).

(25) For instance, the percentage identity between two polypeptide sequences can be determined by the algorithm developed by S. B. Needleman & C. D. Wunsch, J. Mol. Biol. 1970, 48, 443-453, which has been integrated into the GAP program in the GCG software package, using either a BLOSUM62 matrix or a PAM250 matrix, a gap weight of 16, 14, 12, 10, 8, 6 or 4 and a length weight of 1, 2, 3, 4, 5 or 6. The person skilled in the art will recognize that the use of different parameters will lead to slightly different results, but that the percentage identity between two polypeptide overall will not be significantly different. The BLOSUM62 matrix is typically used applying the default settings (gap weight: 12, length weight: 1).

(26) In the context of the present invention, a sequence identity of 60% according to the above algorithm means 60% homology. The same applies to higher sequence identities.

(27) Most preferably, the degree of identity between sequences is determined in the context of the

present invention by the programme “Needle” using the substitution matrix BLOSUM62, the gap opening penalty of 10, and the gap extension penalty of 0.5. The Needle program implements the global alignment algorithm described in S. B. Needleman & C. D. Wunsch, J. Mol. Biol. 1970, 48, 443-453. The substitution matrix used according to the present invention is BLOSUM62, gap opening penalty is 10, and gap extension penalty is 0.5. The preferred version used in the context of this invention is the one presented by F. Madeira, Y. M. Park, J. Lee, N. Buso, T. Gur, N. Madhusoodanan, P. Basutkar, A. R. N. Tivey, S. C. Potter, R. D. Finn, Nucleic Acids Research 2019, 47, W636-W641, Web Server issue (preferred version accessible online on Mar. 4, 2022 via https://www.ebi.ac.uk/Tools/psa/emboss_needle/).

(28) In a particular embodiment, the percentage of identity of an amino acid sequence of a polypeptide with, or to, a reference polypeptide sequence is determined by i) aligning the two amino acid sequences using the Needle program, with the BLOSUM62 substitution matrix, a gap opening penalty of 10, and a gap extension penalty of 0.5; ii) counting the number of exact matches in the alignment; iii) dividing the number of exact matches by the length of the longest of the two amino acid sequences, and iv) converting the result of the division of iii) into percentage.

4.5 First Aspect: Method for Production of a Glufosinate P-Ester

4.5.1 Step (c)

4.5.1.1 Enzymatic Catalysis of Step (c)

(29) The present invention relates in a first aspect to a method for the production of an L-glufosinate P-ester according to formula L-(I):

(30) ##STR00014##

(31) The method according to the first aspect of the invention comprises a step (c).

(32) In step (c), a compound according to formula L-(II) is reacted to give a compound according to formula L-(I). Formula L-(II) has the following formula:

(33) ##STR00015##

(34) The reaction according step (c) is enzymatically catalyzed, namely it is catalyzed by a carbamoylase E.sub.1.

(35) R.sup.1L is a radical selected from the formulae (VIII), (IX), (X), preferably selected from the formulae (IX), (X), more preferably R.sup.1L has formula (X).

(36) R.sup.2L is a radical selected from the formulae (VIII), (IX), (X), preferably selected from the formulae (IX), (X), more preferably R.sup.2L has formula (X).

(37) Formulae (VIII), (IX), (X) are as follows:

(38) ##STR00016##

(39) When R.sup.1L is a radical of the formula (VIII), the radical R.sup.1L may be selected from the formulae L-(VIII) and D-(VIII), preferably L-(VIII).

(40) When R.sup.1L is a radical of the formula (IX), the radical R.sup.1L may be selected from the formulae L-(IX) and D-(IX), preferably L-(IX).

(41) When R.sup.1L is a radical of the formula (X), the radical R.sup.1L may be selected from the formulae L-(X) and D-(X), preferably L-(X).

(42) When R.sup.2L is a radical of the formula (VIII), the radical R.sup.2L may be selected from the formulae L-(VIII) and D-(VIII), preferably L-(VIII).

(43) When R.sup.2L is a radical of the formula (IX), the radical R.sup.2L may be selected from the formulae L-(IX) and D-(IX), preferably L-(IX).

(44) When R.sup.2L is a radical of the formula (X), the radical R.sup.2L may be selected from the formulae L-(X) and D-(X), preferably L-(X).

(45) Formulae L-(VIII), D-(VIII), L-(IX), D-(IX), L-(X), and D-(X) are as follows:

(46) ##STR00017##

4.5.1.2 Enantioselective and Enantiospecific Catalysis of Step (c)

(47) Step (c) of the method according to the first aspect of the present invention is preferably L-enantioselective, even more preferably L-enantiospecific.

(48) In such a preferred embodiment, L-(II) is in particular employed in step (c) as a mixture M.sub.II comprising at least one compound L-(II) and at least one compound D-(II), wherein D-(II) has the following formula:

(49) ##STR00018##

(50) R.sup.2D is a radical selected from the formulae (VIII), (IX), (X), preferably selected from the formulae (IX), (X), more preferably R.sup.2D has formula (X).

(51) When R.sup.2D is a radical of the formula (VIII), the radical R.sup.2D may be selected from the formulae L-(VIII) and D-(VIII), preferably L-(VIII).

(52) When R.sup.2D is a radical of the formula (IX), the radical R.sup.2D may be selected from the formulae L-(IX) and D-(IX), preferably L-(IX).

(53) When R.sup.2D is a radical of the formula (X), the radical R.sup.2D may be selected from the formulae L-(X) and D-(X), preferably L-(X).

(54) Mixture M.sub.II preferably comprises at least one compound L-(II) and at least one compound D-(II), wherein R.sup.2L=R.sup.2D.

(55) For such mixtures Mu, it may be observed that the at least one compound D-(II) undergoes a reaction according to step (c)*, i.e. parallel to step (c). Namely, in the reaction according to step (c)*, the at least one compound D-(II) in mixture M.sub.II is reacted to give a compound according to formula D-(I):

(56) ##STR00019##

(57) R.sup.1D is a radical selected from the above-mentioned formulae (VIII), (IX), (X), preferably selected from the above-mentioned formulae (IX), (X), more preferably R.sup.1D has formula (X).

(58) When R.sup.1D is a radical of the formula (VIII), the radical R.sup.1D may be selected from the above-mentioned formulae L-(VIII) and D-(VIII), preferably L-(VIII).

(59) When R.sup.1D is a radical of the formula (IX), the radical R.sup.1D may be selected from the above-mentioned formulae L-(IX) and D-(IX), preferably L-(IX).

(60) When R.sup.1D is a radical of the formula (X), the radical R.sup.1D may be selected from the above-mentioned formulae L-(X) and D-(X), preferably L-(X).

(61) In case that step (c) is "L-enantioselective", this means that in case a mixture M.sub.II is employed in step (c), then there is either no reaction according to step (c)* of any compound according to formula D-(II) to a compound according to D-(I) in the mixture M.sub.II or, in case there is a reaction according to step (c)*, then there is at least one pair of compounds D-(II)  L-(II)  in the mixture M.sub.III, for which the reaction according to step (c)* of D-(II)  to give D-(I)  is lesser than the rate of the reaction according to step (c) of L-(II)  to give L-(I) .

(62) D-(II)  is a compound according to formula D-(II), and L-(II)  is a compound according to formula L-(II), wherein the radical R.sup.2L in L-(II)  and the radical R.sup.2D in D-(II)  are identical.

(63) D-(I)  is a compound according to formula D-(I), and L-(I)  is a compound according to formula L-(I), wherein the radical R.sup.2L in L-(I)  and the radical R.sup.2D in D-(I)  are identical.

(64) Preferably, the radical R.sup.2L in L-(II)  and the radical R.sup.1L in L-(I)  are identical. Preferably, the radical R.sup.2D in D-(II)  and the radical R.sup.1D in D-(I)  are identical.

(65) Step (c) is "L-enantiospecific", if the rate of reaction according to step (c)* is essentially zero, i.e. there is no reaction according to step (c)* of any compound according to formula D-(II) to a compound according to D-(I) in the mixture M.sub.II.

(66) In a preferred embodiment, the molar ratio of all compounds according to formula L-(II) in the mixture M.sub.II to all compounds according to formula D-(II) in the mixture M.sub.II is essentially 1:1.

(67) In other preferred embodiments, the molar ratio of all compounds according to formula L-(II) in the mixture M.sub.II to all compounds according to formula D-(II) in the mixture M.sub.II is in the range of from 3:2 to 1:99, more preferably in the range of from 1.01:1 to 1:99, more preferably in the range of from 1:1 to 1:99, more preferably in the range of from 1:1.01 to 1:99, more preferably in the range of from 1:1.01 to 1:9, more preferably in the range of from 1:1.01 to 1:8, more preferably in the range of from 1:1.01 to 1:3.

(68) Alternatively, D-(II) is comprised in an excess to L-(II) in mixture Mu, meaning that, while at least one compound according to formula L-(II) is present in the mixture Mu, the molar ratio of all compounds according to formula L-(II) in the mixture M.sub.II to all compounds according to formula D-(II) in the mixture M.sub.II is <1:1, preferably <0.9:1, more preferably <0.75:1, more preferably <0.5:1, more preferably <0.2:1, more preferably <0.1:1, more preferably <0.01:1.

(69) Step (c) is in particular L-enantioselective, if it is catalyzed by an L-carbamoylase E.sub.1, which may be determined by the skilled person as set forth under 4.5.4.3.

(70) In case step (c) is L-enantioselective with respect to a given pair of compounds D-(II)

/L-(II) , the reaction according to step (c) of L-(II)

 to give L-(I)  proceeds preferably at a reaction rate that is at least 2 times greater, preferably at least 10 times greater, more preferably at least 100 times greater, even more preferably at least 103 times greater, even more preferably at least 104 times greater, even more preferably at least 105 times greater than the reaction rate at which step (c)* of D-(II)

 to give D-(I)  proceeds.

(71) To quantify for a given pair of compounds D-(II) /L-(II)  the factor at which the reaction rate of step (c) proceeds compared to the reaction rate of step (c)*, the following test may be carried out: (1) An equimolar mixture [molar ratio of the two compounds L-(II)  and D-(II)  is 1:1] is subjected to the respective

reactions conditions and the development of the two products L-(I)  and D-(I)

 is monitored over time (e.g. by LC-MS as set forth under item 5.4). (2) When

n.sub.LI10=10 mol-% of the initially employed L-(II)  has reacted to the product L-(I) , the amount of D-(I)  that was formed by reaction from D-(II)  [in mol-% relative to the initially employed D-(II)+] is measured

(=n.sub.DI10). (3) The ratio of n.sub.LI10/n.sub.DI10=10/n.sub.DI10 gives the factor at which the reaction rate of step (c) proceeds compared to the reaction rate of step (c)* for the given pair of compounds D-(II) /L-(II) . If n.sub.LI10/n.sub.DI10>1, the reaction rate of step (c) is greater than the reaction rate of step (c)* for the given pair of compounds D-(II) /L-(II) . If n.sub.LI10/n.sub.DI10<1, the reaction rate of step (c) is lesser than the reaction rate of step (c)* for the given pair of compounds D-(II)

/L-(II) . If n.sub.LI10/n.sub.DI10=1, the reaction rates of steps (c) and (c)* are the same for the given pair of compounds D-(II) /L-(II) .

4.5.2 Carbamoylases

(72) The reaction according to step (c) of the method according to the first aspect of the invention is catalyzed by a carbamoylase E.sub.1.

(73) Namely, it was surprisingly found that carbamoylases accept compounds of formula L-(II) as substrates and convert them to products L-(I), and hence can be used to catalyze the reaction according to step (c). This finding is of high scientific and economic value, as it opens new synthetic routes based on new starting materials for the production of L-glufosinate P-esters and L-glufosinate. Even more surprisingly, it was found that L-glufosinate carbamoylate, i.e. the compound according to L-(II), in which R=H, does not undergo reaction by carbamoylases to give L-glufosinate.

(74) In nature, carbamoylases generally catalyze the following reaction <1>, wherein R.sup.X may be an organic residue, e.g. a side chain of one of the naturally occurring amino acids.

(75) ##STR00020##

(76) It was now surprisingly found that carbamoylases also accept substrates in which $R.\sup.X=R.\sup.Y=$

(77) ##STR00021##

wherein $R=R.\sup.2L$.

(78) Surprisingly, they do not accept substrates in which $R.\sup.X=R.\sup.Y=$

(79) ##STR00022##

wherein $R=H$.

(80) In the context of the present invention, a “carbamoylase E.sub.1” is a carbamoylase that catalyzes the following reaction <1A> of a carbamoyl substrate SL to the respective amino acid product P.sub.L, wherein $R.\sup.X=R.\sup.Y=$

(81) ##STR00023##

wherein $R=R.\sup.2L$:

(82) ##STR00024##

(83) In particular, the carbamoylase E.sub.1 is a “L-carbamoylase”, i.e. it has a greater catalytic activity for reaction <1A> than for reaction <1B>, wherein $R.\sup.X$ in the substrate Sp in the reaction <1B> and $R.\sup.X$ in the substrate SL in the reaction <1A> are identical, and $R.\sup.X=R.\sup.Y=$

(84) ##STR00025##

wherein $R=R.\sup.2L$

(85) ##STR00026##

(86) As an L-carbamoylase has a higher catalytic activity for reaction <1A> than for reaction <1B>, it is “L-enantioselective”. An L-carbamoylase that has no catalytic activity for reaction <1B> and thus only has catalytic activity for reaction <1A> is “L-enantiospecific”.

(87) A “D-carbamoylase” is defined as a carbamoylase which is “D-enantioselective”, i.e. it has a higher catalytic activity for reaction <1B> than for reaction <1A>. A D-carbamoylase that does not catalyze reaction <1A> and thus only has catalytic activity for reaction <1B> is “D-enantiospecific”.

(88) A carbamoylase which has the same catalytic activity for reaction <1B> as for reaction <1A>, is referred to as “non-enantioselective carbamoylase”.

(89) For determination whether a carbamoylase may be denoted as “L-carbamoylase”, “D-carbamoylase” or “non-enantioselective carbamoylase” in the context of the present invention, the procedure according to Assay B (item 4.5.4) may preferably be used.

(90) The carbamoylase E.sub.1, in particular the L-carbamoylase E.sub.1, that may be used in step (c) of the method according to the first aspect of the invention may originate from *Achromobacter* sp., in particular *Achromobacter xylosoxidans*; *Agrobacterium* sp., in particular *Agrobacterium tumefaciens*; *Arthrobacter* sp., in particular *Arthrobacter crystallopoietes*, *Arthrobacter aurescens*, *Arthrobacter* sp. BT801; *Bacillus* sp., in particular *Bacillus fordii*; *Blastobacter* sp.; *Bradyrhizobium* sp., in particular *Bradyrhizobium japonicum*; *Brevibacillus* sp., in particular *Brevibacillus reuszeri*; *Comamonas* sp.; *Ensifer* sp., in particular *Ensifer adhaerens*; *Flavobacterium* sp. *Geobacillus* sp., in particular *Geobacillus kaustophilus*, *Geobacillus stearothermophilus*; *Microbacterium* sp., in particular *Microbacterium liquefaciens* strain AJ3912; *Paenarthrobacter* sp., in particular *Paenarthrobacter aurescens*; *Pasteurella* sp.; *Pseudomonas* sp.; *Ralstonia* sp., in particular *Ralstonia pickettii*; *Sinorhizobium* sp., in particular *Sinorhizobium meliloti*.

(91) An L-carbamoylase E.sub.1 suitable for the method according to the present invention may be the enzyme HyuC, which originates from *Arthrobacter*. Other enzymes are AmaB, AtcC, Inc, SinmeB_2280.

(92) WO 01/23582 A1 discloses an example of an enzyme having carbamoylase activity according to the invention.

(93) The carbamoylase E.sub.1 that may be used in step (c) of the method according to the present invention may be an L-carbamoylase categorized in the EC class EC 3.5.1.87.

(94) L-carbamoylase enzymes are for example described by J. Ogawa, H. Miyake, S. Shimizu, *Appl. Microbiol. Biotechnol.* 1995 43, 1039-1043 and in WO 01/23582 A1.

(95) A L-carbamoylase E.sub.1 that may preferably be used in step (c) according to the first aspect of the invention may originate from *Arthrobacter* sp., in particular *Arthrobacter crystallopoietes*, *Arthrobacter aurescens*, *Arthrobacter* sp. BT801, *Arthrobacter aurescens* DSM 3747; *Bacillus* sp., in particular *Bacillus fordii*; *Geobacillus* sp., in particular *Geobacillus stearothermophilus*, *Geobacillus kaustophilus*; *Microbacterium* sp., in particular *Microbacterium liquefaciens* strain AJ3912; *Paenarthrobacter* sp., in particular *Paenarthrobacter aurescens*; *Pseudomonas* sp., in particular *Pseudomonas* sp. QR-101; *Sinorhizobium* sp., in particular *Sinorhizobium meliloti*. Even more preferably, the L-carbamoylase E.sub.1 that may preferably be used in step (c) according to the first aspect of the invention may originate from *Arthrobacter* sp., in particular *Arthrobacter crystallopoietes*, *Arthrobacter aurescens*, *Arthrobacter* sp. BT801, *Arthrobacter aurescens* DSM 3747, most preferably from *Arthrobacter aurescens* DSM 3747.

(96) The respective sequences can be derived from databases such as the Braunschweig Enzyme Database (BRENDA, Germany, available under www.brenda-enzymes.org/index.php), the National Center for Biotechnological Information (NCBI, available under www.ncbi.nlm.nih.gov/) or the Kyoto Encyclopedia of Genes and Genomes (KEGG, Japan, available under www.genome.jp/kegg/).

(97) The following table 1 gives preferred examples for polypeptide sequences of L-carbamoylases E.sub.1 that may be preferably used in step (c) of the method according to the first aspect of the invention. The genes encoding the respective L-carbamoylase E.sub.1 and the respective accession code are indicated as far as known.

(98) TABLE-US-00001 TABLE 1 L-Carbamoylases (EC 3.5.1.87) GenBank/UniProt SEQ ID NO: of the Strain Gene name accession polypeptide *Arthrobacter aurescens* DSM hyuC SEQ ID NO: 1 3747 *Geobacillus amaB* Q53389 SEQ ID NO: 2 *stearothermophilus Pseudomonas* sp. QR-101 atcC H9B8T5 SEQ ID NO: 3 *Geobacillus kaustophilus* Inc Q8GQG5 SEQ ID NO: 4 *Paenarthrobacter aurescens* hyuC Q9F464 SEQ ID NO: 5 *Sinorhizobium meliloti* SinmeB_2280 A0A0E0UEY4 SEQ ID NO: 6 *Bacillus fordii* strain MH602 not assigned ABL14248 SEQ ID NO: 7 *Arthrobacter* sp. BT801 hyuC AAL55413 SEQ ID NO: 8 *Microbacterium liquefaciens* not assigned SEQ ID NO: 9 strain AJ3912

(99) In a preferred embodiment of the method according to the first aspect of the present invention, the reaction according to step (c) is catalyzed by an L-carbamoylases E.sub.1, wherein the polypeptide sequence of E.sub.1 is selected from the group consisting of SEQ ID NO: 1 and variants thereof, SEQ ID NO: 2 and variants thereof, SEQ ID NO: 3 and variants thereof, SEQ ID NO: 4 and variants thereof, SEQ ID NO: 5 and variants thereof, SEQ ID NO: 6 and variants thereof, SEQ ID NO: 7 and variants thereof, SEQ ID NO: 8 and variants thereof, SEQ ID NO: 9 and variants thereof, preferably SEQ ID NO: 1 and variants thereof.

4.5.3 Assays A.SUB.L .and A.SUB.D .for Determining Carbamoylase Activity

(100) The skilled person is aware of carbamoylases, in particular L-carbamoylases, that may be used in step (c) of the method according to the first aspect of the invention.

(101) In particular, Assay A.sub.L, described in the following, may be used to determine carbamolyase and L-carbamoylase activity of a given enzyme E.sub.X and may advantageously be used according to the invention to determine carbamoylase and L-carbamoylase activity in variants of SEQ ID NO: 1, variants of SEQ ID NO: 2, variants of SEQ ID NO: 3, variants of SEQ ID NO: 4, variants of SEQ ID NO: 5, variants of SEQ ID NO: 6, variants of SEQ ID NO: 7, variants of SEQ ID NO: 8, variants of SEQ ID NO: 9.

(102) For comparative reasons, Assay A.sub.D may be used to determine D-carbamoylase activity of a given enzyme E.sub.X.

(103) For the purpose of Assay A.sub.L and Assay A.sub.D, the molar mass of the enzyme E.sub.X to be tested is calculated as the molar mass of the polypeptide sequence of E.sub.X.

(104) Moreover, a substrate according to formula L-(II), wherein $R_{sup.2L}=n$ -butyl, is used in Assay A.sub.L so that there is only one chiral carbon atom in the substrate. The compound according to formula L-(II), wherein $R_{sup.2L}=n$ -butyl, is denoted as “L-(II)+”.

(105) A product according to formula L-(I), wherein $R_{sup.1L}=n$ -butyl, is detected in Assay A.sub.L.

(106) The compound according to formula L-(I), wherein $R_{sup.1L}=n$ -butyl, is denoted as “L-(I) ”.

(107) Moreover, a substrate according to formula D-(II), wherein $R_{sup.2D}=n$ -butyl, is used in Assay A.sub.D so that there is only one chiral carbon atom in the substrate. The compound according to formula D-(II), wherein $R_{sup.2D}=n$ -butyl, is denoted as “D-(II) ”.

(108) A product according to formula D-(I), wherein $R_{sup.1D}=n$ -butyl, is detected in Assay A.sub.D.

(109) The compound according to formula D-(I), wherein $R_{sup.1D}=n$ -butyl, is denoted as “D-(I) ”.

(110) 4.5.3.1 Assay A.sub.L:

(111) To 0.9 ml of an aqueous reaction solution (phosphate buffer, pH 7.2, 10 mM $MnCl_{sub.2}$), containing 50 mM of a compound according to formula L-(II)  are added 400 nmol of E.sub.X in 0.1 ml aqueous phosphate buffer (pH 7.2, 10 mM $MnCl_{sub.2}$). The resulting solution is incubated at 25° C., and the pH is held at pH 7.2 by addition of 0.5 M NaOH. After 300 minutes, the reaction is stopped by addition of 2 M HCl to achieve a pH of 2.5, and the molar amount of the respective compound according to formula L-(I)  is determined. L-(I)  may be detected by the LC-MS method as described in the example section (item 5.4) for detection of LGA.

(112) 4.5.3.2 Assay A.sub.D:

(113) To 0.9 ml of an aqueous reaction solution (phosphate buffer, pH 7.2, 10 mM $MnCl_{sub.2}$), containing 50 mM of a compound according to formula D-(II)  are added 400 nmol of E.sub.X in 0.1 ml aqueous phosphate buffer (pH 7.2, 10 mM $MnCl_{sub.2}$). The resulting solution is incubated at 25° C., and the pH is held at pH 7.2 by addition of 0.5 M NaOH. After 300 minutes, the reaction is stopped by addition of 2 M HCl to achieve a pH of 2.5, and the molar amount of the respective compound according to formula D-(I)  is determined. D-(I)  may be detected by the LC-MS method described in the example section (item 5.4) for detection of LGA.

4.5.4 Assay B for Identifying Carbamoylases, L-Carbamoylases, D-Carbamoylases, L- and D-Enantiospecificity

(114) The carbamoylase E.sub.1 according to the invention is preferably an L-carbamoylase, more preferably L-enantiospecific.

(115) Whether a given enzyme E.sub.x may be considered a carbamoylase E.sub.1, in particular an L-carbamoylase, may be determined in the context of the present invention by the following Assay B.

(116) 4.5.4.1 Assay B:

(117) B-1. Firstly, Assay A.sub.L as set forth under item 4.5.3.1 is conducted, and the obtained molar amount of the compound of the formula L-(I)  is determined according to Assay A.sub.L.

(118) B-2. Secondly, Assay A.sub.D as set forth under item 4.5.3.2 is conducted, and the obtained molar amount of the compound of the formula D-(I)  is determined according to Assay A.sub.D.

(119) B-3. Then, step B-1 is repeated, except that instead of the addition of 400 nmol E.sub.x in 0.1 ml aqueous phosphate buffer (pH 7.2, 10 mM $MnCl_{sub.2}$), 0.1 ml aqueous phosphate buffer (pH

7.2, 10 mM MnCl.sub.2) without E.sub.x is added.

(120) B-4. Then, step B-2 is repeated, except that instead of the addition of 400 nmol E.sub.x in 0.1 ml aqueous phosphate buffer (pH 7.2, 10 mM MnCl.sub.2), 0.1 ml aqueous phosphate buffer (pH 7.2, 10 mM MnCl.sub.2) without E.sub.x is added.

(121) 4.5.4.2 Carbamoylase Activity

(122) If the molar amount of the compound of the formula L-(I)  that is determined in step B-1, is greater than the molar amount of the compound of the formula L-(I)  that is determined in step B-3, then E.sub.x is deemed to have carbamoylase activity, and hence may be considered a carbamoylase E.sub.1 in the context of the invention.

(123) 4.5.4.3 L-Carbamoylase Activity

(124) 4.5.4.3.1 L-Carbamoylases

(125) (i) If the molar amount of the compound of formula L-(I)  that is determined in step B-1, is greater than the molar amount of the compound of formula L-(I)  that is determined in step B-3, and (ii) if the molar amount of the compound of formula D-(I)  that is determined in step B-2, is greater than or the same as the molar amount of the compound of formula D-(I)  that is determined in step B-4, and (iii) if, in addition, the molar amount of the compound of formula of L-(I)  that is determined in step B-1 is greater than the molar amount of the compound of formula of D-(I)  that is determined in step B-2, then E.sub.x is deemed to have L-carbamoylase activity, and hence may be considered an L-carbamoylase in the context of the invention. In this case, E.sub.x is deemed to be “L-enantioselective” in the context of the invention.

(126) For the sake of clarity, it is pointed out that condition (iii) is automatically fulfilled in those cases in which the molar amount of the compound of formula D-(I)  that is determined in step B-2, is the same as the molar amount of the compound of formula D-(I)  that is determined in step B-4. (i) If the molar amount of the compound of formula L-(I)  that is determined in step B-1, is greater than the molar amount of the compound of formula L-(I)  that is determined in step B-3, and (ii) if the molar amount of the compound of formula D-(I)  that is determined in step B-2 is the same as the molar amount of the compound of formula D-(I)  that is determined in step B-4, then E.sub.x is deemed to have L-carbamoylase activity, and hence may be considered an L-carbamoylase in the context of the invention. In this case, E.sub.x is not only “L-enantioselective”, but also “L-enantiospecific” in the context of the invention.

(127) For L-carbamoylases E.sub.x that are not L-enantiospecific, the L-enantioselectivity may then be quantified by dividing the molar amount of the compound of formula L-(I)  that is determined in step B-1, by the molar amount of the compound of formula D-(I)  that is determined in step B-2, and then multiplying the obtained value by 100, giving the L-enantioselectivity of E.sub.X in %.

(128) 4.5.4.3.2 D-Carbamoylases

(129) (i) If the molar amount of the compound of formula L-(I)  that is determined in step B-1, is greater than or the same as the molar amount of the compound of formula L-(I)  that is determined in step B-3, and (ii) if the molar amount of the compound of formula D-(I)  that is determined in step B-2, is greater than the molar amount of the compound of formula D-(I)  that is determined in step B-4, and (iii) if, in addition, the molar amount of the compound of formula D-(I)  that is determined in step B-2 is greater than the molar amount of the compound of formula L-(I)  that is determined in step B-1, then E.sub.x is deemed to have D-carbamoylase activity, and hence may be considered a D-carbamoylase. In this case, E.sub.x is “D-enantioselective” in the context of the invention.

(130) For the sake of clarity it is pointed out that condition (iii) is automatically fulfilled in those cases in which the molar amount of the compound of formula L-(I)  that is

determined in step B-1 is the same as the molar amount of the compound of formula L-(I)  custom character that is determined in step B-3. (i) If the molar amount of the compound of formula L-(I)  custom character that is determined in step B-1, is the same as the molar amount of the compound of formula L-(I)  custom character that is determined in step B-3, and (ii) if the molar amount of the compound of formula D-(I)  custom character that is determined in step B-2, is greater than the molar amount of the compound of formula D-(I)  custom character that is determined in step B-4, then E.sub.x is deemed to have D-carbamoylase activity, and hence may be considered to be a D-carbamoylase. In this case, E.sub.x is not only “D-enantioselective”, but also “D-enantiospecific” in the context of the invention.

(131) For D-carbamoylases E.sub.x, that are not D-enantiospecific, the D-enantioselectivity may then be quantified by dividing the molar amount of the compound of formula D-(I)  custom character that is determined in step B-2 by the molar amount of the compound of formula L-(I)  custom character that is determined in step B-1 and then multiplying the obtained value by 100, giving the D-enantioselectivity of E.sub.x in %.

(132) 4.5.4.3.3 Non-Enantioselective Carbamoylases

(133) (i) If the molar amount of the compound of formula L-(I)  custom character that is determined in step B-1, is greater than the molar amount of the compound of formula L-(I)  custom character that is determined in step B-3, and (ii) if the molar amount of the compound of formula D-(I)  custom character that is determined in step B-2 is greater than the molar amount of the compound of formula D-(I)  custom character that is determined in step B-4, and (iii) if, in addition, the molar amount of the compound of formula D-(I)  custom character that is determined in step B-2 is the same as the molar amount of the compound of formula L-(I)  custom character that is determined in step B-1, then E.sub.x is deemed to be a non-enantioselective carbamoylase in the context of the invention. In this case, E.sub.x is “non-enantioselective”.

4.5.5 Assay C for Identifying Preferred Carbamoylase Variants of SEQ ID NOs: 1-9

(134) 4.5.5.1 L- and D-Enantioselective Carbamoylase Variants

(135) An enzyme, the polypeptide sequence of which is selected from the group consisting of SEQ ID NO: 1, SEQ ID NO: 2, SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 9 has carbamoylase, and L-carbamoylase activity.

(136) In a preferred embodiment of the method according to the first aspect of the invention, the polypeptide sequence of the L-carbamoylase E.sub.1 is selected from the group consisting of SEQ ID NO: 1 and variants thereof, SEQ ID NO: 2 and variants thereof, SEQ ID NO: 3 and variants thereof, SEQ ID NO: 4, and variants thereof, SEQ ID NO: 5 and variants thereof, SEQ ID NO: 6 and variants thereof, SEQ ID NO: 7, and variants thereof, SEQ ID NO: 8 and variants thereof, SEQ ID NO: 9 and variants thereof, more preferably SEQ ID NO: 1 and variants thereof.

(137) The term “variant” is defined under item 4.3.

(138) In the context of the invention, an enzyme E.sub.1, the polypeptide sequence of which is a variant of one of SEQ ID NO: 1, SEQ ID NO: 2, SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 9 has carbamoylase activity, preferably L-carbamoylase activity, more preferably is L-enantiospecific.

(139) Whether a given enzyme E.sub.X, the polypeptide sequence of which is a variant of one of SEQ ID NO: 1, SEQ ID NO: 2, SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 9, has carbamoylase activity, L carbamoylase activity and/or is L-enantiospecific may be determined as set forth under items 4.5.4.2 and 4.5.4.3.1, respectively.

(140) The carbamoylase activity of a given L-carbamoylase E.sub.1V, the polypeptide sequence of which is a variant of one of one of SEQ ID NO: 1, SEQ ID NO: 2, SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 9, relative to the carbamoylase activity of an L-carbamoylase E.sub.1S, wherein the polypeptide sequence of E.sub.1S is selected from of one of SEQ ID NO: 1, SEQ ID NO: 2, SEQ ID NO: 3, SEQ ID NO: 4,

SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 9, may be quantified in the context of the present invention by the following Assay C:

(141) 4.5.5.2 Assay C

(142) C-1 Assay A.sub.L as set forth under item 4.5.3.1 is conducted, wherein E.sub.1S is the enzyme to be tested. The obtained molar amount of the compound according to formula L-(I)  is determined according to Assay A.sub.L.

(143) C-2 Step C-1 is repeated, except that, instead of E.sub.1S, E.sub.1V is used as the enzyme to be tested.

(144) C-3. Then, the molar amount of the compound according to formula L-(I)  that is determined in step C-2, is divided by the molar amount of the compound according to formula L-(I)  that is determined in step C-1, and the obtained ratio is multiplied by 100, giving the carbamoylase activity of L-carbamoylase E.sub.1V, relative to the carbamoylase activity of the L-carbamoylase E.sub.1S, in %.

4.5.6 Preferred Carbamoylase Variants of SEQ ID NOs: 1-9

(145) In the context of the invention, L-carbamoylases E.sub.1, the polypeptide sequence of which is selected from the group consisting of SEQ ID NO: 1, SEQ ID NO: 2, SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 9, are generally denoted as “E.sub.1S”.

(146) L-carbamoylases E.sub.1, the polypeptide sequence of which is selected from variants of a sequence selected from the group consisting of SEQ ID NO: 1, SEQ ID NO: 2, SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 9, are generally denoted as “E.sub.1V”.

(147) In a preferred embodiment of the method according to the first aspect of the present invention, the reaction in step (c) is catalyzed by an L-carbamoylase E.sub.1, and the polypeptide sequence of the L-carbamoylase E.sub.1 is selected from the group consisting of SEQ ID NO: 1 and variants thereof, SEQ ID NO: 2 and variants thereof, SEQ ID NO: 3 and variants thereof, SEQ ID NO: 4 and variants thereof, SEQ ID NO: 5 and variants thereof, SEQ ID NO: 6 and variants thereof, SEQ ID NO: 7 and variants thereof, SEQ ID NO: 8 and variants thereof, SEQ ID NO: 9 and variants thereof. More preferably, the reaction in step (c) is catalyzed by an L-carbamoylase E.sub.1, and the polypeptide sequence of the L-carbamoylase E.sub.1 is selected from the group consisting of SEQ ID NO: 1 and variants of SEQ ID NO: 1.

(148) 4.5.6.1 Preferred Variants of SEQ ID NO: 1

(149) According to the invention, the polypeptide sequence of the L-carbamoylase E.sub.1 may also be a variant of SEQ ID NO: 1.

(150) The L-carbamoylase E.sub.1, the polypeptide sequence of which is SEQ ID NO: 1, is denoted as “E.sub.101S”. L-carbamoylases E.sub.1, the polypeptide sequence of which is selected from variants of SEQ ID NO: 1, are generally denoted as “E.sub.101V”.

(151) A variant of the polypeptide sequence of SEQ ID NO: 1 is a polypeptide with sequence identity of at least 60%, preferably $\geq 65\%$, more preferably $\geq 70\%$, more preferably $\geq 75\%$, more preferably $\geq 80\%$, more preferably $\geq 85\%$, more preferably $\geq 90\%$, more preferably $\geq 91\%$, more preferably $\geq 92\%$, more preferably $\geq 93\%$, more preferably $\geq 94\%$, more preferably $\geq 95\%$, more preferably $\geq 96\%$, more preferably $\geq 97\%$, more preferably $\geq 98\%$, more preferably $\geq 99\%$, more preferably $\geq 99.9\%$ sequence identity to polypeptide sequence SEQ ID NO: 1.

(152) The polypeptide sequence of a variant of the polypeptide sequence SEQ ID NO: 1 is not identical to SEQ ID NO: 1.

(153) According to the invention, an L-carbamoylase E.sub.101V has carbamoylase activity and L-carbamoylase activity, determined as described under items 4.5.4.2 and 4.5.4.3.1.

(154) According to the invention, an L-carbamoylase E.sub.101V preferably has carbamoylase activity of at least 1%, preferably of at least 10%, more preferably of at least 20%, more preferably of at least 30%, more preferably of at least 40%, more preferably of at least 50%, more preferably

of at least 60%, more preferably of at least 70%, more preferably of at least 80%, more preferably of at least 90%, more preferably of at least 99%, more preferably of at least 100% the carbamoylase activity of the L-carbamoylase E.sub.101S, wherein the carbamoylase activity of E.sub.101V, relative to the carbamoylase activity of E.sub.101S, is determined by Assay C described under item 4.5.5.2.

(155) It is even more preferable according to the invention, that an L-carbamoylase E.sub.101V has carbamoylase activity in the range of 1 to 1000%, preferably in the range of 5 to 500%, more preferably in the range of 10 to 400%, more preferably in the range of 40 to 200%, more preferably in the range of 50 to 150%, more preferably in the range of 60 to 140%, more preferably in the range of 70 to 130%, more preferably in the range of 80 to 120%, more preferably in the range of 90 to 110%, more preferably 100% the carbamoylase activity of the L-carbamoylase E.sub.101S, wherein the carbamoylase activity of E.sub.101V, relative to the carbamoylase activity of E.sub.101S, is determined by Assay C described under item 4.5.5.2.

(156) 4.5.6.2 Preferred Variants of SEQ ID NO: 2

(157) According to the invention, the polypeptide sequence of the L-carbamoylase E.sub.1 may also be a variant of SEQ ID NO: 2.

(158) The L-carbamoylase E.sub.1, the polypeptide sequence of which is SEQ ID NO: 2, is denoted as "E.sub.102S". L-carbamoylases E.sub.1, the polypeptide sequence of which is selected from variants of SEQ ID NO: 2, are generally denoted as "E.sub.102V".

(159) A variant of the polypeptide sequence of SEQ ID NO: 2 is a polypeptide with sequence identity of at least 60%, preferably $\geq 65\%$, more preferably $\geq 70\%$, more preferably $\geq 75\%$, more preferably $\geq 80\%$, more preferably $\geq 85\%$, more preferably $\geq 90\%$, more preferably $\geq 91\%$, more preferably $\geq 92\%$, more preferably $\geq 93\%$, more preferably $\geq 94\%$, more preferably $\geq 95\%$, more preferably $\geq 96\%$, more preferably $\geq 97\%$, more preferably $\geq 98\%$, more preferably $\geq 99\%$, more preferably $\geq 99.9\%$ sequence identity to polypeptide sequence SEQ ID NO: 2.

(160) The polypeptide sequence of a variant of the polypeptide sequence SEQ ID NO: 2 is not identical to SEQ ID NO: 2.

(161) According to the invention, an L-carbamoylase E.sub.102V has carbamoylase activity and L-carbamoylase activity, determined as described under items 4.5.4.2 and 4.5.4.3.1.

(162) According to the invention, an L-carbamoylase E.sub.102V preferably has carbamoylase activity of at least 1%, preferably of at least 10%, more preferably of at least 20%, more preferably of at least 30%, more preferably of at least 40%, more preferably of at least 50%, more preferably of at least 60%, more preferably of at least 70%, more preferably of at least 80%, more preferably of at least 90%, more preferably of at least 99%, more preferably of at least 100% the carbamoylase activity of the L-carbamoylase E.sub.102S, wherein the carbamoylase activity of E.sub.102V, relative to the carbamoylase activity of E.sub.102S, is determined by Assay C described under item 4.5.5.2.

(163) It is even more preferable according to the invention, that an L-carbamoylase E.sub.102V has carbamoylase activity in the range of 1 to 1000%, preferably in the range of 5 to 500%, more preferably in the range of 10 to 400%, more preferably in the range of 40 to 200%, more preferably in the range of 50 to 150%, more preferably in the range of 60 to 140%, more preferably in the range of 70 to 130%, more preferably in the range of 80 to 120%, more preferably in the range of 90 to 110%, more preferably 100% the carbamoylase activity of the L-carbamoylase E.sub.102S, wherein the carbamoylase activity of E.sub.102V, relative to the carbamoylase activity of E.sub.102S, is determined by Assay C described under item 4.5.5.2.

(164) 4.5.6.3 Preferred Variants of SEQ ID NO: 3

(165) According to the invention, the polypeptide sequence of the L-carbamoylase E.sub.1 may also be a variant of SEQ ID NO: 3.

(166) The L-carbamoylase E.sub.1, the polypeptide sequence of which is SEQ ID NO: 3, is denoted as "E.sub.103S". L-carbamoylases E.sub.1, the polypeptide sequence of which is selected

from variants of SEQ ID NO: 3, are generally denoted as “E.sub.103V”.

(167) A variant of the polypeptide sequence of SEQ ID NO: 3 is a polypeptide with sequence identity of at least 60%, preferably $\geq 65\%$, more preferably $\geq 70\%$, more preferably $\geq 75\%$, more preferably $\geq 80\%$, more preferably $\geq 85\%$, more preferably $\geq 90\%$, more preferably $\geq 91\%$, more preferably $\geq 92\%$, more preferably $\geq 93\%$, more preferably $\geq 94\%$, more preferably $\geq 95\%$, more preferably $\geq 96\%$, more preferably $\geq 97\%$, more preferably $\geq 98\%$, more preferably $\geq 99\%$, more preferably $\geq 99.9\%$ sequence identity to polypeptide sequence SEQ ID NO: 3.

(168) The polypeptide sequence of a variant of the polypeptide sequence SEQ ID NO: 3 is not identical to SEQ ID NO: 3.

(169) According to the invention, an L-carbamoylase E.sub.103V has carbamoylase activity and L-carbamoylase activity, determined as described under items 4.5.4.2 and 4.5.4.3.1.

(170) According to the invention, an L-carbamoylase E.sub.103V preferably has carbamoylase activity of at least 1%, preferably of at least 10%, more preferably of at least 20%, more preferably of at least 30%, more preferably of at least 40%, more preferably of at least 50%, more preferably of at least 60%, more preferably of at least 70%, more preferably of at least 80%, more preferably of at least 90%, more preferably of at least 99%, more preferably of at least 100% the carbamoylase activity of the L-carbamoylase E.sub.103S, wherein the carbamoylase activity of E.sub.103V, relative to the carbamoylase activity of E.sub.103S, is determined by Assay C described under item 4.5.5.2.

(171) It is even more preferable according to the invention, that an L-carbamoylase E.sub.103V has carbamoylase activity in the range of 1 to 1000%, preferably in the range of 5 to 500%, more preferably in the range of 10 to 400%, more preferably in the range of 40 to 200%, more preferably in the range of 50 to 150%, more preferably in the range of 60 to 140%, more preferably in the range of 70 to 130%, more preferably in the range of 80 to 120%, more preferably in the range of 90 to 110%, more preferably 100% the carbamoylase activity of the L-carbamoylase E.sub.103S, wherein the carbamoylase activity of E.sub.103V, relative to the carbamoylase activity of E.sub.103S, is determined by Assay C described under item 4.5.5.2.

(172) 4.5.6.4 Preferred Variants of SEQ ID NO: 4

(173) According to the invention, the polypeptide sequence of the L-carbamoylase E.sub.1 may also be a variant of SEQ ID NO: 4.

(174) The L-carbamoylase E.sub.1, the polypeptide sequence of which is SEQ ID NO: 4, is denoted as “E.sub.104S”. L-carbamoylases E.sub.1, the polypeptide sequence of which is selected from variants of SEQ ID NO: 4, are generally denoted as “E.sub.104V”.

(175) A variant of the polypeptide sequence of SEQ ID NO: 4 is a polypeptide with sequence identity of at least 60%, preferably $\geq 65\%$, more preferably $\geq 70\%$, more preferably $\geq 75\%$, more preferably $\geq 80\%$, more preferably $\geq 85\%$, more preferably $\geq 90\%$, more preferably $\geq 91\%$, more preferably $\geq 92\%$, more preferably $\geq 93\%$, more preferably $\geq 94\%$, more preferably $\geq 95\%$, more preferably $\geq 96\%$, more preferably $\geq 97\%$, more preferably $\geq 98\%$, more preferably $\geq 99\%$, more preferably $\geq 99.9\%$ sequence identity to polypeptide sequence SEQ ID NO: 4.

(176) The polypeptide sequence of a variant of the polypeptide sequence SEQ ID NO: 4 is not identical to SEQ ID NO: 4.

(177) According to the invention, an L-carbamoylase E.sub.104V has carbamoylase activity and L-carbamoylase activity, determined as described under items 4.5.4.2 and 4.5.4.3.1.

(178) According to the invention, an L-carbamoylase E.sub.104V preferably has carbamoylase activity of at least 1%, preferably of at least 10%, more preferably of at least 20%, more preferably of at least 30%, more preferably of at least 40%, more preferably of at least 50%, more preferably of at least 60%, more preferably of at least 70%, more preferably of at least 80%, more preferably of at least 90%, more preferably of at least 99%, more preferably of at least 100% the carbamoylase activity of the L-carbamoylase E.sub.104S, wherein the carbamoylase activity of E.sub.104V, relative to the carbamoylase activity of E.sub.104S, is determined by Assay C

described under item 4.5.5.2.

(179) It is even more preferable according to the invention, that an L-carbamoylase E.sub.104V has carbamoylase activity in the range of 1 to 1000%, preferably in the range of 5 to 500%, more preferably in the range of 10 to 400%, more preferably in the range of 40 to 200%, more preferably in the range of 50 to 150%, more preferably in the range of 60 to 140%, more preferably in the range of 70 to 130%, more preferably in the range of 80 to 120%, more preferably in the range of 90 to 110%, more preferably 100% the carbamoylase activity of the L-carbamoylase E.sub.104S, wherein the carbamoylase activity of E.sub.104V, relative to the carbamoylase activity of E.sub.104S, is determined by Assay C described under item 4.5.5.2.

(180) 4.5.6.5 Preferred Variants of SEQ ID NO: 5

(181) According to the invention, the polypeptide sequence of the L-carbamoylase E.sub.1 may also be a variant of SEQ ID NO: 5.

(182) The L-carbamoylase E.sub.1, the polypeptide sequence of which is SEQ ID NO: 5, is denoted as "E.sub.105S". L-carbamoylases E.sub.1, the polypeptide sequence of which is selected from variants of SEQ ID NO: 5, are generally denoted as "E.sub.105V".

(183) A variant of the polypeptide sequence of SEQ ID NO: 5 is a polypeptide with sequence identity of at least 60%, preferably $\geq 65\%$, more preferably $\geq 70\%$, more preferably $\geq 75\%$, more preferably $\geq 80\%$, more preferably $\geq 85\%$, more preferably $\geq 90\%$, more preferably $\geq 91\%$, more preferably $\geq 92\%$, more preferably $\geq 93\%$, more preferably $\geq 94\%$, more preferably $\geq 95\%$, more preferably $\geq 96\%$, more preferably $\geq 97\%$, more preferably $\geq 98\%$, more preferably $\geq 99\%$, more preferably $\geq 99.9\%$ sequence identity to polypeptide sequence SEQ ID NO: 5.

(184) The polypeptide sequence of a variant of the polypeptide sequence SEQ ID NO: 5 is not identical to SEQ ID NO: 5.

(185) According to the invention, an L-carbamoylase E.sub.105V has carbamoylase activity and L-carbamoylase activity, determined as described under items 4.5.4.2 and 4.5.4.3.1.

(186) According to the invention, an L-carbamoylase E.sub.105V preferably has carbamoylase activity of at least 1%, preferably of at least 10%, more preferably of at least 20%, more preferably of at least 30%, more preferably of at least 40%, more preferably of at least 50%, more preferably of at least 60%, more preferably of at least 70%, more preferably of at least 80%, more preferably of at least 90%, more preferably of at least 99%, more preferably of at least 100% the carbamoylase activity of the L-carbamoylase E.sub.105S, wherein the carbamoylase activity of E.sub.105V, relative to the carbamoylase activity of E.sub.105S, is determined by Assay C described under item 4.5.5.2.

(187) It is even more preferable according to the invention, that an L-carbamoylase E.sub.105V has carbamoylase activity in the range of 1 to 1000%, preferably in the range of 5 to 500%, more preferably in the range of 10 to 400%, more preferably in the range of 40 to 200%, more preferably in the range of 50 to 150%, more preferably in the range of 60 to 140%, more preferably in the range of 70 to 130%, more preferably in the range of 80 to 120%, more preferably in the range of 90 to 110%, more preferably 100% the carbamoylase activity of the L-carbamoylase E.sub.105S, wherein the carbamoylase activity of E.sub.105V, relative to the carbamoylase activity of E.sub.105S, is determined by Assay C described under item 4.5.5.2.

(188) 4.5.6.6 Preferred Variants of SEQ ID NO: 6

(189) According to the invention, the polypeptide sequence of the L-carbamoylase E.sub.1 may also be a variant of SEQ ID NO: 6.

(190) The L-carbamoylase E.sub.1, the polypeptide sequence of which is SEQ ID NO: 6, is denoted as "E.sub.106S". L-carbamoylases E.sub.1, the polypeptide sequence of which is selected from variants of SEQ ID NO: 6, are generally denoted as "E.sub.106V".

(191) A variant of the polypeptide sequence of SEQ ID NO: 6 is a polypeptide with sequence identity of at least 60%, preferably $\geq 65\%$, more preferably $\geq 70\%$, more preferably $\geq 75\%$, more preferably $\geq 80\%$, more preferably $\geq 85\%$, more preferably $\geq 90\%$, more preferably $\geq 91\%$, more

preferably $\geq 92\%$, more preferably $\geq 93\%$, more preferably $\geq 94\%$, more preferably $\geq 95\%$, more preferably $\geq 96\%$, more preferably $\geq 97\%$, more preferably $\geq 98\%$, more preferably $\geq 99\%$, more preferably $\geq 99.9\%$ sequence identity to polypeptide sequence SEQ ID NO: 6.

(192) The polypeptide sequence of a variant of the polypeptide sequence SEQ ID NO: 6 is not identical to SEQ ID NO: 6.

(193) According to the invention, an L-carbamoylase E.sub.106V has carbamoylase activity and L-carbamoylase activity, determined as described under items 4.5.4.2 and 4.5.4.3.1.

(194) According to the invention, an L-carbamoylase E.sub.106V preferably has carbamoylase activity of at least 1%, preferably of at least 10%, more preferably of at least 20%, more preferably of at least 30%, more preferably of at least 40%, more preferably of at least 50%, more preferably of at least 60%, more preferably of at least 70%, more preferably of at least 80%, more preferably of at least 90%, more preferably of at least 99%, more preferably of at least 100% the carbamoylase activity of the L-carbamoylase E.sub.106S, wherein the carbamoylase activity of E.sub.106V, relative to the carbamoylase activity of E.sub.106S, is determined by Assay C described under item 4.5.5.2.

(195) It is even more preferable according to the invention, that an L-carbamoylase E.sub.106V has carbamoylase activity in the range of 1 to 1000%, preferably in the range of 5 to 500%, more preferably in the range of 10 to 400%, more preferably in the range of 40 to 200%, more preferably in the range of 50 to 150%, more preferably in the range of 60 to 140%, more preferably in the range of 70 to 130%, more preferably in the range of 80 to 120%, more preferably in the range of 90 to 110%, more preferably 100% the carbamoylase activity of the L-carbamoylase E.sub.106S, wherein the carbamoylase activity of E.sub.106V, relative to the carbamoylase activity of E.sub.106S, is determined by Assay C described under item 4.5.5.2.

(196) 4.5.6.7 Preferred Variants of SEQ ID NO: 7

(197) According to the invention, the polypeptide sequence of the L-carbamoylase E.sub.1 may also be a variant of SEQ ID NO: 7.

(198) The L-carbamoylase E.sub.1, the polypeptide sequence of which is SEQ ID NO: 7, is denoted as "E.sub.107S". L-carbamoylases E.sub.1, the polypeptide sequence of which is selected from variants of SEQ ID NO: 7, are generally denoted as "E.sub.107V".

(199) A variant of the polypeptide sequence of SEQ ID NO: 7 is a polypeptide with sequence identity of at least 60%, preferably $\geq 65\%$, more preferably $\geq 70\%$, more preferably $\geq 75\%$, more preferably $\geq 80\%$, more preferably $\geq 85\%$, more preferably $\geq 90\%$, more preferably $\geq 91\%$, more preferably $\geq 92\%$, more preferably $\geq 93\%$, more preferably $\geq 94\%$, more preferably $\geq 95\%$, more preferably $\geq 96\%$, more preferably $\geq 97\%$, more preferably $\geq 98\%$, more preferably $\geq 99\%$, more preferably $\geq 99.9\%$ sequence identity to polypeptide sequence SEQ ID NO: 7.

(200) The polypeptide sequence of a variant of the polypeptide sequence SEQ ID NO: 7 is not identical to SEQ ID NO: 7.

(201) According to the invention, an L-carbamoylase E.sub.107V has carbamoylase activity and L-carbamoylase activity, determined as described under items 4.5.4.2 and 4.5.4.3.1.

(202) According to the invention, an L-carbamoylase E.sub.107V preferably has carbamoylase activity of at least 1%, preferably of at least 10%, more preferably of at least 20%, more preferably of at least 30%, more preferably of at least 40%, more preferably of at least 50%, more preferably of at least 60%, more preferably of at least 70%, more preferably of at least 80%, more preferably of at least 90%, more preferably of at least 99%, more preferably of at least 100% the carbamoylase activity of the L-carbamoylase E.sub.107S, wherein the carbamoylase activity of E.sub.107V, relative to the carbamoylase activity of E.sub.107S, is determined by Assay C described under item 4.5.5.2.

(203) It is even more preferable according to the invention, that an L-carbamoylase E.sub.107V has carbamoylase activity in the range of 1 to 1000%, preferably in the range of 5 to 500%, more preferably in the range of 10 to 400%, more preferably in the range of 40 to 200%, more preferably

in the range of 50 to 150%, more preferably in the range of 60 to 140%, more preferably in the range of 70 to 130%, more preferably in the range of 80 to 120%, more preferably in the range of 90 to 110%, more preferably 100% the carbamoylase activity of the L-carbamoylase E.sub.107S, wherein the carbamoylase activity of E.sub.107V, relative to the carbamoylase activity of E.sub.107S, is determined by Assay C described under item 4.5.5.2.

(204) 4.5.6.8 Preferred Variants of SEQ ID NO: 8

(205) According to the invention, the polypeptide sequence of the L-carbamoylase E.sub.1 may also be a variant of SEQ ID NO: 8.

(206) The L-carbamoylase E.sub.1, the polypeptide sequence of which is SEQ ID NO: 8, is denoted as "E.sub.108S". L-carbamoylases E.sub.1, the polypeptide sequence of which is selected from variants of SEQ ID NO: 8, are generally denoted as "E.sub.108V".

(207) A variant of the polypeptide sequence of SEQ ID NO: 8 is a polypeptide with sequence identity of at least 60%, preferably $\geq 65\%$, more preferably $\geq 70\%$, more preferably $\geq 75\%$, more preferably $\geq 80\%$, more preferably $\geq 85\%$, more preferably $\geq 90\%$, more preferably $\geq 91\%$, more preferably $\geq 92\%$, more preferably $\geq 93\%$, more preferably $\geq 94\%$, more preferably $\geq 95\%$, more preferably $\geq 96\%$, more preferably $\geq 97\%$, more preferably $\geq 98\%$, more preferably $\geq 99\%$, more preferably $\geq 99.9\%$ sequence identity to polypeptide sequence SEQ ID NO: 8.

(208) The polypeptide sequence of a variant of the polypeptide sequence SEQ ID NO: 8 is not identical to SEQ ID NO: 8.

(209) According to the invention, an L-carbamoylase E.sub.108V has carbamoylase activity and L-carbamoylase activity, determined as described under items 4.5.4.2 and 4.5.4.3.1.

(210) According to the invention, an L-carbamoylase E.sub.108V preferably has carbamoylase activity of at least 1%, preferably of at least 10%, more preferably of at least 20%, more preferably of at least 30%, more preferably of at least 40%, more preferably of at least 50%, more preferably of at least 60%, more preferably of at least 70%, more preferably of at least 80%, more preferably of at least 90%, more preferably of at least 99%, more preferably of at least 100% the carbamoylase activity of the L-carbamoylase E.sub.108S, wherein the carbamoylase activity of E.sub.108V, relative to the carbamoylase activity of E.sub.108S, is determined by Assay C described under item 4.5.5.2.

(211) It is even more preferable according to the invention, that an L-carbamoylase E.sub.108V has carbamoylase activity in the range of 1 to 1000%, preferably in the range of 5 to 500%, more preferably in the range of 10 to 400%, more preferably in the range of 40 to 200%, more preferably in the range of 50 to 150%, more preferably in the range of 60 to 140%, more preferably in the range of 70 to 130%, more preferably in the range of 80 to 120%, more preferably in the range of 90 to 110%, more preferably 100% the carbamoylase activity of the L-carbamoylase E.sub.108S, wherein the carbamoylase activity of E.sub.108, relative to the carbamoylase activity of E.sub.108S, is determined by Assay C described under item 4.5.5.2.

(212) 4.5.6.9 Preferred Variants of SEQ ID NO: 9

(213) According to the invention, the polypeptide sequence of the L-carbamoylase E.sub.1 may also be a variant of SEQ ID NO: 9.

(214) The L-carbamoylase E.sub.1, the polypeptide sequence of which is SEQ ID NO: 9, is denoted as "E.sub.109S". L-carbamoylases E.sub.1, the polypeptide sequence of which is selected from variants of SEQ ID NO: 9, are generally denoted as "E.sub.109V".

(215) A variant of the polypeptide sequence of SEQ ID NO: 9 is a polypeptide with sequence identity of at least 60%, preferably $\geq 65\%$, more preferably $\geq 70\%$, more preferably $\geq 75\%$, more preferably $\geq 80\%$, more preferably $\geq 85\%$, more preferably $\geq 90\%$, more preferably $\geq 91\%$, more preferably $\geq 92\%$, more preferably $\geq 93\%$, more preferably $\geq 94\%$, more preferably $\geq 95\%$, more preferably $\geq 96\%$, more preferably $\geq 97\%$, more preferably $\geq 98\%$, more preferably $\geq 99\%$, more preferably $\geq 99.9\%$ sequence identity to polypeptide sequence SEQ ID NO: 9.

(216) The polypeptide sequence of a variant of the polypeptide sequence SEQ ID NO: 9 is not

identical to SEQ ID NO: 9.

(217) According to the invention, an L-carbamoylase E.sub.109V has carbamoylase activity and L-carbamoylase activity, determined as described under items 4.5.4.2 and 4.5.4.3.1.

(218) According to the invention, an L-carbamoylase E.sub.109V preferably has carbamoylase activity of at least 1%, preferably of at least 10%, more preferably of at least 20%, more preferably of at least 30%, more preferably of at least 40%, more preferably of at least 50%, more preferably of at least 60%, more preferably of at least 70%, more preferably of at least 80%, more preferably of at least 90%, more preferably of at least 99%, more preferably of at least 100% the carbamoylase activity of the L-carbamoylase E.sub.109S, wherein the carbamoylase activity of E.sub.109V, relative to the carbamoylase activity of E.sub.109S, is determined by Assay C described under item 4.5.5.2.

(219) It is even more preferable according to the invention, that an L-carbamoylase E.sub.109V has carbamoylase activity in the range of 1 to 1000%, preferably in the range of 5 to 500%, more preferably in the range of 10 to 400%, more preferably in the range of 40 to 200%, more preferably in the range of 50 to 150%, more preferably in the range of 60 to 140%, more preferably in the range of 70 to 130%, more preferably in the range of 80 to 120%, more preferably in the range of 90 to 110%, more preferably 100% the carbamoylase activity of the L-carbamoylase E.sub.109S, wherein the carbamoylase activity of E.sub.109V, relative to the carbamoylase activity of E.sub.109S, is determined by Assay C described under item 4.5.5.2.

4.5.7 Preferred Method Conditions in Step (c)

(220) The reaction in step (c) of the method according to the first aspect of the present invention may be carried out under conditions known to the skilled person.

(221) The reaction medium is preferably aqueous, more preferably an aqueous buffer.

(222) Exemplary buffers commonly used in biotransformation reactions and advantageously used herein include Tris, phosphate, or any of Good's buffers, such as 2-(N-morpholino) ethanesulfonic acid ("MES"), N-(2-acetamido)iminodiacetic acid ("ADA"), piperazine-N,N'-bis(2-ethanesulfonic acid) ("PIPES"), N-(2-acetamido)-2-aminoethanesulfonic acid ("ACES"), P-hydroxy-4-morpholinepropanesulfonic acid ("MOPSO"), choline chloride, 3-(N-morpholino) propanesulfonic acid ("MOPS"), N,N-Bis(2-hydroxyethyl)-2-aminoethanesulfonic acid ("BES"), 2-[[1,3-dihydroxy-2-(hydroxymethyl) propan-2-yl]amino]ethanesulfonic acid ("TES"), 4-(2-hydroxyethyl)-1-piperazineethanesulfonic acid ("HEPES"), 3-(Bis(2-hydroxyethyl)amino)-2-hydroxypropane-1-sulfonic acid ("DIPSO"), acetamidoglycine, 3-(N-Tris(hydroxymethyl)methylamino(-2-hydroxypropane) sulfonic acid ("TAPSO"), piperazine-N,N'-bis(2-hydroxypropanesulfonic acid) ("POPSO"), 4-(2-Hydroxyethyl) piperazine-1-(2-hydroxypropanesulfonic acid) ("HEPPSO"), 3-[4-(2-Hydroxyethyl)-1-piperazinyl]propanesulfonic acid ("HEPPS"), tricine, glycine, bicine, or 3-[[1,3-dihydroxy-2-(hydroxymethyl) propan-2-yl]amino]propane-1-sulfonic acid ("TAPS").

(223) In some embodiments, ammonium can act as a buffer. One or more organic solvents can also be added to the reaction.

(224) The buffer preferably contains metal salts, more preferably metal salts such as halogenides of metals, preferably halogenides of monovalent or bivalent or trivalent metals, preferably chlorides of monovalent or bivalent metals, preferably CoCl.sub.2 or MnCl.sub.2, preferably CoCl.sub.2.

(225) The concentration of these metal salts in the reaction medium is preferably in the range from 1 μ M to 1 M, more preferably 1 mM to 100 mM, even more preferably 1 to 10 mM.

(226) Preferably, step (c) of the method according to the first aspect of the invention is carried out in a phosphate buffer.

(227) The pH of the reaction medium in step (c) of the method according to the first aspect of the invention is preferably in the range of from 2 to 10, more preferably in the range of from 5 to 8, more preferably 7.2 to 7.5, most preferably 7.5.

(228) Preferably, step (c) of the method according to the first aspect of the invention is carried out

at a temperature in the range of from 20° C. to 70° C., more preferably in the range of from 30° C. to 55° C., most preferably 50° C.

(229) Preferably, the total concentration of all carbamoylases E.sub.1 in the reaction solution in step (c) is in the range of from 1 μM to 10 mM, preferably 10 μM to 1 mM, more preferably 0.1 mM to 0.5 mM, most preferably 0.4 mM.

(230) In alternative preferred embodiments, the total concentration of all carbamoylases E.sub.1 in the reaction solution in step (c) is in the range of from 1 μg/l to 10 g/l, preferably 0.1 mg/l to 5 g/l, more preferably 1 mg/l to 1 g/l, more preferably 5 mg/l to 500 mg/l.

(231) Preferably, the initial concentration of all the compounds according to formula L-(II) in the reaction medium in step (c) is in the range of from 1 μM to 1 M, preferably of from 10 μM to 0.5 M, more preferably of from 0.1 mM to 0.1 M, more preferably of from 1 mM to 10 mM, most preferably 1.25 mM.

(232) If compounds according to formula D-(II) are present in the reaction medium in step (c), the initial concentration of all the compounds according to formula D-(II) in the reaction medium is preferably from 1% to 100% the concentration of all the compounds according to formula L-(II), more preferably 10% to 100% the concentration, even more preferably 50 to 100%, even more preferably 100% the concentration of all the compounds according to formula L-(II).

(233) "Initial concentration of all the compounds according to formula L-(II)/D-(II)" refers to the concentration of the respective compound L-(II) or D-(II) respectively, in the reaction medium when the respective compounds are employed in step (c).

4.5.8 Step (b)

(234) 4.5.8.1 Enzymatic Catalysis of Step (b)

(235) In a preferred embodiment of the method according to the first aspect of the invention, the compound according to formula L-(II) is obtained by a step (b) in which a compound according to formula L-(III) is reacted to give a compound according to formula L-(II):

(236) ##STR00027##

(237) Step (b) gives the starting material for step (c).

(238) R_{sup}.3L is a radical selected from the above-mentioned formulae (VIII), (IX), (X), preferably selected from the formulae (VIII), (IX), more preferably R_{sup}.3L has formula (VIII).

(239) When R_{sup}.3L is a radical of the formula (VIII), the radical R_{sup}.3L may be selected from the above-mentioned formulae L-(VIII) and D-(VIII), preferably L-(VIII).

(240) When R_{sup}.3L is a radical of the formula (IX), the radical R_{sup}.3L may be selected from the above-mentioned formulae L-(IX) and D-(IX), preferably L-(IX).

(241) When R_{sup}.3L is a radical of the formula (X), the radical R_{sup}.3L may be selected from the above-mentioned formulae L-(X) and D-(X), preferably L-(X).

(242) The reaction according step (b) is enzymatically catalyzed, namely it is catalyzed by a hydantoinase E.sub.2.

(243) 4.5.8.2 Enantioselective and Enantiospecific Catalysis of Step (b)

(244) Step (b) of the method according to the first aspect of the present invention is preferably L-enantioselective, even more preferably L-enantiospecific.

(245) In such a preferred embodiment, L-(III) is in particular employed in step (b) as a mixture M.sub.III comprising at least one compound L-(III) and at least one compound D-(III), wherein D-(III) has the following formula:

(246) ##STR00028##

(247) R_{sup}.3D is a radical selected from the above-mentioned formulae (VIII), (IX), (X), preferably selected from the formulae (VIII), (IX), more preferably R_{sup}.3D has formula (VIII).

(248) When R_{sup}.3D is a radical of the formula (VIII), the radical R_{sup}.3D may be selected from the above-mentioned formulae L-(VIII) and D-(VIII), preferably L-(VIII).

(249) When R_{sup}.3D is a radical of the formula (IX), the radical R_{sup}.3D may be selected from the above-mentioned formulae L-(IX) and D-(IX), preferably L-(IX).

(250) When R.sup.3D is a radical of the formula (X), the radical R.sup.3D may be selected from the above-mentioned formulae L-(X) and D-(X), preferably L-(X).

(251) Mixture M.sub.II preferably comprises at least one compound L-(III) and at least one compound D-(III), wherein R.sup.3L=R.sup.3D.

(252) For such mixtures Mu, it may be observed that the at least one compound D-(II) undergoes a reaction according to step (b)*, i.e. parallel to step (b).

(253) Namely, in the reaction according to step (b)*, the at least one compound D-(III) in mixture M.sub.II is reacted to give a compound according to formula D-(II), wherein compounds according to formula D-(II) are explained above (in particular under item 4.5.1.2).

(254) In case that step (b) is “L-enantioselective”, this means that in case a mixture Mil is employed in step (b), then there is either no reaction according to step (b)* of any compound according to formula D-(III) to a compound according to D-(II) in the mixture M.sub.II or, in case there is a reaction according to step (b)*, then there is at least one pair of compounds D-(III)  L-(III)  in the mixture Mu, for which the reaction according to step (b)* of D-(III)  to give D-(II)  is lesser than the rate of the reaction according to step (b) of L-(III)  to give L-(II) .

(255) D-(III)  is a compound according to formula D-(III), and L-(III)  is a compound according to formula L-(III), wherein the radical R.sup.3L in L-(III)  and the radical R.sup.3D in D-(III)+ are identical.

(256) D-(II)  is a compound according to formula D-(II), and L-(II)  is a compound according to formula L-(II), wherein the radical R.sup.2L in L-(II)  and the radical R.sup.2D in D-(II)+ are identical.

(257) Preferably, the radical R.sup.2L in L-(II)+ and the radical R.sup.3L in L-(III)+ are identical. Preferably, the radical R.sup.2D in D-(II)  and the radical R.sup.3D in D-(III)+ are identical.

(258) Step (b) is “L-enantiospecific”, if the rate of reaction according to step (b)* is essentially zero, i.e. there is no reaction according to step (b)* of any compound according to formula D-(III) to a compound according to D-(II) in the mixture M.sub.III.

(259) In a preferred embodiment, the molar ratio of all compounds according to formula L-(III) in the mixture M.sub.III to all compounds according to formula D-(III) in the mixture M.sub.III is essentially 1:1.

(260) In other preferred embodiments, the molar ratio of all compounds according to formula L-(III) in the mixture M.sub.III to all compounds according to formula D-(III) in the mixture M.sub.III is in the range of from 3:2 to 1:99, more preferably in the range of from 1.01:1 to 1:99, more preferably in the range of from 1:1 to 1:99, more preferably in the range of from 1:1.01 to 1:99, more preferably in the range of from 1:1.01 to 1:9, more preferably in the range of from 1:1.01 to 1:8, more preferably in the range of from 1:1.01 to 1:3.

(261) Alternatively, D-(III) is comprised in an excess to L-(III) in mixture Mm, meaning that, while at least one compound according to formula L-(III) is in the mixture M.sub.III, the molar ratio of all compounds according to formula L-(III) is in the mixture M.sub.III to all compounds according to formula D-(III) in the mixture M.sub.III is <1:1, preferably <0.9:1, more preferably <0.75:1, more preferably <0.5:1, more preferably <0.2:1, more preferably <0.1:1, more preferably <0.01:1.

(262) Step (b) is in particular L-enantioselective, if it is preferably catalyzed by an L-hydantoinase E.sub.2, which may be determined by the skilled person as set forth under 4.5.10.3.

(263) In case step (b) is L-enantioselective with respect to a given pair of compounds D-(III)  L-(III)+, the reaction according to step (b) of L-(III)  to give L-(II)  proceeds preferably at a reaction rate that is at least 2 times greater, preferably at least 10 times greater, more preferably at least 100 times greater, even more preferably at least 103 times greater, even more preferably at least 104 times greater, even more preferably at least 105 times greater than the reaction rate at which step (b)* of D-(III)

custom character to give D-(II) custom character proceeds.

(264) To quantify for a given pair of compounds D-(III) custom character/L-(III)

custom character the factor at which the reaction rate of step (b) proceeds compared to the reaction rate of step (b)*, the following test may be carried out: (1) An equimolar mixture [molar ratio of the two compounds L-(III) custom character and D-(III) custom character is 1:1] is subjected to the respective reactions conditions and the development of the two products L-(I) custom character and D-(I) custom character is monitored over time (e.g. by LC-MS as set forth under item 5.4). (2) When n.sub.LII10=10 mol-% of the initially employed L-(III) custom character has reacted to the product L-(II) custom character, the molar amount of D-(II) custom character that was formed by reaction from D-(III)+ [in mol-% relative to the initially employed D-(II) custom character] is measured (=n.sub.DII10). (3) The ratio of n.sub.DII10/n.sub.LII10=10/n.sub.DII10 gives the factor by which the reaction rate of step (b) proceeds compared to the reaction rate of step (b)* for the given pair of compounds D-(III) custom character/L-(III) custom character. If n.sub.LII10/n.sub.DII10>1, the reaction rate of step (b) is greater than the reaction rate of step (b)* for the given pair of compounds D-(III) custom character/L-(III)+. If n.sub.LII10/n.sub.DII10<1, the reaction rate of step (b) is lesser than the reaction rate of step (b)* for the given pair of compounds D-(III) custom character/L-(III) custom character. If n.sub.LII10/n.sub.DII10=1, the reaction rates of steps (b) and (b)* are the same for the given pair of compounds D-(III) custom character/L-(III) custom character.

(265) The reaction according to step (b) of the preferred embodiment of the first aspect of the invention is catalyzed by a hydantoinase (“dihydropyrimidinase”) E.sub.2.

(266) Namely, it was surprisingly found that hydantoinases accept compounds of formula L-(III) as substrates and convert them to products according to formulae L-(II), and hence catalyze the reaction according to step (b). This finding is of high scientific and economic value, as it further broadens the scope of synthetic routes based on new starting materials for the production of L-glufosinate P-esters and L-glufosinate. Even more surprisingly, it is suggested that L-glufosinate hydantoin, i.e. the compound according to formula L-(III), in which R=H, does not undergo reaction by hydantoinases to give the respective LGA carbamoylate.

(267) In nature, hydantoinases (“dihydropyrimidinases”) generally catalyze the reaction of 5,6-dihydrouracil to produce ureidopropionate (see the following reaction <2A>):

(268) ##STR00029##

(269) They also catalyze the ring opening of monosubstituted hydantoins to give the respective carbamoyl amino acid according to the following reaction <2B>, wherein R* may be an organic residue, e.g. a side chain of one of the naturally occurring amino acids.

(270) ##STR00030##

(271) Chapter 1.3 (pages 7 to 11) of the dissertation “Untersuchungen zur Substratspezifität und Enantioselektivität mikrobieller Hydantoinasen/Investigations of substrate specificity and enantioselectivity of microbial hydantoinases” by T. Waniek, University of Stuttgart, 2000 (available under: <https://elib.uni-stuttgart.de/bitstream/11682/1511/1/Diss.pdf>) gives an overview over hydantoinases.

(272) It was now surprisingly found that hydantoinases also accept substrates in which $R^*=R.\text{sup.}Y=$

(273) ##STR00031##

wherein $R=R.\text{sup.}3L$.

(274) Surprisingly, they supposedly do not accept substrates in which $R^*=R.\text{sup.}Y=$

(275) ##STR00032##

wherein R=H.

(276) In the context of the present invention, a “hydantoinase E.sub.2” is a hydantoinase that catalyzes the following reaction <2C> of a carbamoyl substrate S', to the respective amino acid product P'L, wherein $R^*=R.\text{sup.}Y=$

(277) ##STR00033##

wherein R=R.sup.3L:

(278) ##STR00034##

(279) In particular, the hydantoinase E.sub.2 is a "L-hydantoinase", i.e. it has a greater catalytic activity for reaction <2C> than for reaction <2D>, wherein R* in the substrate S'D in the reaction <2D> and R* in the substrate S'L in the reaction <2C> are identical and R*=R.sup.Y=

(280) ##STR00035##

wherein R=R.sup.3L.

(281) ##STR00036##

(282) As an L-hydantoinase has a higher catalytic activity for reaction <2C> than for reaction <2D>, it is "L-enantioselective". An L-hydantoinase that has no catalytic activity for reaction <2D> and thus only has catalytic activity for reaction <2C> is "L-enantiospecific".

(283) A "D-hydantoinase" is defined as a hydantoinase which is "D-enantioselective", i.e. it has a higher catalytic activity for reaction <2D> than for reaction <2C>. A D-carbamoylase that does not catalyze reaction <2C> and thus only has catalytic activity for reaction <2D> is "D-enantiospecific".

(284) A hydantoinase which has the same catalytic activity for reaction <2C> as for reaction <2D>, is referred to as "non enantioselective hydantoinase".

(285) For determination whether a hydantoinase may be denoted as "L-hydantoinase", "D-hydantoinase" or "non-enantioselective hydantoinase" in the context of the present invention, the procedure according to Assay E (item 4.5.10) may preferably be used.

(286) The hydantoinase E.sub.2, in particular the L-hydantoinase E.sub.2, that may be used in step (b) of the preferred embodiment of the method according to the first aspect of the invention may originate from *Arthrobacter* sp., in particular *Arthrobacter crystallopoietes*, *Arthrobacter aurescens*, *Arthrobacter* sp. BT801; *Alcaligenes* sp., in particular *Alcaligenes faecalis* subsp. *faecalis*; *Bacillus* sp., in particular *Bacillus fordii*; *Microbacterium* sp., in particular *Microbacterium liquefaciens* strain AJ3912; *Pseudomonas* sp., in particular *Pseudomonas fluorescens*, *Pseudomonas aeruginosa*.

(287) A hydantoinase E.sub.2, in particular an L-hydantoinase E.sub.2 suitable for the method according to the present invention may be the enzyme HyuH, which originates from *Arthrobacter*. Another enzyme may be Dht.

(288) Even more preferably, the hydantoinase E.sub.2, in particular the L-hydantoinase E.sub.2, that may preferably be used in step (b) according to the first aspect of the invention may originate from *Arthrobacter* sp., in particular *Arthrobacter crystallopoietes*, *Arthrobacter aurescens*, *Arthrobacter* sp. BT801, *Arthrobacter aurescens* DSM 9771, most preferably from *Arthrobacter aurescens* DSM 9771.

(289) A hydantoinase suitable for the method according to the present invention is described e.g. in WO 01/23582 A1 and by J. M. Clemente-Jiménez, S. Martínez-Rodríguez, F. Rodríguez-Vico, F. J. L. Heras-Vázquez, Recent Pat. Biotechnology 2008, 2, 35-46; G. Latacz, E. Pekala, K. Kiec-Kononowicz, Biotechnologia 2006, 2, 189-205.

(290) Further suitable hydantoinases are described by K. Yokozeki, H. Yoshiteru, K. Kubota, Agric. Biol. Chem. 1987, 51, 737-746.

(291) The hydantoinase E.sub.2 that may be used in preferred step (b) of the method according to the present invention may be a hydantoinase categorized in the EC class EC 3.5.2.2.

(292) The following table 2 gives preferred examples for polypeptide sequences of hydantoinases E.sub.2 that may be preferably used in step (b) of the preferred embodiment of the method according to the first aspect of the invention. The genes encoding the respective hydantoinase E.sub.2 and the respective accession code are indicated as far as known.

(293) TABLE-US-00002 TABLE 2 Hydantoinases (EC 3.5.2.2) GenBank/UniProt SEQ ID NO: of the Strain Gene name accession polypeptide *Arthrobacter aurescens* DSM hyuH SEQ ID NO: 10

9771 *Pseudomonas fluorescens* not assigned S5MPT0 SEQ ID NO: 11 *Pseudomonas aeruginosa* dht Q9I676 SEQ ID NO: 12 *Bacillus fordii* MH602 not assigned ABL14245 SEQ ID NO: 13 *Arthrobacter* sp. BT801 hyuH AAL55412 SEQ ID NO: 14 *Alcaligenes faecalis* subsp. not assigned SEQ ID NO: 15 *faecalis Microbacterium liquefaciens* not assigned SEQ ID NO: 16 strain AJ3912 (294) In a preferred embodiment of the method according to the first aspect of the present invention, the reaction according to step (b) is catalyzed by an hydantoinase E.sub.2, wherein the polypeptide sequence of E.sub.2 is selected from the group consisting of SEQ ID NO: 10 and variants thereof, SEQ ID NO: 11 and variants thereof, SEQ ID NO: 12 and variants thereof, SEQ ID NO: 13 and variants thereof, SEQ ID NO: 14 and variants thereof, SEQ ID NO: 15 and variants thereof, SEQ ID NO: 16 and variants thereof, preferably SEQ ID NO: 10 and variants thereof.

4.5.9 Assays D.SUB.L .and D.SUB.D .for Determining Hydantoinase Activity

(295) The skilled person is aware of hydantoinases, in particular L-hydantoinases, that may be used in step (b) of the preferred method according to the first aspect of the invention.

(296) In particular, Assay D.sub.L, described in the following, may be used to determine hydantoinase and L-hydantoinase activity of a given enzyme E.sub.Y and may advantageously be used according to the invention to determine hydantoinase and L-hydantoinase activity in variants of SEQ ID NO: 10, variants of SEQ ID NO: 11, variants of SEQ ID NO: 12, variants of SEQ ID NO: 13, variants of SEQ ID NO: 14, variants of SEQ ID NO: 15, variants of SEQ ID NO: 16.

(297) For comparative reasons, Assay D.sub.D may be used to determine D-hydantoinase activity of a given enzyme E.sub.Y.

(298) For the purpose of Assay D.sub.L and Assay D.sub.D, the molar mass of the enzyme E.sub.Y to be tested is calculated as the molar mass of the polypeptide sequence of E.sub.Y.

(299) Moreover, a substrate according to formula L-(III), wherein R.sup.3L=n-butyl, is used in Assay D.sub.L so that there is only one chiral carbon atom in the substrate. The compound according to formula L-(III), wherein R.sup.3L=n-butyl, is denoted as “L-(III)+”.

(300) A product according to formula L-(II), wherein R.sup.2L=n-butyl, is detected in Assay A.sub.L.

(301) The compound according to formula L-(II), wherein R.sup.2L=n-butyl, is denoted as “L-(II)+”.

(302) Moreover, a substrate according to formula D-(III), wherein R.sup.3D=n-butyl, is used in Assay A.sub.D so that there is only one chiral carbon atom in the substrate. The compound according to formula D-(III), wherein R.sup.3D=n-butyl, is denoted as “D-(III)+”.

(303) A product according to formula D-(II), wherein R.sup.2D=n-butyl, is detected in Assay A.sub.D.

(304) The compound according to formula D-(II), wherein R.sup.2D=n-butyl, is denoted as “D-(II)+”.

(305) 4.5.9.1 Assay D.sub.L:

(306) To 0.9 ml of an aqueous reaction solution (phosphate buffer, pH 7.2, 10 mM MnCl.sub.2), containing 50 mM of a compound according to formula L-(III)  custom character are added 400 nmol of E.sub.Y in 0.1 ml aqueous phosphate buffer (pH 7.2, 10 mM MnCl.sub.2). The resulting solution is incubated at 25° C., and the pH is held at pH 7.2 by addition of 1 M NaOH. After 300 minutes, the reaction is stopped by addition of 2 M HCl to achieve a pH of 2.5, and the molar amount of the compound according to formula L-(II)  custom character is determined. L-(II)  custom character may be detected by the LC-MS method as described in the example section (item 5.4) for detection of LGA.

(307) 4.5.9.2 Assay D.sub.D:

(308) To 0.9 ml of an aqueous reaction solution (phosphate buffer, pH 7.2, 10 mM MnCl.sub.2), containing 50 mM of a compound according to formula D-(III)  custom character are added 400 nmol of E.sub.Y in 0.1 ml aqueous phosphate buffer (pH 7.2, 10 mM MnCl.sub.2). The resulting solution is incubated at 25° C., and the pH is held at pH 7.2 by addition of 1 M NaOH. After 300

minutes, the reaction is stopped by addition of 2 M HCl to achieve a pH of 2.5, and the molar amount of the compound according to formula D-(II)  is determined. D-(II)  may be detected by the LC-MS method as described in the example section (item 5.4) for detection of LGA.

4.5.10 Assay E for Identifying Hydantoinases, L-Hydantoinases, D-Hydantoinases, L- and D-Enantiospecificity

(309) The hydantoinase E.sub.2 according to the invention is preferably an L-hydantoinase, more preferably L-enantiospecific.

(310) Whether a given enzyme E.sub.Y may be considered a hydantoinase E.sub.2, in particular an L-hydantoinase, may be determined in the context of the present invention by the following Assay E:

(311) 4.5.10.1 Assay E:

(312) E-1. Firstly, Assay D.sub.L as set forth under item 4.5.9.1 is conducted, and the obtained molar amount of the compound of the formula L-(II)  is determined according to Assay D.sub.L.

(313) E-2 Secondly, Assay D.sub.D as set forth under item 4.5.9.2 is conducted, and the obtained molar amount of the compound of the formula D-(II)  is determined according to Assay D.sub.D.

(314) E-3. Then, step E-1 is repeated, except that instead of the addition of 400 nmol E.sub.Y in 0.1 ml aqueous phosphate buffer (pH 7.2, 10 mM MnCl.sub.2), 0.1 ml aqueous phosphate buffer (pH 7.2, 10 mM MnCl.sub.2) without E.sub.Y is added.

(315) E-4 Then, step E-2 is repeated, except that instead of the addition of 400 nmol E.sub.Y in 0.1 ml aqueous phosphate buffer (pH 7.2, 10 mM MnCl.sub.2), 0.1 ml aqueous phosphate buffer (pH 7.2, 10 mM MnCl.sub.2) without E.sub.Y is added.

(316) 4.5.10.2 Hydantoinase Activity:

(317) If the molar amount of the compound of the formula L-(II)  that is determined in step E-1, is greater than the molar amount of the compound of formula L-(II)  that is determined in step E-3, then E.sub.Y is deemed to have hydantoinase activity, and hence may be considered a hydantoinase E.sub.2 in the context of the invention.

(318) 4.5.10.3 L-Hydantoinase Activity

(319) 4.5.10.3.1 L-Hydantoinases

(320) (i) If the molar amount of the compound of the formula L-(II)  that is determined in step E-1, is greater than the molar amount of the compound of the formula L-(II)  that is determined in step E-3, and (ii) if the molar amount of the compound of the formula D-(II)  that is determined in step E-2, is greater than or the same as the molar amount of the compound of the formula D-(II)  that is determined in step E-4, and (iii) if, in addition, the molar amount of the compound of the formula L-(II)  that is determined in step E-1 is greater than the molar amount of the compound of the formula D-(II)  that is determined in step E-2, then E.sub.Y is deemed to have L-hydantoinase activity, and hence may be considered an L-hydantoinase in the context of the invention. In this case, E.sub.Y is deemed to be “L-enantioselective” in the context of the invention.

(321) For the sake of clarity, it is pointed out that condition (iii) is automatically fulfilled in those cases in which the molar amount of the compound of formula D-(II)  that is determined in step E-2 is the same as the molar amount of the compound of formula D-(II)  that is determined in step E-4. (i) If the molar amount of the compound of formula L-(II)  that is determined in step E-1, is greater than the molar amount of the compound of formula L-(II)  that is determined in step E-3, and (ii) if the molar amount of the compound of the formula D-(II)  that is determined in step E-2, is the same as the molar amount of the compound of the formula D-(II)  that

is determined in step E-4, then E.sub.Y is deemed to have L-hydantoinase activity, and hence may be considered a L-hydantoinase in the context of the invention. In this case, E.sub.Y is deemed to be not only “L-enantioselective”, but also “L-enantiospecific” in the context of the invention.

(322) For L-hydantoinases E.sub.Y that are not L-enantiospecific, the L-enantioselectivity may then be quantified by dividing the molar amount of the compound of formula L-(II)

 that is determined in E-1, by the molar amount of the compound of formula D-(II)  that is determined in E-2, and then multiplying the obtained value by 100, giving the L-enantioselectivity of E.sub.Y in %.

(323) 4.5.10.3.2 D-Carbamoylases

(324) (i) If the molar amount of the compound of formula L-(II)  that is determined in step E-1, is greater than or the same as the molar amount of the compound of formula L-(II)  that is determined in step E-3, and (ii) if the molar amount of the compound of formula D-(II)  that is determined in step E-2 is greater than the molar amount of the compound of formula D-(II)  that is determined in step E-4, and (iii) if, in addition, the molar amount of the compound of formula D-(II)  that is determined in step E-2, is greater than the molar amount of the compound of formula L-(II)  that is determined in step E-1, then E.sub.Y is deemed to have D-hydantoinase activity, and hence may be considered a D-hydantoinase. In this case, E.sub.Y is “D-enantioselective” in the context of the invention.

(325) For the sake of clarity it is pointed out that condition (iii) is automatically fulfilled in those cases in which the molar amount of the compound of formula L-(II)  that is determined in step E-1 is the same as the molar amount of the compound of formula L-(II)

 that is determined in step E-3. (i) If the molar amount of the compound of formula L-(II)  that is determined in step E-1, is the same as the molar amount of the compound of formula L-(II)  that is determined in step E-3, and (ii) if the molar amount of the compound of formula D-(II)  that is determined in step E-2, is greater than the molar amount of the compound of formula D-(II)  that is determined in step E-4, then E.sub.Y is deemed to have D-hydantoinase activity, and hence may be considered a D-hydantoinase in the context of the invention. In this case, E.sub.Y is deemed to be not only “D-enantioselective”, but also “D-enantiospecific” in the context of the invention.

(326) For D-hydantoinases E.sub.Y that are not D-enantiospecific, the D-enantioselectivity may then be quantified by dividing the molar amount of the compound of formula D-(II)

 that is determined in step E-2 by the molar amount of the compound of formula L-(II)  that is determined in step E-1 and then multiplying the obtained value by 100, giving the D-enantioselectivity of E.sub.Y in %.

(327) 4.5.10.3.3 Non-Enantioselective Carbamoylases

(328) (i) If the molar amount of the compound of formula L-(II)  that is determined in step E-1, is greater than the molar amount of the compound of formula L-(II)  that is determined in step E-3, and (ii) if the molar amount of the compound of formula D-(II)  that is determined in step E-2, is greater than the molar amount of the compound of formula D-(II)  that is determined in step E-4, and (iii) if, in addition, the molar amount of the compound of formula D-(II)  that is determined in step E-2 is the same as the molar amount of the compound of formula L-(II)  that is determined in step E-1, then E.sub.Y is deemed to be a non-enantioselective hydantoinase in the context of the invention. In this case, E.sub.Y is “non-enantioselective”.

4.5.11 Assay F for Identifying Preferred Hydantoinases Variants of SEQ ID NOs: 10-16

(329) 4.5.11.1 L- and D-Enantioselective Hydantoinase Variants

(330) An enzyme, the polypeptide sequence of which is selected from the group consisting of SEQ ID NO: 10, SEQ ID NO: 11, SEQ ID NO: 12, SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15,

SEQ ID NO: 16 has hydantoinase, in particular L-hydantoinase activity.

(331) In a preferred embodiment of the method according to the first aspect of the invention, the polypeptide sequence of the hydantoinase, in particular the L-hydantoinase E.sub.2 is selected from the group consisting SEQ ID NO: 10 and variants thereof, SEQ ID NO: 11 and variants thereof, SEQ ID NO: 12 and variants thereof, SEQ ID NO: 13 and variants thereof, SEQ ID NO: 14, and variants thereof, SEQ ID NO: 15 and variants thereof, SEQ ID NO: 16 and variants thereof, more preferably SEQ ID NO: 10 and variants thereof.

(332) The term “variant” is defined under item 4.3.

(333) In the context of the invention, an enzyme, the polypeptide sequence of which is a variant of one of SEQ ID NO: 10, SEQ ID NO: 11, SEQ ID NO: 12, SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15, SEQ ID NO: 16, has hydantoinase activity, preferably L-hydantoinase activity, more preferably is L-enantiospecific.

(334) Whether a given enzyme E.sub.Y, the polypeptide sequence of which is a variant of one of SEQ ID NO: 10, SEQ ID NO: 11, SEQ ID NO: 12, SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15, SEQ ID NO: 16, has hydantoinase activity, L-hydantoinase activity and is L-enantiospecific may be determined as set forth under items 4.5.10.2 and 4.5.10.3.1, respectively.

(335) The hydantoinase activity of a given hydantoinase E.sub.2V, the polypeptide sequence of which is a variant of one of SEQ ID NO: 10, SEQ ID NO: 11, SEQ ID NO: 12, SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15, SEQ ID NO: 16, relative to the hydantoinase activity of an hydantoinase E.sub.2S, wherein the polypeptide sequence of E.sub.2S is selected from SEQ ID NO: 10, SEQ ID NO: 11, SEQ ID NO: 12, SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15, SEQ ID NO: 16, may be determined in the context of the present invention by the following Assay F:

(336) 4.5.11.2 Assay F

(337) F-1 Assay D.sub.L as set forth under item 4.5.9.1 is conducted, wherein E.sub.2S is the enzyme to be tested. The obtained molar amount of the compound according to formula L-(II)  is determined according to Assay D.sub.L.

(338) F-2 Step F-1 is repeated, except that, instead of E.sub.2S, E.sub.2V is used as the enzyme to be tested.

(339) F-3. Then, the molar amount of the compound according to formula L-(II)  that is determined in step F-2, is divided by the molar amount of the compound according to formula L-(II)  that is determined in step F-1, and the obtained ratio is multiplied by 100, giving the hydantoinase activity of hydantoinase E.sub.2V relative to the hydantoinase activity of the hydantoinase E.sub.2S, in %.

4.5.12 Preferred Hydantoinase Variants of SEQ ID NOs: 10-16

(340) In the context of the invention, hydantoinases E.sub.2, the polypeptide sequence of which is selected from the group consisting of SEQ ID NO: 10, SEQ ID NO: 11; SEQ ID NO: 12, SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15, SEQ ID NO: 16, are generally denoted as “E.sub.2S”. Hydantoinases E.sub.2, the polypeptide sequence of which is selected from variants of a sequence selected from the group consisting of SEQ ID NO: 10, SEQ ID NO: 11; SEQ ID NO: 12, SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15, SEQ ID NO: 16, are generally denoted as “E.sub.2V”.

(341) In a preferred embodiment of the method according to the first aspect of the present invention, the reaction in step (b) is catalyzed by a hydantoinase E.sub.2, preferably an L-hydantoinase E.sub.2, and the polypeptide sequence of the hydantoinase E.sub.2, preferably the L-hydantoinase E.sub.2, is selected from the group consisting of SEQ ID NO: 10 and variants thereof, SEQ ID NO: 11 and variants thereof, SEQ ID NO: 12 and variants thereof, SEQ ID NO: 13 and variants thereof, SEQ ID NO: 14 and variants thereof, SEQ ID NO: 15 and variants thereof, SEQ ID NO: 16 and variants thereof. More preferably, the reaction in step (b) is catalyzed by a hydantoinase E.sub.2, preferably an L-hydantoinase E.sub.2, and the polypeptide sequence of the hydantoinase E.sub.2, preferably the L-hydantoinase E.sub.2, is selected from the group consisting of SEQ ID NO: 10 and variants of SEQ ID NO: 10.

(342) 4.5.12.1 Preferred Variants of SEQ ID NO: 10

(343) According to the invention, the polypeptide sequence of the hydantoinase E.sub.2, preferably the polypeptide sequence of the L-hydantoinase E.sub.2, may also be a variant of SEQ ID NO: 10.

(344) The hydantoinase E.sub.2, the polypeptide sequence of which is SEQ ID NO: 10, is denoted as "E.sub.210S". The hydantoinase E.sub.2, the polypeptide sequence of which is selected from variants of SEQ ID NO: 10, are generally denoted as "E.sub.210V".

(345) A variant of the polypeptide sequence of SEQ ID NO: 10 is a polypeptide with sequence identity of at least 60%, preferably $\geq 65\%$, more preferably $\geq 70\%$, more preferably $\geq 75\%$, more preferably $\geq 80\%$, more preferably $\geq 85\%$, more preferably $\geq 90\%$, more preferably $\geq 91\%$, more preferably $\geq 92\%$, more preferably $\geq 93\%$, more preferably $\geq 94\%$, more preferably $\geq 95\%$, more preferably $\geq 96\%$, more preferably $\geq 97\%$, more preferably $\geq 98\%$, more preferably $\geq 99\%$, more preferably $\geq 99.9\%$ sequence identity to polypeptide sequence SEQ ID NO: 10.

(346) The polypeptide sequence of a variant of the polypeptide sequence SEQ ID NO: 10 is not identical to SEQ ID NO: 10.

(347) According to the invention, a hydantoinase E.sub.210V has hydantoinase activity and preferably L-hydantoinase activity, determined as described under items 4.5.10.2 and 4.5.10.3.1.

(348) According to the invention, a hydantoinase E.sub.210V preferably has hydantoinase activity of at least 1%, preferably of at least 10%, more preferably of at least 20%, more preferably of at least 30%, more preferably of at least 40%, more preferably of at least 50%, more preferably of at least 60%, more preferably of at least 70%, more preferably of at least 80%, more preferably of at least 90%, more preferably of at least 99%, more preferably of at least 100% the hydantoinase activity of the hydantoinase E.sub.210S, wherein the hydantoinase activity of E.sub.210V, relative to the hydantoinase activity of E.sub.210S is determined by Assay F described under item 4.5.11.2.

(349) It is even more preferably according to the invention, that a hydantoinase E.sub.210V has hydantoinase activity in the range of 1 to 1000%, preferably in the range of 5 to 500%, more preferably in the range of 10 to 400%, more preferably in the range of 40 to 200%, more preferably in the range of 50 to 150%, more preferably in the range of 60 to 140%, more preferably in the range of 70 to 130%, more preferably in the range of 80 to 120%, more preferably in the range of 90 to 110%, more preferably 100% the hydantoinase activity of the hydantoinase E.sub.210S, wherein the hydantoinase activity of E.sub.210V, relative to the hydantoinase activity E.sub.210S is determined by Assay F described under item 4.5.11.2.

(350) 4.5.12.2 Preferred Variants of SEQ ID NO: 11

(351) According to the invention, the polypeptide sequence of the hydantoinase E.sub.2, preferably the polypeptide sequence of the L-hydantoinase E.sub.2, may also be a variant of SEQ ID NO: 11.

(352) The hydantoinase E.sub.2, the polypeptide sequence of which is SEQ ID NO: 11, is denoted as "E.sub.211S". The hydantoinase E.sub.2, the polypeptide sequence of which is selected from variants of SEQ ID NO: 11, are generally denoted as "E.sub.211V".

(353) A variant of the polypeptide sequence of SEQ ID NO: 11 is a polypeptide with sequence identity of at least 60%, preferably $\geq 65\%$, more preferably $\geq 70\%$, more preferably $\geq 75\%$, more preferably $\geq 80\%$, more preferably $\geq 85\%$, more preferably $\geq 90\%$, more preferably $\geq 91\%$, more preferably $\geq 92\%$, more preferably $\geq 93\%$, more preferably $\geq 94\%$, more preferably $\geq 95\%$, more preferably $\geq 96\%$, more preferably $\geq 97\%$, more preferably $\geq 98\%$, more preferably $\geq 99\%$, more preferably $\geq 99.9\%$ sequence identity to polypeptide sequence SEQ ID NO: 11.

(354) The polypeptide sequence of a variant of the polypeptide sequence SEQ ID NO: 11 is not identical to SEQ ID NO: 11.

(355) According to the invention, a hydantoinase E.sub.211V has hydantoinase activity and preferably L-hydantoinase activity, determined as described under items 4.5.10.2 and 4.5.10.3.1.

(356) According to the invention, a hydantoinase E.sub.211V preferably has hydantoinase activity of at least 1%, preferably of at least 10%, more preferably of at least 20%, more preferably of at least 30%, more preferably of at least 40%, more preferably of at least 50%, more preferably of at

least 60%, more preferably of at least 70%, more preferably of at least 80%, more preferably of at least 90%, more preferably of at least 99%, more preferably of at least 100% the hydantoinase activity of the hydantoinase E.sub.211S, wherein the hydantoinase activity of E.sub.211V, relative to the hydantoinase activity of E.sub.211S is determined by Assay F described under item 4.5.11.2. (357) It is even more preferably according to the invention, that a hydantoinase E.sub.211V has hydantoinase activity in the range of 1 to 1000%, preferably in the range of 5 to 500%, more preferably in the range of 10 to 400%, more preferably in the range of 40 to 200%, more preferably in the range of 50 to 150%, more preferably in the range of 60 to 140%, more preferably in the range of 70 to 130%, more preferably in the range of 80 to 120%, more preferably in the range of 90 to 110%, more preferably 100% the hydantoinase activity of the hydantoinase E.sub.211S, wherein the hydantoinase activity of E.sub.211V, relative to the hydantoinase activity E.sub.211S is determined by Assay F described under item 4.5.11.2.

(358) 4.5.12.3 Preferred Variants of SEQ ID NO: 12

(359) According to the invention, the polypeptide sequence of the hydantoinase E.sub.2, preferably the polypeptide sequence of the L-hydantoinase E.sub.2, may also be a variant of SEQ ID NO: 12.

(360) The hydantoinase E.sub.2, the polypeptide sequence of which is SEQ ID NO: 12, is denoted as "E.sub.212S". The hydantoinase E.sub.2, the polypeptide sequence of which is selected from variants of SEQ ID NO: 12, are generally denoted as "E.sub.212V".

(361) A variant of the polypeptide sequence of SEQ ID NO: 12 is a polypeptide with sequence identity of at least 60%, preferably $\geq 65\%$, more preferably $\geq 70\%$, more preferably $\geq 75\%$, more preferably $\geq 80\%$, more preferably $\geq 85\%$, more preferably $\geq 90\%$, more preferably $\geq 91\%$, more preferably $\geq 92\%$, more preferably $\geq 93\%$, more preferably $\geq 94\%$, more preferably $\geq 95\%$, more preferably $\geq 96\%$, more preferably $\geq 97\%$, more preferably $\geq 98\%$, more preferably $\geq 99\%$, more preferably $\geq 99.9\%$ sequence identity to polypeptide sequence SEQ ID NO: 12.

(362) The polypeptide sequence of a variant of the polypeptide sequence SEQ ID NO: 12 is not identical to SEQ ID NO: 12.

(363) According to the invention, a hydantoinase E.sub.212V has hydantoinase activity and preferably L-hydantoinase activity, determined as described under items 4.5.10.2 and 4.5.10.3.1.

(364) According to the invention, a hydantoinase E.sub.212V preferably has hydantoinase activity of at least 1%, preferably of at least 10%, more preferably of at least 20%, more preferably of at least 30%, more preferably of at least 40%, more preferably of at least 50%, more preferably of at least 60%, more preferably of at least 70%, more preferably of at least 80%, more preferably of at least 90%, more preferably of at least 99%, more preferably of at least 100% the hydantoinase activity of the hydantoinase E.sub.212S, wherein the hydantoinase activity of E.sub.212V, relative to the hydantoinase activity of E.sub.212S is determined by Assay F described under item 4.5.11.2.

(365) It is even more preferably according to the invention, that a hydantoinase E.sub.212V has hydantoinase activity in the range of 1 to 1000%, preferably in the range of 5 to 500%, more preferably in the range of 10 to 400%, more preferably in the range of 40 to 200%, more preferably in the range of 50 to 150%, more preferably in the range of 60 to 140%, more preferably in the range of 70 to 130%, more preferably in the range of 80 to 120%, more preferably in the range of 90 to 110%, more preferably 100% the hydantoinase activity of the hydantoinase E.sub.212S, wherein the hydantoinase activity of E.sub.212V, relative to the hydantoinase activity E.sub.212S is determined by Assay F described under item 4.5.11.2.

(366) 4.5.12.4 Preferred Variants of SEQ ID NO: 13

(367) According to the invention, the polypeptide sequence of the hydantoinase E.sub.2, preferably the polypeptide sequence of the L-hydantoinase E.sub.2, may also be a variant of SEQ ID NO: 13.

(368) The hydantoinase E.sub.2, the polypeptide sequence of which is SEQ ID NO: 13, is denoted as "E.sub.213S". The hydantoinase E.sub.2, the polypeptide sequence of which is selected from variants of SEQ ID NO: 13, are generally denoted as "E.sub.213V".

(369) A variant of the polypeptide sequence of SEQ ID NO: 13 is a polypeptide with sequence

identity of at least 60%, preferably $\geq 65\%$, more preferably $\geq 70\%$, more preferably $\geq 75\%$, more preferably $\geq 80\%$, more preferably $\geq 85\%$, more preferably $\geq 90\%$, more preferably $\geq 91\%$, more preferably $\geq 92\%$, more preferably $\geq 93\%$, more preferably $\geq 94\%$, more preferably $\geq 95\%$, more preferably $\geq 96\%$, more preferably $\geq 97\%$, more preferably $\geq 98\%$, more preferably $\geq 99\%$, more preferably $\geq 99.9\%$ sequence identity to polypeptide sequence SEQ ID NO: 13.

(370) The polypeptide sequence of a variant of the polypeptide sequence SEQ ID NO: 13 is not identical to SEQ ID NO: 13.

(371) According to the invention, a hydantoinase E.sub.213V has hydantoinase activity and preferably L-hydantoinase activity, determined as described under items 4.5.10.2 and 4.5.10.3.1.

(372) According to the invention, a hydantoinase E.sub.213V preferably has hydantoinase activity of at least 1%, preferably of at least 10%, more preferably of at least 20%, more preferably of at least 30%, more preferably of at least 40%, more preferably of at least 50%, more preferably of at least 60%, more preferably of at least 70%, more preferably of at least 80%, more preferably of at least 90%, more preferably of at least 99%, more preferably of at least 100% the hydantoinase activity of the hydantoinase E.sub.213S, wherein the hydantoinase activity of E.sub.213V, relative to the hydantoinase activity of E.sub.213S is determined by Assay F described under item 4.5.11.2.

(373) It is even more preferably according to the invention, that a hydantoinase E.sub.213V has hydantoinase activity in the range of 1 to 1000%, preferably in the range of 5 to 500%, more preferably in the range of 10 to 400%, more preferably in the range of 40 to 200%, more preferably in the range of 50 to 150%, more preferably in the range of 60 to 140%, more preferably in the range of 70 to 130%, more preferably in the range of 80 to 120%, more preferably in the range of 90 to 110%, more preferably 100% the hydantoinase activity of the hydantoinase E.sub.213S, wherein the hydantoinase activity of E.sub.213V, relative to the hydantoinase activity E.sub.213S is determined by Assay F described under item 4.5.11.2.

(374) 4.5.12.5 Preferred Variants of SEQ ID NO: 14

(375) According to the invention, the polypeptide sequence of the hydantoinase E.sub.2, preferably the polypeptide sequence of the L-hydantoinase E.sub.2, may also be a variant of SEQ ID NO: 14.

(376) The hydantoinase E.sub.2, the polypeptide sequence of which is SEQ ID NO: 14, is denoted as "E.sub.214S". The hydantoinase E.sub.2, the polypeptide sequence of which is selected from variants of SEQ ID NO: 14, are generally denoted as "E.sub.214V".

(377) A variant of the polypeptide sequence of SEQ ID NO: 14 is a polypeptide with sequence identity of at least 60%, preferably $\geq 65\%$, more preferably $\geq 70\%$, more preferably $\geq 75\%$, more preferably $\geq 80\%$, more preferably $\geq 85\%$, more preferably $\geq 90\%$, more preferably $\geq 91\%$, more preferably $\geq 92\%$, more preferably $\geq 93\%$, more preferably $\geq 94\%$, more preferably $\geq 95\%$, more preferably $\geq 96\%$, more preferably $\geq 97\%$, more preferably $\geq 98\%$, more preferably $\geq 99\%$, more preferably $\geq 99.9\%$ sequence identity to polypeptide sequence SEQ ID NO: 14.

(378) The polypeptide sequence of a variant of the polypeptide sequence SEQ ID NO: 14 is not identical to SEQ ID NO: 14.

(379) According to the invention, a hydantoinase E.sub.214V has hydantoinase activity and preferably L-hydantoinase activity, determined as described under items 4.5.10.2 and 4.5.10.3.1.

(380) According to the invention, a hydantoinase E.sub.214V preferably has hydantoinase activity of at least 1%, preferably of at least 10%, more preferably of at least 20%, more preferably of at least 30%, more preferably of at least 40%, more preferably of at least 50%, more preferably of at least 60%, more preferably of at least 70%, more preferably of at least 80%, more preferably of at least 90%, more preferably of at least 99%, more preferably of at least 100% the hydantoinase activity of the hydantoinase E.sub.214S, wherein the hydantoinase activity of E.sub.214V, relative to the hydantoinase activity of E.sub.214S is determined by Assay F described under item 4.5.11.2.

(381) It is even more preferably according to the invention, that a hydantoinase E.sub.214V has hydantoinase activity in the range of 1 to 1000%, preferably in the range of 5 to 500%, more preferably in the range of 10 to 400%, more preferably in the range of 40 to 200%, more preferably

in the range of 50 to 150%, more preferably in the range of 60 to 140%, more preferably in the range of 70 to 130%, more preferably in the range of 80 to 120%, more preferably in the range of 90 to 110%, more preferably 100% the hydantoinase activity of the hydantoinase E.sub.214S, wherein the hydantoinase activity of E.sub.214V, relative to the hydantoinase activity E.sub.214S is determined by Assay F described under item 4.5.11.2.

(382) 4.5.12.6 Preferred Variants of SEQ ID NO: 15

(383) According to the invention, the polypeptide sequence of the hydantoinase E.sub.2, preferably the polypeptide sequence of the L-hydantoinase E.sub.2, may also be a variant of SEQ ID NO: 15.

(384) The hydantoinase E.sub.2, the polypeptide sequence of which is SEQ ID NO: 15, is denoted as "E.sub.215S". The hydantoinase E.sub.2, the polypeptide sequence of which is selected from variants of SEQ ID NO: 15, are generally denoted as "E.sub.215V".

(385) A variant of the polypeptide sequence of SEQ ID NO: 15 is a polypeptide with sequence identity of at least 60%, preferably $\geq 65\%$, more preferably $\geq 70\%$, more preferably $\geq 75\%$, more preferably $\geq 80\%$, more preferably $\geq 85\%$, more preferably $\geq 90\%$, more preferably $\geq 91\%$, more preferably $\geq 92\%$, more preferably $\geq 93\%$, more preferably $\geq 94\%$, more preferably $\geq 95\%$, more preferably $\geq 96\%$, more preferably $\geq 97\%$, more preferably $\geq 98\%$, more preferably $\geq 99\%$, more preferably $\geq 99.9\%$ sequence identity to polypeptide sequence SEQ ID NO: 15.

(386) The polypeptide sequence of a variant of the polypeptide sequence SEQ ID NO: 15 is not identical to SEQ ID NO: 15.

(387) According to the invention, a hydantoinase E.sub.215V has hydantoinase activity and preferably L-hydantoinase activity, determined as described under items 4.5.10.2 and 4.5.10.3.1.

(388) According to the invention, a hydantoinase E.sub.215V preferably has hydantoinase activity of at least 1%, preferably of at least 10%, more preferably of at least 20%, more preferably of at least 30%, more preferably of at least 40%, more preferably of at least 50%, more preferably of at least 60%, more preferably of at least 70%, more preferably of at least 80%, more preferably of at least 90%, more preferably of at least 99%, more preferably of at least 100% the hydantoinase activity of the hydantoinase E.sub.215S, wherein the hydantoinase activity of E.sub.215V, relative to the hydantoinase activity of E.sub.215S is determined by Assay F described under item 4.5.11.2.

(389) It is even more preferably according to the invention, that a hydantoinase E.sub.215V has hydantoinase activity in the range of 1 to 1000%, preferably in the range of 5 to 500%, more preferably in the range of 10 to 400%, more preferably in the range of 40 to 200%, more preferably in the range of 50 to 150%, more preferably in the range of 60 to 140%, more preferably in the range of 70 to 130%, more preferably in the range of 80 to 120%, more preferably in the range of 90 to 110%, more preferably 100% the hydantoinase activity of the hydantoinase E.sub.215S, wherein the hydantoinase activity of E.sub.215V, relative to the hydantoinase activity E.sub.215S is determined by Assay F described under item 4.5.11.2.

(390) 4.5.12.7 Preferred Variants of SEQ ID NO: 16

(391) According to the invention, the polypeptide sequence of the hydantoinase E.sub.2, preferably the polypeptide sequence of the L-hydantoinase E.sub.2, may also be a variant of SEQ ID NO: 16.

(392) The hydantoinase E.sub.2, the polypeptide sequence of which is SEQ ID NO: 16, is denoted as "E.sub.216S". The hydantoinase E.sub.2, the polypeptide sequence of which is selected from variants of SEQ ID NO: 16, are generally denoted as "E.sub.216V".

(393) A variant of the polypeptide sequence of SEQ ID NO: 16 is a polypeptide with sequence identity of at least 60%, preferably $\geq 65\%$, more preferably $\geq 70\%$, more preferably $\geq 75\%$, more preferably $\geq 80\%$, more preferably $\geq 85\%$, more preferably $\geq 90\%$, more preferably $\geq 91\%$, more preferably $\geq 92\%$, more preferably $\geq 93\%$, more preferably $\geq 94\%$, more preferably $\geq 95\%$, more preferably $\geq 96\%$, more preferably $\geq 97\%$, more preferably $\geq 98\%$, more preferably $\geq 99\%$, more preferably $\geq 99.9\%$ sequence identity to polypeptide sequence SEQ ID NO: 16.

(394) The polypeptide sequence of a variant of the polypeptide sequence SEQ ID NO: 16 is not identical to SEQ ID NO: 16.

(395) According to the invention, a hydantoinase E.sub.216V has hydantoinase activity and preferably L-hydantoinase activity, determined as described under items 4.5.10.2 and 4.5.10.3.1.

(396) According to the invention, a hydantoinase E.sub.216V preferably has hydantoinase activity of at least 1%, preferably of at least 10%, more preferably of at least 20%, more preferably of at least 30%, more preferably of at least 40%, more preferably of at least 50%, more preferably of at least 60%, more preferably of at least 70%, more preferably of at least 80%, more preferably of at least 90%, more preferably of at least 99%, more preferably of at least 100% the hydantoinase activity of the hydantoinase E.sub.216S, wherein the hydantoinase activity of E.sub.216V, relative to the hydantoinase activity of E.sub.216S is determined by Assay F described under item 4.5.11.2.

(397) It is even more preferably according to the invention, that a hydantoinase E.sub.216V has hydantoinase activity in the range of 1 to 1000%, preferably in the range of 5 to 500%, more preferably in the range of 10 to 400%, more preferably in the range of 40 to 200%, more preferably in the range of 50 to 150%, more preferably in the range of 60 to 140%, more preferably in the range of 70 to 130%, more preferably in the range of 80 to 120%, more preferably in the range of 90 to 110%, more preferably 100% the hydantoinase activity of the hydantoinase E.sub.216S, wherein the hydantoinase activity of E.sub.216V, relative to the hydantoinase activity E.sub.216S is determined by Assay F described under item 4.5.11.2.

4.5.13 Preferred Method Conditions in Step (b)

(398) The reaction in step (b) of the method according to the first aspect of the present invention may be carried out under conditions known to the skilled person.

(399) The reaction medium is preferably aqueous, more preferably an aqueous buffer.

(400) Exemplary buffers commonly used in biotransformation reactions and advantageously used herein include Tris, phosphate, or any of Good's buffers, such as 2-(N-morpholino) ethanesulfonic acid ("MES"), N-(2-acetamido)iminodiacetic acid ("ADA"), piperazine-N,N'-bis(2-ethanesulfonic acid) ("PIPES"), N-(2-acetamido)-2-aminoethanesulfonic acid ("ACES"), P-hydroxy-4-morpholinepropanesulfonic acid ("MOPSO"), choline chloride, 3-(N-morpholino) propanesulfonic acid ("MOPS"), N, N-Bis(2-hydroxyethyl)-2-aminoethanesulfonic acid ("BES"), 2-[[1,3-dihydroxy-2-(hydroxymethyl) propan-2-yl]amino]ethanesulfonic acid ("TES"), 4-(2-hydroxyethyl)-1-piperazineethanesulfonic acid ("HEPES"), 3-(Bis(2-hydroxyethyl)amino)-2-hydroxypropane-1-sulfonic acid ("DIPSO"), acetamidoglycine, 3-(N-Tris(hydroxymethyl)methylamino(-2-hydroxypropane) sulfonic acid ("TAPSO"), piperazine-N,N'-bis(2-hydroxypropanesulfonic acid) ("POPSO"), 4-(2-Hydroxyethyl) piperazine-1-(2-hydroxypropanesulfonic acid) ("HEPPSO"), 3-[4-(2-Hydroxyethyl)-1-piperazinyl]propanesulfonic acid ("HEPPS"), tricine, glycine, bicine, or 3-[[1,3-dihydroxy-2-(hydroxymethyl) propan-2-yl]amino]propane-1-sulfonic acid ("TAPS").

(401) In some embodiments, ammonium can act as a buffer. One or more organic solvents can also be added to the reaction.

(402) The buffer preferably contains metal salts, more preferably metal salts such as halogenides of metals, preferably halogenides of monovalent or bivalent or trivalent metals, preferably chlorides of monovalent or bivalent metals, preferably CoCl.sub.2 or MnCl.sub.2, preferably CoCl.sub.2.

(403) The concentration of these metal salts in the reaction medium is preferably in the range from 1 μ M to 1 M, more preferably 1 mM to 100 mM, even more preferably 1 to 10 mM.

(404) Preferably, step (b) of the method according to the first aspect of the invention is carried out in a phosphate buffer.

(405) The pH of the reaction medium in step (b) of the method according to the first aspect of the invention is preferably in the range of from 2 to 10, more preferably in the range of from 5 to 8, more preferably 7.2 to 7.5, most preferably 7.5.

(406) Preferably, step (b) of the method according to the first aspect of the invention is carried out at a temperature in the range of from 20° C. to 70° C., more preferably in the range of from 30° C. to 55° C., most preferably 50° C.

(407) Preferably, the total concentration of all hydantoinases E.sub.2 in the reaction solution in step (b) is in the range of from 1 μ M to 10 mM, preferably 10 μ M to 1 mM, more preferably 0.1 mM to 0.5 mM, most preferably 0.4 mM.

(408) In alternative preferred embodiments, the total concentration of all hydantoinases E.sub.2 in the reaction solution in step (b) is in the range of from 1 μ g/l to 10 g/l, preferably 0.1 mg/l to 5 g/l, more preferably 1 mg/l to 1 g/l, more preferably 5 mg/l to 500 mg/l.

(409) Preferably, the initial concentration of all the compounds according to formula L-(III) in the reaction medium in step (b) is in the range of from 1 μ M to 1 M, preferably of from 10 μ M to 0.5 M, more preferably of from 0.1 mM to 0.1 M, more preferably of from 1 mM to 10 mM, most preferably 1.25 mM.

(410) If compounds according to formula D-(III) are present in the reaction medium in step (b), the initial concentration of all the compounds according to formula D-(III) in the reaction medium is preferably from 1% to 100% the concentration of all the compounds according to formula L-(III), more preferably 10% to 100% the concentration, even more preferably 50 to 100%, even more preferably 100% the concentration of all the compounds according to formula L-(III).

(411) Preferably, step (b) is carried out in the same reaction medium in which step (c) is carried out.

(412) In this case, preferably, the initial concentration of all the compounds according to formula L-(III) in the reaction medium in step (b) is in the range of from 1 μ M to 1 M, preferably of from 10 μ M to 0.5 M, more preferably of from 0.1 mM to 0.1 M, more preferably of from 1 mM to 10 mM, most preferably 1.25 mM.

(413) "Initial concentration of all the compounds according to formula L-(III)/D-(III)" refers to the concentration of the respective compound L-(III) or D-(III), respectively, in the reaction medium when the respective compounds are employed in step (b).

4.5.14 Step (a)

(414) In a preferred embodiment of the method according to the first aspect of the invention, the compound according to formula L-(III) is obtained by a step (a) in which a compound according to formula D-(III) is reacted to give a compound according to formula L-(III):

(415) ##STR00037##

(416) Step (a) gives the starting material for step (b).

(417) R.sup.3D is a radical selected from the above-mentioned formulae (VIII), (IX), (X), preferably selected from the formulae (VIII), (IX), more preferably R.sup.3D has formula (VIII).

(418) When R.sup.3D is a radical of the formula (VIII), the radical R.sup.3D may be selected from the above-mentioned formulae L-(VIII) and D-(VIII), preferably L-(VIII).

(419) When R.sup.3D is a radical of the formula (IX), the radical R.sup.3D may be selected from the above-mentioned formulae L-(IX) and D-(IX), preferably L-(IX).

(420) When R.sup.3D is a radical of the formula (X), the radical R.sup.3D may be selected from the above-mentioned formulae L-(X) and D-(X), preferably L-(X).

(421) The reaction according step (a) may be carried out enzymatically or non-enzymatically, preferably enzymatically. More preferably, the reaction according step (a) is catalyzed by a hydantoin racemase E.sub.3.

(422) In preferred step (a), the compound according to formula D-(III) is employed in step (a) as a mixture M.sub.III of at least one compound D-(III) with at least one compound L-(III), wherein R.sup.3L=R.sup.3D.

(423) In a preferred embodiment, the molar ratio of all compounds according to formula L-(III) in the mixture M.sub.III to all compounds according to formula D-(III) in the mixture M.sub.III is essentially 1:1.

(424) In other preferred embodiments, the molar ratio of all compounds according to formula L-(III) in the mixture M.sub.II to all compounds according to formula D-(III) in the mixture M.sub.III is in the range of from 3:2 to 1:99, more preferably in the range of from 1.01:1 to 1:99, more

preferably in the range of from 1:1 to 1:99, more preferably in the range of from 1:1.01 to 1:99, more preferably in the range of from 1:1.01 to 1:9, more preferably in the range of from 1:1.01 to 1:8, more preferably in the range of from 1:1.01 to 1:3.

(425) Alternatively, D-(III) is comprised in an excess to L-(III) in mixture Mm, meaning that, while at least one compound according to formula L-(III) in the mixture M.sub.III to all compounds according to formula D-(III) in the mixture M.sub.III is <1:1, preferably <0.9:1, more preferably <0.75:1, more preferably <0.5:1, more preferably <0.2:1, more preferably <0.1:1, more preferably <0.01:1.

(426) 4.5.14.1 Step (a) without Enzymatic Catalysis

(427) Step (a) may be carried out non-enzymatically, i.e. without the use of an enzyme. The reaction of compounds according to step D-(III) to compounds according to L-(III) proceeds in alkaline solution, as known to the skilled person and as described by Slomka et al., M. Bovarnick & H. T. Clarke, Journal of the American Chemical Society 1938, 60, 2426-2430, by R. A. Lazarus, J. Org. Chem. 1990, 55, 4755-4757, and by A. S. Bommarius, M. Kottenhahn, H. Klenk, K. Drauz: "A direct route from hydantoins to D-amino acids employing a resting cell biocatalyst with D-hydantoinase and D-carbamoylase activity" on page 164 and 167 in "Microbial Reagents in Organic Synthesis" Series C: Mathematical and Physical Sciences-Vol. 381, S. Servi (Ed.), 1992, Springer Science+Business Media, B. V., Dordrecht.

(428) Therefore, if step (a) is carried out non-enzymatically, the conditions that are preferably applied in the reaction medium in which non-enzymatic step (a) is carried out are preferably those that are described for the preferred conditions for step (b) (item 4.5.13), except that the pH is ≥ 8 , preferably in the range of 8 to 12, more preferably 8 to 11, more preferably 8 to 10, even more preferably 8 to 9. As in these embodiments, the preferred conditions in step (a) and (b) with respect to the pH ranges are different, it is preferable that the reaction media in steps (a) and (b) are different.

(429) 4.5.14.2 Enzymatic Conversion of D-(III) into L-(III)

(430) Step (a) is preferably carried out enzymatically, i.e. the reaction according to step (a) is preferably catalyzed by a hydantoin racemase E.sub.3.

(431) Namely, it was surprisingly found that hydantoin racemases accept compounds of formula D-(III) as substrates and convert them to products according to formulae L-(III), and hence catalyze the reaction according to step (a), while there is no catalysis of the corresponding compound according to formula D-(III), in which R=H.

(432) This finding is of high scientific and economic value, as it further expands the scope of new starting materials for the production of L-glufosinate alkyl esters and L-glufosinate via new synthetic routes. Moreover, this finding also opens new possibilities of enantioselective production of LGA from racemic mixtures of L-(III) and D-(III), as embodied in the method according to the second aspect of the invention.

(433) In nature, hydantoin racemases catalyze the conversion of one of the two hydantoin enantiomers H.sub.L and H.sub.R into the other (see the following reaction <3>):

(434) ##STR00038##

(435) It was now surprisingly found that hydantoin racemases also accept substrates in which $R^{**}=R_{sup}.Y=$

(436) ##STR00039##

wherein $R=R_{sup}.3D$.

(437) Surprisingly, they do not accept substrates in which $R^{**}=R_{sup}.Y=$

(438) ##STR00040##

wherein R=H.

(439) Suitable hydantoin racemases are described e.g. in WO 01/23582 A1 and by U. Engel, J. Rudat, C. Sylatk in "The hydantoinase process: recent developments for the production of non-canonical amino acids" in the book "Industrial biocatalysis" by P. Grunwald (Ed.), Pan Stanford

Series on Biocatalysis, 2015, pages 817-862, and by F. J. Las Heras-Vazquez, J. M. Clemente-Jimenez, S. Martinez-Rodriguez, F. Rodriguez-Vico in "Hydantoin racemase: the key enzyme for the production of optically pure α -amino acids" in chapter 12 of the book "Modern Biocatalysis: Stereoselective and environmentally friendly reactions" by W. Fessner, T. Anthonsen (Eds), Weinheim: WILEY-VCH Verlag GmbH & Co, 2009, pages 173-193.

(440) A hydantoin racemase E.sub.3 that may be used in optional step (a) of the method according to the first aspect of the invention may originate from *Agrobacterium* sp., in particular *Agrobacterium* strain IP_I-671; *Arthrobacter* sp., in particular *Arthrobacter aurescens*, more in particular *Arthrobacter aurescens* DSM 3745 or *Arthrobacter* sp. BT801; *Flavobacterium* sp., in particular *Flavobacterium* sp. AJ 11199; *Microbacterium* sp., in particular *Microbacterium liquefaciens*, preferably *Microbacterium liquefaciens* AJ 3912; *Pasteurella* sp., in particular *Pasteurella* sp. AJ11221; *Pseudomonas* sp., in particular *Pseudomonas* sp. NS671; *Pyrococcus* sp., in particular *Pyrococcus horikoshii* OT3; *Rhodococcus* sp., in particular *Rhodococcus* R04; *Sinorhizobium* sp., in particular *Sinorhizobium meliloti*, more in particular *Sinorhizobium meliloti* CECT 4114, most preferably from *Arthrobacter aurescens* DSM 3745.

(441) A hydantoin racemase E.sub.3 suitable for the method according to the present invention may be the enzyme HyuR, which originates from *Arthrobacter aurescens* DSM 3745. Another enzyme may be selected from HyuE, Hyu2, HRase, HyuA, PH1054.

(442) The hydantoin racemase E.sub.3 that may preferably be used in preferred step (a) of the method according to the first aspect of the invention may be categorized in the EC class 5.1.99.5.

(443) The following table 3 gives preferred examples for polypeptide sequences of hydantoin racemase E.sub.3 that may be preferably used in step (a) of the method according to the first aspect of the invention. The genes encoding the respective hydantoin racemase E.sub.3 and the respective accession code are indicated as far as known.

(444) TABLE-US-00003 TABLE 3 Hydantoin Racemases (EC 5.1.99.5) GenBank/UniProt SEQ ID NO: of Strain Gene name accession the polypeptide *Arthrobacter hyuR* SEQ ID NO: 17 *aurescens* DSM 3745 *Pseudomonas* sp. hyuE Q00924 SEQ ID NO: 18 NS671 *Rhodococcus* R04 hyu2 SEQ ID NO: 19 *Microbacterium* "HRase" SEQ ID NO: 20 *liquefaciens* AJ 3912 *Sinorhizobium* hyuA Q6TMG4 SEQ ID NO: 21 *meliloti* CECT 4114 *Flavobacterium* sp. "HRase" SEQ ID NO: 22 AJ11199 *Agrobacterium* hyuA SEQ ID NO: 23 IP_I-671 *Pasteurella* not assigned SEQ ID NO: 24 sp. AJ 11221 *Pyrococcus* PH1054 O58781 SEQ ID NO: 25 *horikoshii* OT3 *Arthrobacter* sp. hyuA AAL55411 SEQ ID NO: 26 BT801

(445) In a preferred embodiment of the preferred method according to the second aspect of the present invention, the reaction according to step (a) is catalyzed by a hydantoin racemase E.sub.3, wherein the polypeptide sequence of E.sub.3 is selected from the group consisting of SEQ ID NO: 17 and variants thereof, SEQ ID NO: 18 and variants thereof, SEQ ID NO: 19 and variants thereof, SEQ ID NO: 20 and variants thereof, SEQ ID NO: 21 and variants thereof, SEQ ID NO: 22 and variants thereof, SEQ ID NO: 23 and variants thereof, SEQ ID NO: 24 and variants thereof, SEQ ID NO: 25 and variants thereof, SEQ ID NO: 26 and variants thereof.

(446) 4.5.14.3 Assay G for Determining Hydantoin Racemase Activity

(447) The skilled person is aware of hydantoin racemases, that may be used in preferred step (a) of the method according to the first aspect of the invention.

(448) In particular, Assay G, described in the following, may be used to determine hydantoin racemase activity of a given enzyme E.sub.Z and may advantageously be used according to the invention to determine carbamoylase and L-carbamoylase activity in variants of SEQ ID NO: 17, variants of variants of SEQ ID NO: 18, variants of SEQ ID NO: 19, variants of SEQ ID NO: 20, variants of SEQ ID NO: 21, variants of SEQ ID NO: 22, variants of variants of SEQ ID NO: 23, variants of SEQ ID NO: 24, variants of SEQ ID NO: 25, variants of SEQ ID NO: 26.

(449) For the purpose of Assay G, the molar mass of the enzyme E.sub.Z to be tested is calculated as the molar mass of the polypeptide sequence of E.sub.Z.

(450) Moreover, a substrate according to formula D-(III), wherein R^{sup.3D}=n-butyl, is used in Assay G so that there is only one chiral carbon atom in the substrate. The compound according to formula D-(III), wherein R^{sup.3D}=n-butyl, is denoted as “D-(III)+”.

(451) A product according to formula L-(III), wherein R^{sup.3L}=n-butyl, is detected in Assay G.

(452) The compound according to formula L-(III), wherein R^{sup.3L}=n-butyl, is denoted as “L-(III)+”.

(453) Assay G:

(454) To 0.9 ml of an aqueous reaction solution (phosphate buffer, pH 7.2, 10 mM Mg_{sub.2}Cl), containing 50 mM of the pure D-enantiomer of an n-butyl ester of hydantoin glufosinate of the formula D-(III)  are added 400 nmol of E_{sub.Z} in 0.1 ml aqueous phosphate buffer (pH 7.2, 10 mM MnCl_{sub.2}). The resulting solution is incubated at 25° C., and the pH is held at pH 7.2 by addition of 0.5 M. After 300 minutes, the reaction is stopped by addition of 2 M HCl to achieve a pH of 2.5. The molar amount of the L-enantiomer of formula L-(III)

 is measured, at least every 3 minutes (determination by LC-MS, e.g. by the LC-MS method described in the example section, item 5.4, for the detection of LGA).

(455) 4.5.14.4 Assay H for Identifying Hydantoin Racemases

(456) Whether a given enzyme E_{sub.Z} may be considered a hydantoin racemase E_{sub.3}, may be determined in the context of the present invention by the following Assay H:

(457) H-1 Firstly, Assay G as set forth under item 4.5.14.3 is conducted, and the obtained molar amount of L-(III)  is determined according to Assay G.

(458) H-2 Then, step H-1 is repeated, except that instead of the addition of 400 nmol E_{sub.Z} in 0.1 ml aqueous phosphate buffer (pH 7.2, 10 mM MnCl_{sub.2}), 0.1 ml aqueous phosphate buffer (pH 7.2, 10 mM MnCl_{sub.2}) without E_{sub.Z} is added.

(459) 4.5.14.5 Hydantoin Racemase Activity

(460) If the molar amount of the compound of formula L-(III)  that is determined in step H-1 is greater than the molar amount of the compound of formula L-(III)  that is determined in step H-2, then E_{sub.Z} is deemed to have hydantoin racemase activity, and hence may be considered a hydantoin racemase E_{sub.3} in the context of the invention.

(461) 4.5.14.6 Assay J for Identifying Preferred Hydantoin Racemase Variants of SEQ ID NO: 17-26

(462) An enzyme, the polypeptide sequence of which is selected from the group consisting of SEQ ID NO: 17, SEQ ID NO: 18, SEQ ID NO: 19, SEQ ID NO: 20, SEQ ID NO: 21, SEQ ID NO: 22, SEQ ID NO: 23, SEQ ID NO: 24, SEQ ID NO: 25, SEQ ID NO: 26, has hydantoin racemase activity.

(463) In a preferred embodiment of the method according to the first aspect of the invention, the polypeptide sequence of the hydantoin racemase E_{sub.3} is selected from the group consisting of SEQ ID NO: 17 and variants thereof, SEQ ID NO: 18 and variants thereof, SEQ ID NO: 19 and variants thereof, SEQ ID NO: 20 and variants thereof, SEQ ID NO: 21 and variants thereof, SEQ ID NO: 22 and variants thereof, SEQ ID NO: 23 and variants thereof, SEQ ID NO: 24 and variants thereof, SEQ ID NO: 25 and variants thereof, SEQ ID NO: 26 and variants thereof.

(464) The term “variant” is defined under item 4.3.

(465) In the context of the invention, an enzyme, the polypeptide sequence of which is a variant of one of SEQ ID NO: 17, SEQ ID NO: 18, SEQ ID NO: 19, SEQ ID NO: 20, SEQ ID NO: 21, SEQ ID NO: 22, SEQ ID NO: 23, SEQ ID NO: 24, SEQ ID NO: 25, SEQ ID NO: 26, has hydantoin racemase activity.

(466) Whether a given enzyme E_{sub.Z}, the polypeptide sequence of which is a variant of one SEQ ID NO: 17, SEQ ID NO: 18, SEQ ID NO: 19, SEQ ID NO: 20, SEQ ID NO: 21, SEQ ID NO: 22, SEQ ID NO: 23, SEQ ID NO: 24, SEQ ID NO: 25, SEQ ID NO: 26, has hydantoin racemase activity may be determined as set forth under items 4.5.14.4 and 4.5.14.5.

(467) The hydantoin racemase activity of a given hydantoin racemase E_{sub.3V}, the polypeptide

sequence of which is a variant of one of SEQ ID NO: 17, SEQ ID NO: 18, SEQ ID NO: 19, SEQ ID NO: 20, SEQ ID NO: 21, SEQ ID NO: 22, SEQ ID NO: 23, SEQ ID NO: 24, SEQ ID NO: 25, SEQ ID NO: 26, relative to the hydantoin racemase activity of a hydantoin racemase E.sub.3S, wherein the polypeptide sequence of E.sub.3S is selected from SEQ ID NO: 17, SEQ ID NO: 18, SEQ ID NO: 19, SEQ ID NO: 20, SEQ ID NO: 21, SEQ ID NO: 22, SEQ ID NO: 23, SEQ ID NO: 24, SEQ ID NO: 25, SEQ ID NO: 26, may be quantified in the context of the present invention by the following Assay J.

(468) Assay J:

(469) J-1 Firstly, Assay G as set forth under item 4.5.14.3 is conducted, wherein E.sub.3S is the enzyme to be tested. The molar amount of the compound according to formula L-(III)

 is determined according to Assay G.

(470) J-2 Step J-1 is repeated, except that, instead of E.sub.3S, E.sub.3V is used as the enzyme to be tested.

(471) J-3. Then, the molar amount of the compound according to formula L-(III)  is determined in step J-2, is divided by the molar amount of the compound according to formula L-(III), wherein R=n-butyl, is determined in step J-1, and the obtained ratio is multiplied by 100, giving the hydantoin racemase activity of hydantoin racemase E.sub.3V, relative to the hydantoin racemase activity of the hydantoin racemase E.sub.3S, in %.

(472) 4.5.14.7 Preferred Hydantoin Racemase Variants SEQ ID NO: 17-26

(473) In the context of the present invention, hydantoin racemase E.sub.3, the polypeptide sequence of which is selected from the group consisting of SEQ ID NO: 17, SEQ ID NO: 18, SEQ ID NO: 19, SEQ ID NO: 20, SEQ ID NO: 21, SEQ ID NO: 22, SEQ ID NO: 23, SEQ ID NO: 24, SEQ ID NO: 25, SEQ ID NO: 26, are generally denoted as “E.sub.3S”.

(474) Hydantoin racemases E.sub.3, the polypeptide sequence of which is selected from variants of a sequence selected from the group consisting of f SEQ ID NO: 17, SEQ ID NO: 18, SEQ ID NO: 19, SEQ ID NO: 20, SEQ ID NO: 21, SEQ ID NO: 22, SEQ ID NO: 23, SEQ ID NO: 24, SEQ ID NO: 25, SEQ ID NO: 26, are generally denoted as “E.sub.3V”.

(475) In a preferred embodiment of the method according to the first aspect of the present invention, the reaction in step (a) is catalyzed by a hydantoin racemase E.sub.3, and the polypeptide sequence of the hydantoin racemase E.sub.3 is selected from the group consisting of SEQ ID NO: 17 and variants thereof, SEQ ID NO: 18 and variants thereof, SEQ ID NO: 19 and variants thereof, SEQ ID NO: 20 and variants thereof, SEQ ID NO: 21 and variants thereof, SEQ ID NO: 22 and variants thereof, SEQ ID NO: 23 and variants thereof, SEQ ID NO: 24 and variants thereof, SEQ ID NO: 25 and variants thereof, SEQ ID NO: 26 and variants thereof. More preferably, the reaction in step (a) is catalyzed by a hydantoin racemase E.sub.3, and the polypeptide sequence of the hydantoin racemase E.sub.3 is selected from the group consisting of SEQ ID NO: 1 and variants of SEQ ID NO: 1.

(476) 4.5.14.7.1 Preferred Variants of SEQ ID NO: 17

(477) According to the invention, the polypeptide sequence of the hydantoin racemase E.sub.3 may also be a variant of SEQ ID NO: 17.

(478) The hydantoin racemase E.sub.3, the polypeptide sequence of which is SEQ ID NO: 17, is denoted as “E.sub.317S”. Hydantoin racemases E.sub.3, the polypeptide sequence of which is selected from variants of SEQ ID NO: 17, are generally denoted as “E.sub.317V”.

(479) A variant of the polypeptide sequence of SEQ ID NO: 17 is a polypeptide with sequence identity of at least 60%, preferably $\geq 65\%$, more preferably $\geq 70\%$, more preferably $\geq 75\%$, more preferably $\geq 80\%$, more preferably $\geq 85\%$, more preferably $\geq 90\%$, more preferably $\geq 91\%$, more preferably $\geq 92\%$, more preferably $\geq 93\%$, more preferably $\geq 94\%$, more preferably $\geq 95\%$, more preferably $\geq 96\%$, more preferably $\geq 97\%$, more preferably $\geq 98\%$, more preferably $\geq 99\%$, more preferably $\geq 99.9\%$ sequence identity to polypeptide sequence SEQ ID NO: 17.

(480) The polypeptide sequence of a variant of the polypeptide sequence SEQ ID NO: 17 is not

identical to SEQ ID NO: 17.

(481) According to the invention, a hydantoin racemase E.sub.317V has hydantoin racemase activity, determined as described under item 4.5.14.4 and 4.5.14.5.

(482) According to the invention, a hydantoin racemase E.sub.317V preferably has hydantoin racemase activity of at least 1%, preferably of at least 10%, more preferably of at least 20%, more preferably of at least 30%, more preferably of at least 40%, more preferably of at least 50%, more preferably of at least 60%, more preferably of at least 70%, more preferably of at least 80%, more preferably of at least 90%, more preferably of at least 99%, more preferably of at least 100% the hydantoin racemase of the hydantoin racemase E.sub.317S, wherein the hydantoin racemase activity of E.sub.317V, relative to the hydantoin racemase activity E.sub.317S is determined by Assay J described under item 4.5.14.6.

(483) It is even more preferably according to the invention, that a hydantoin racemase E.sub.317V has hydantoin racemase activity in the range of 1 to 1000%, preferably in the range of 5 to 500%, more preferably in the range of 10 to 400%, more preferably in the range of 40 to 200%, more preferably in the range of 50 to 150%, more prefab in the range of 60 to 140%, more preferably in the range of 70 to 130%, more preferably in the range of 80 to 120%, more preferably in the range of 90 to 110%, more preferably 100% the hydantoin racemase activity of the hydantoin racemase E.sub.317S, wherein the hydantoin racemase activity of E.sub.317V, relative to the hydantoin racemase activity of E.sub.317S is determined by Assay J described under item 4.5.14.6.

(484) 4.5.14.7.2 Preferred Variants of SEQ ID NO: 18

(485) According to the invention, the polypeptide sequence of the hydantoin racemase E.sub.3 may also be a variant of SEQ ID NO: 18.

(486) The hydantoin racemase E.sub.3, the polypeptide sequence of which is SEQ ID NO: 18, is denoted as "E.sub.318S". Hydantoin racemases E.sub.3, the polypeptide sequence of which is selected from variants of SEQ ID NO: 18, are generally denoted as "E.sub.318V".

(487) A variant of the polypeptide sequence of SEQ ID NO: 18 is a polypeptide with sequence identity of at least 60%, preferably $\geq 65\%$, more preferably $\geq 70\%$, more preferably $\geq 75\%$, more preferably $\geq 80\%$, more preferably $\geq 85\%$, more preferably $\geq 90\%$, more preferably $\geq 91\%$, more preferably $\geq 92\%$, more preferably $\geq 93\%$, more preferably $\geq 94\%$, more preferably $\geq 95\%$, more preferably $\geq 96\%$, more preferably $\geq 97\%$, more preferably $\geq 98\%$, more preferably $\geq 99\%$, more preferably $\geq 99.9\%$ sequence identity to polypeptide sequence SEQ ID NO: 18.

(488) The polypeptide sequence of a variant of the polypeptide sequence SEQ ID NO: 18 is not identical to SEQ ID NO: 18.

(489) According to the invention, a hydantoin racemase E.sub.318V has hydantoin racemase activity, determined as described under item 4.5.14.4 and 4.5.14.5.

(490) According to the invention, a hydantoin racemase E.sub.318V preferably has hydantoin racemase activity of at least 1%, preferably of at least 10%, more preferably of at least 20%, more preferably of at least 30%, more preferably of at least 40%, more preferably of at least 50%, more preferably of at least 60%, more preferably of at least 70%, more preferably of at least 80%, more preferably of at least 90%, more preferably of at least 99%, more preferably of at least 100% the hydantoin racemase of the hydantoin racemase E.sub.318S, wherein the hydantoin racemase activity of E.sub.318V, relative to the hydantoin racemase activity E.sub.318S is determined by Assay J described under item 4.5.14.6.

(491) It is even more preferably according to the invention, that a hydantoin racemase E.sub.318V has hydantoin racemase activity in the range of 1 to 1000%, preferably in the range of 5 to 500%, more preferably in the range of 10 to 400%, more preferably in the range of 40 to 200%, more preferably in the range of 50 to 150%, more preferably in the range of 60 to 140%, more preferably in the range of 70 to 130%, more preferably in the range of 80 to 120%, more preferably in the range of 90 to 110%, more preferably 100% the hydantoin racemase activity of the hydantoin racemase E.sub.318S, wherein the hydantoin racemase activity of E.sub.318V, relative to the

hydantoin racemase activity of E.sub.318S is determined by Assay J described under item 4.5.14.6.

(492) 4.5.14.7.3 Preferred Variants of SEQ ID NO: 19

(493) According to the invention, the polypeptide sequence of the hydantoin racemase E.sub.3 may also be a variant of SEQ ID NO: 19.

(494) The hydantoin racemase E.sub.3, the polypeptide sequence of which is SEQ ID NO: 19, is denoted as “E.sub.319S”. Hydantoin racemases E.sub.3, the polypeptide sequence of which is selected from variants of SEQ ID NO: 19, are generally denoted as “E.sub.319V”.

(495) A variant of the polypeptide sequence of SEQ ID NO: 19 is a polypeptide with sequence identity of at least 60%, preferably $\geq 65\%$, more preferably $\geq 70\%$, more preferably $\geq 75\%$, more preferably $\geq 80\%$, more preferably $\geq 85\%$, more preferably $\geq 90\%$, more preferably $\geq 91\%$, more preferably $\geq 92\%$, more preferably $\geq 93\%$, more preferably $\geq 94\%$, more preferably $\geq 95\%$, more preferably $\geq 96\%$, more preferably $\geq 97\%$, more preferably $\geq 98\%$, more preferably $\geq 99\%$, more preferably $\geq 99.9\%$ sequence identity to polypeptide sequence SEQ ID NO: 19.

(496) The polypeptide sequence of a variant of the polypeptide sequence SEQ ID NO: 19 is not identical to SEQ ID NO: 19.

(497) According to the invention, a hydantoin racemase E.sub.319V has hydantoin racemase activity, determined as described under item 4.5.14.4 and 4.5.14.5.

(498) According to the invention, a hydantoin racemase E.sub.319V preferably has hydantoin racemase activity of at least 1%, preferably of at least 10%, more preferably of at least 20%, more preferably of at least 30%, more preferably of at least 40%, more preferably of at least 50%, more preferably of at least 60%, more preferably of at least 70%, more preferably of at least 80%, more preferably of at least 90%, more preferably of at least 99%, more preferably of at least 100% the hydantoin racemase of the hydantoin racemase E.sub.319S, wherein the hydantoin racemase activity of E.sub.319V, relative to the hydantoin racemase activity E.sub.319S is determined by Assay J described under item 4.5.14.6.

(499) It is even more preferably according to the invention, that a hydantoin racemase E.sub.319V has hydantoin racemase activity in the range of 1 to 1000%, preferably in the range of 5 to 500%, more preferably in the range of 10 to 400%, more preferably in the range of 40 to 200%, more preferably in the range of 50 to 150%, more preferably in the range of 60 to 140%, more preferably in the range of 70 to 130%, more preferably in the range of 80 to 120%, more preferably in the range of 90 to 110%, more preferably 100% the hydantoin racemase activity of the hydantoin racemase E.sub.319S, wherein the hydantoin racemase activity of E.sub.319V, relative to the hydantoin racemase activity of E.sub.319S is determined by Assay J described under item 4.5.14.6.

(500) 4.5.14.7.4 Preferred Variants of SEQ ID NO: 20

(501) According to the invention, the polypeptide sequence of the hydantoin racemase E.sub.3 may also be a variant of SEQ ID NO: 20.

(502) The hydantoin racemase E.sub.3, the polypeptide sequence of which is SEQ ID NO: 20, is denoted as “E.sub.320S”. Hydantoin racemases E.sub.3, the polypeptide sequence of which is selected from variants of SEQ ID NO: 20, are generally denoted as “E.sub.320V”.

(503) A variant of the polypeptide sequence of SEQ ID NO: 20 is a polypeptide with sequence identity of at least 60%, preferably $\geq 65\%$, more preferably $\geq 70\%$, more preferably $\geq 75\%$, more preferably $\geq 80\%$, more preferably $\geq 85\%$, more preferably $\geq 90\%$, more preferably $\geq 91\%$, more preferably $\geq 92\%$, more preferably $\geq 93\%$, more preferably $\geq 94\%$, more preferably $\geq 95\%$, more preferably $\geq 96\%$, more preferably $\geq 97\%$, more preferably $\geq 98\%$, more preferably $\geq 99\%$, more preferably $\geq 99.9\%$ sequence identity to polypeptide sequence SEQ ID NO: 20.

(504) The polypeptide sequence of a variant of the polypeptide sequence SEQ ID NO: 20 is not identical to SEQ ID NO: 20.

(505) According to the invention, a hydantoin racemase E.sub.320V has hydantoin racemase activity, determined as described under item 4.5.14.4 and 4.5.14.5.

(506) According to the invention, a hydantoin racemase E.sub.320V preferably has hydantoin

racemase activity of at least 1%, preferably of at least 10%, more preferably of at least 20%, more preferably of at least 30%, more preferably of at least 40%, more preferably of at least 50%, more preferably of at least 60%, more preferably of at least 70%, more preferably of at least 80%, more preferably of at least 90%, more preferably of at least 99%, more preferably of at least 100% the hydantoin racemase of the hydantoin racemase E.sub.320S, wherein the hydantoin racemase activity of E.sub.320V, relative to the hydantoin racemase activity E.sub.320S is determined by Assay J described under item 4.5.14.6.

(507) It is even more preferably according to the invention, that a hydantoin racemase E.sub.320V has hydantoin racemase activity in the range of 1 to 1000%, preferably in the range of 5 to 500%, more preferably in the range of 10 to 400%, more preferably in the range of 40 to 200%, more preferably in the range of 50 to 150%, more preferably in the range of 60 to 140%, more preferably in the range of 70 to 130%, more preferably in the range of 80 to 120%, more preferably in the range of 90 to 110%, more preferably 100% the hydantoin racemase activity of the hydantoin racemase E.sub.320S, wherein the hydantoin racemase activity of E.sub.320V, relative to the hydantoin racemase activity of E.sub.320S is determined by Assay J described under item 4.5.14.6.

(508) 4.5.14.7.5 Preferred Variants of SEQ ID NO: 21

(509) According to the invention, the polypeptide sequence of the hydantoin racemase E.sub.3 may also be a variant of SEQ ID NO: 21.

(510) The hydantoin racemase E.sub.3, the polypeptide sequence of which is SEQ ID NO: 21, is denoted as "E.sub.321S". Hydantoin racemases E.sub.3, the polypeptide sequence of which is selected from variants of SEQ ID NO: 21, are generally denoted as "E.sub.321V".

(511) A variant of the polypeptide sequence of SEQ ID NO: 21 is a polypeptide with sequence identity of at least 60%, preferably $\geq 65\%$, more preferably $\geq 70\%$, more preferably $\geq 75\%$, more preferably $\geq 80\%$, more preferably $\geq 85\%$, more preferably $\geq 90\%$, more preferably $\geq 91\%$, more preferably $\geq 92\%$, more preferably $\geq 93\%$, more preferably $\geq 94\%$, more preferably $\geq 95\%$, more preferably $\geq 96\%$, more preferably $\geq 97\%$, more preferably $\geq 98\%$, more preferably $\geq 99\%$, more preferably $\geq 99.9\%$ sequence identity to polypeptide sequence SEQ ID NO: 21.

(512) The polypeptide sequence of a variant of the polypeptide sequence SEQ ID NO: 21 is not identical to SEQ ID NO: 21.

(513) According to the invention, a hydantoin racemase E.sub.321V has hydantoin racemase activity, determined as described under item 4.5.14.4 and 4.5.14.5.

(514) According to the invention, a hydantoin racemase E.sub.321V preferably has hydantoin racemase activity of at least 1%, preferably of at least 10%, more preferably of at least 20%, more preferably of at least 30%, more preferably of at least 40%, more preferably of at least 50%, more preferably of at least 60%, more preferably of at least 70%, more preferably of at least 80%, more preferably of at least 90%, more preferably of at least 99%, more preferably of at least 100% the hydantoin racemase of the hydantoin racemase E.sub.321S, wherein the hydantoin racemase activity of E.sub.321V, relative to the hydantoin racemase activity E.sub.321S is determined by Assay J described under item 4.5.14.6.

(515) It is even more preferably according to the invention, that a hydantoin racemase E.sub.321V has hydantoin racemase activity in the range of 1 to 1000%, preferably in the range of 5 to 500%, more preferably in the range of 10 to 400%, more preferably in the range of 40 to 200%, more preferably in the range of 50 to 150%, more preferably in the range of 60 to 140%, more preferably in the range of 70 to 130%, more preferably in the range of 80 to 120%, more preferably in the range of 90 to 110%, more preferably 100% the hydantoin racemase activity of the hydantoin racemase E.sub.321S, wherein the hydantoin racemase activity of E.sub.321V, relative to the hydantoin racemase activity of E.sub.321S is determined by Assay J described under item 4.5.14.6.

(516) 4.5.14.7.6 Preferred Variants of SEQ ID NO: 22

(517) According to the invention, the polypeptide sequence of the hydantoin racemase E.sub.3 may also be a variant of SEQ ID NO: 22.

(518) The hydantoin racemase E.sub.3, the polypeptide sequence of which is SEQ ID NO: 22, is denoted as “E.sub.322S”. Hydantoin racemases E.sub.3, the polypeptide sequence of which is selected from variants of SEQ ID NO: 22, are generally denoted as “E.sub.322V”.

(519) A variant of the polypeptide sequence of SEQ ID NO: 22 is a polypeptide with sequence identity of at least 60%, preferably $\geq 65\%$, more preferably $\geq 70\%$, more preferably $\geq 75\%$, more preferably $\geq 80\%$, more preferably $\geq 85\%$, more preferably $\geq 90\%$, more preferably $\geq 91\%$, more preferably $\geq 92\%$, more preferably $\geq 93\%$, more preferably $\geq 94\%$, more preferably $\geq 95\%$, more preferably $\geq 96\%$, more preferably $\geq 97\%$, more preferably $\geq 98\%$, more preferably $\geq 99\%$, more preferably $\geq 99.9\%$ sequence identity to polypeptide sequence SEQ ID NO: 22.

(520) The polypeptide sequence of a variant of the polypeptide sequence SEQ ID NO: 22 is not identical to SEQ ID NO: 22.

(521) According to the invention, a hydantoin racemase E.sub.322V has hydantoin racemase activity, determined as described under item 4.5.14.4 and 4.5.14.5.

(522) According to the invention, a hydantoin racemase E.sub.322V preferably has hydantoin racemase activity of at least 1%, preferably of at least 10%, more preferably of at least 20%, more preferably of at least 30%, more preferably of at least 40%, more preferably of at least 50%, more preferably of at least 60%, more preferably of at least 70%, more preferably of at least 80%, more preferably of at least 90%, more preferably of at least 99%, more preferably of at least 100% the hydantoin racemase of the hydantoin racemase E.sub.322S, wherein the hydantoin racemase activity of E.sub.322V, relative to the hydantoin racemase activity E.sub.322S is determined by Assay J described under item 4.5.14.6.

(523) It is even more preferably according to the invention, that a hydantoin racemase E.sub.322V has hydantoin racemase activity in the range of 1 to 1000%, preferably in the range of 5 to 500%, more preferably in the range of 10 to 400%, more preferably in the range of 40 to 200%, more preferably in the range of 50 to 150%, more preferably in the range of 60 to 140%, more preferably in the range of 70 to 130%, more preferably in the range of 80 to 120%, more preferably in the range of 90 to 110%, more preferably 100% the hydantoin racemase activity of the hydantoin racemase E.sub.322S, wherein the hydantoin racemase activity of E.sub.322V, relative to the hydantoin racemase activity of E.sub.322S is determined by Assay J described under item 4.5.14.6.

(524) 4.5.14.7.7 Preferred Variants of SEQ ID NO: 23

(525) According to the invention, the polypeptide sequence of the hydantoin racemase E.sub.3 may also be a variant of SEQ ID NO: 23.

(526) The hydantoin racemase E.sub.3, the polypeptide sequence of which is SEQ ID NO: 23, is denoted as “E.sub.323S”. Hydantoin racemases E.sub.3, the polypeptide sequence of which is selected from variants of SEQ ID NO: 23, are generally denoted as “E.sub.323V”.

(527) A variant of the polypeptide sequence of SEQ ID NO: 23 is a polypeptide with sequence identity of at least 60%, preferably $\geq 65\%$, more preferably $\geq 70\%$, more preferably $\geq 75\%$, more preferably $\geq 80\%$, more preferably $\geq 85\%$, more preferably $\geq 90\%$, more preferably $\geq 91\%$, more preferably $\geq 92\%$, more preferably $\geq 93\%$, more preferably $\geq 94\%$, more preferably $\geq 95\%$, more preferably $\geq 96\%$, more preferably $\geq 97\%$, more preferably $\geq 98\%$, more preferably $\geq 99\%$, more preferably $\geq 99.9\%$ sequence identity to polypeptide sequence SEQ ID NO: 23.

(528) The polypeptide sequence of a variant of the polypeptide sequence SEQ ID NO: 23 is not identical to SEQ ID NO: 23.

(529) According to the invention, a hydantoin racemase E.sub.323V has hydantoin racemase activity, determined as described under item 4.5.14.4 and 4.5.14.5.

(530) According to the invention, a hydantoin racemase E.sub.323V preferably has hydantoin racemase activity of at least 1%, preferably of at least 10%, more preferably of at least 20%, more preferably of at least 30%, more preferably of at least 40%, more preferably of at least 50%, more preferably of at least 60%, more preferably of at least 70%, more preferably of at least 80%, more preferably of at least 90%, more preferably of at least 99%, more preferably of at least 100% the

hydantoin racemase of the hydantoin racemase E.sub.323S, wherein the hydantoin racemase activity of E.sub.323V, relative to the hydantoin racemase activity E.sub.323S is determined by Assay J described under item 4.5.14.6.

(531) It is even more preferably according to the invention, that a hydantoin racemase E.sub.323V has hydantoin racemase activity in the range of 1 to 1000%, preferably in the range of 5 to 500%, more preferably in the range of 10 to 400%, more preferably in the range of 40 to 200%, more preferably in the range of 50 to 150%, more preferably in the range of 60 to 140%, more preferably in the range of 70 to 130%, more preferably in the range of 80 to 120%, more preferably in the range of 90 to 110%, more preferably 100% the hydantoin racemase activity of the hydantoin racemase E.sub.323S, wherein the hydantoin racemase activity of E.sub.323V, relative to the hydantoin racemase activity of E.sub.323S is determined by Assay J described under item 4.5.14.6.

(532) 4.5.14.7.8 Preferred Variants of SEQ ID NO: 24

(533) According to the invention, the polypeptide sequence of the hydantoin racemase E.sub.3 may also be a variant of SEQ ID NO: 24.

(534) The hydantoin racemase E.sub.3, the polypeptide sequence of which is SEQ ID NO: 24, is denoted as "E.sub.324S". Hydantoin racemases E.sub.3, the polypeptide sequence of which is selected from variants of SEQ ID NO: 24, are generally denoted as "E.sub.324V".

(535) A variant of the polypeptide sequence of SEQ ID NO: 24 is a polypeptide with sequence identity of at least 60%, preferably $\geq 65\%$, more preferably $\geq 70\%$, more preferably $\geq 75\%$, more preferably $\geq 80\%$, more preferably $\geq 85\%$, more preferably $\geq 90\%$, more preferably $\geq 91\%$, more preferably $\geq 92\%$, more preferably $\geq 93\%$, more preferably $\geq 94\%$, more preferably $\geq 95\%$, more preferably $\geq 96\%$, more preferably $\geq 97\%$, more preferably $\geq 98\%$, more preferably $\geq 99\%$, more preferably $\geq 99.9\%$ sequence identity to polypeptide sequence SEQ ID NO: 24.

(536) The polypeptide sequence of a variant of the polypeptide sequence SEQ ID NO: 24 is not identical to SEQ ID NO: 24.

(537) According to the invention, a hydantoin racemase E.sub.324V has hydantoin racemase activity, determined as described under item 4.5.14.4 and 4.5.14.5.

(538) According to the invention, a hydantoin racemase E.sub.324V preferably has hydantoin racemase activity of at least 1%, preferably of at least 10%, more preferably of at least 20%, more preferably of at least 30%, more preferably of at least 40%, more preferably of at least 50%, more preferably of at least 60%, more preferably of at least 70%, more preferably of at least 80%, more preferably of at least 90%, more preferably of at least 99%, more preferably of at least 100% the hydantoin racemase of the hydantoin racemase E.sub.324S, wherein the hydantoin racemase activity of E.sub.324V, relative to the hydantoin racemase activity E.sub.324S is determined by Assay J described under item 4.5.14.6.

(539) It is even more preferably according to the invention, that a hydantoin racemase E.sub.324V has hydantoin racemase activity in the range of 1 to 1000%, preferably in the range of 5 to 500%, more preferably in the range of 10 to 400%, more preferably in the range of 40 to 200%, more preferably in the range of 50 to 150%, more preferably in the range of 60 to 140%, more preferably in the range of 70 to 130%, more preferably in the range of 80 to 120%, more preferably in the range of 90 to 110%, more preferably 100% the hydantoin racemase activity of the hydantoin racemase E.sub.324S, wherein the hydantoin racemase activity of E.sub.324V, relative to the hydantoin racemase activity of E.sub.324S is determined by Assay J described under item 4.5.14.6.

(540) 4.5.14.7.9 Preferred Variants of SEQ ID NO: 25

(541) According to the invention, the polypeptide sequence of the hydantoin racemase E.sub.3 may also be a variant of SEQ ID NO: 25.

(542) The hydantoin racemase E.sub.3, the polypeptide sequence of which is SEQ ID NO: 25, is denoted as "E.sub.325S". Hydantoin racemases E.sub.3, the polypeptide sequence of which is selected from variants of SEQ ID NO: 25, are generally denoted as "E.sub.325V".

(543) A variant of the polypeptide sequence of SEQ ID NO: 25 is a polypeptide with sequence

identity of at least 60%, preferably $\geq 65\%$, more preferably $\geq 70\%$, more preferably $\geq 75\%$, more preferably $\geq 80\%$, more preferably $\geq 85\%$, more preferably $\geq 90\%$, more preferably $\geq 91\%$, more preferably $\geq 92\%$, more preferably $\geq 93\%$, more preferably $\geq 94\%$, more preferably $\geq 95\%$, more preferably $\geq 96\%$, more preferably $\geq 97\%$, more preferably $\geq 98\%$, more preferably $\geq 99\%$, more preferably $\geq 99.9\%$ sequence identity to polypeptide sequence SEQ ID NO: 25.

(544) The polypeptide sequence of a variant of the polypeptide sequence SEQ ID NO: 25 is not identical to SEQ ID NO: 25.

(545) According to the invention, a hydantoin racemase E.sub.325V has hydantoin racemase activity, determined as described under item 4.5.14.4 and 4.5.14.5.

(546) According to the invention, a hydantoin racemase E.sub.325V preferably has hydantoin racemase activity of at least 1%, preferably of at least 10%, more preferably of at least 20%, more preferably of at least 30%, more preferably of at least 40%, more preferably of at least 50%, more preferably of at least 60%, more preferably of at least 70%, more preferably of at least 80%, more preferably of at least 90%, more preferably of at least 99%, more preferably of at least 100% the hydantoin racemase of the hydantoin racemase E.sub.325S, wherein the hydantoin racemase activity of E.sub.325V, relative to the hydantoin racemase activity E.sub.325S is determined by Assay J described under item 4.5.14.6.

(547) It is even more preferably according to the invention, that a hydantoin racemase E.sub.325V has hydantoin racemase activity in the range of 1 to 1000%, preferably in the range of 5 to 500%, more preferably in the range of 10 to 400%, more preferably in the range of 40 to 200%, more preferably in the range of 50 to 150%, more preferably in the range of 60 to 140%, more preferably in the range of 70 to 130%, more preferably in the range of 80 to 120%, more preferably in the range of 90 to 110%, more preferably 100% the hydantoin racemase activity of the hydantoin racemase E.sub.325S, wherein the hydantoin racemase activity of E.sub.325V, relative to the hydantoin racemase activity of E.sub.325S is determined by Assay J described under item 4.5.14.6.

(548) 4.5.14.7.10 Preferred Variants of SEQ ID NO: 26

(549) According to the invention, the polypeptide sequence of the hydantoin racemase E.sub.3 may also be a variant of SEQ ID NO: 26.

(550) The hydantoin racemase E.sub.3, the polypeptide sequence of which is SEQ ID NO: 26, is denoted as “E.sub.326S”. Hydantoin racemases E.sub.3, the polypeptide sequence of which is selected from variants of SEQ ID NO: 26, are generally denoted as “E.sub.326V”.

(551) A variant of the polypeptide sequence of SEQ ID NO: 26 is a polypeptide with sequence identity of at least 60%, preferably $\geq 65\%$, more preferably $\geq 70\%$, more preferably $\geq 75\%$, more preferably $\geq 80\%$, more preferably $\geq 85\%$, more preferably $\geq 90\%$, more preferably $\geq 91\%$, more preferably $\geq 92\%$, more preferably $\geq 93\%$, more preferably $\geq 94\%$, more preferably $\geq 95\%$, more preferably $\geq 96\%$, more preferably $\geq 97\%$, more preferably $\geq 98\%$, more preferably $\geq 99\%$, more preferably $\geq 99.9\%$ sequence identity to polypeptide sequence SEQ ID NO: 26.

(552) The polypeptide sequence of a variant of the polypeptide sequence SEQ ID NO: 26 is not identical to SEQ ID NO: 26.

(553) According to the invention, a hydantoin racemase E.sub.326V has hydantoin racemase activity, determined as described under item 4.5.14.4 and 4.5.14.5.

(554) According to the invention, a hydantoin racemase E.sub.326V preferably has hydantoin racemase activity of at least 1%, preferably of at least 10%, more preferably of at least 20%, more preferably of at least 30%, more preferably of at least 40%, more preferably of at least 50%, more preferably of at least 60%, more preferably of at least 70%, more preferably of at least 80%, more preferably of at least 90%, more preferably of at least 99%, more preferably of at least 100% the hydantoin racemase of the hydantoin racemase E.sub.326S, wherein the hydantoin racemase activity of E.sub.326V, relative to the hydantoin racemase activity E.sub.326S is determined by Assay J described under item 4.5.14.6.

(555) It is even more preferably according to the invention, that a hydantoin racemase E.sub.326V

has hydantoin racemase activity in the range of 1 to 1000%, preferably in the range of 5 to 500%, more preferably in the range of 10 to 400%, more preferably in the range of 40 to 200%, more preferably in the range of 50 to 150%, more preferably in the range of 60 to 140%, more preferably in the range of 70 to 130%, more preferably in the range of 80 to 120%, more preferably in the range of 90 to 110%, more preferably 100% the hydantoin racemase activity of the hydantoin racemase E.sub.326S, wherein the hydantoin racemase activity of E.sub.326V, relative to the hydantoin racemase activity of E.sub.326S is determined by Assay J described under item 4.5.14.6.

4.5.15 Preferred Method Conditions in Step (a)

(556) In this preferred embodiment, in which step (a) is catalyzed by a hydantoin racemase E.sub.3, the reaction in step (a) may be carried out under conditions known to the skilled person.

(557) The reaction medium is preferably aqueous, more preferably an aqueous buffer.

(558) Exemplary buffers commonly used in biotransformation reactions and advantageously used herein include Tris, phosphate, or any of Good's buffers, such as 2-(N-morpholino) ethanesulfonic acid ("MES"), N-(2-acetamido)iminodiacetic acid ("ADA"), piperazine-N,N'-bis(2-ethanesulfonic acid) ("PIPES"), N-(2-acetamido)-2-aminoethanesulfonic acid ("ACES"), P-hydroxy-4-morpholinepropanesulfonic acid ("MOPSO"), choline chloride, 3-(N-morpholino) propanesulfonic acid ("MOPS"), N, N-Bis(2-hydroxyethyl)-2-aminoethanesulfonic acid ("BES"), 2-[[1,3-dihydroxy-2-(hydroxymethyl) propan-2-yl]amino]ethanesulfonic acid ("TES"), 4-(2-hydroxyethyl)-1-piperazineethanesulfonic acid ("HEPES"), 3-(Bis(2-hydroxyethyl)amino)-2-hydroxypropane-1-sulfonic acid ("DIPSO"), acetamidoglycine, 3-(N-Tris(hydroxymethyl)methylamino(-2-hydroxypropane) sulfonic acid ("TAPSO"), piperazine-N,N'-bis(2-hydroxypropanesulfonic acid) ("POPSO"), 4-(2-Hydroxyethyl) piperazine-1-(2-hydroxypropanesulfonic acid) ("HEPPSO"), 3-[4-(2-Hydroxyethyl)-1-piperazinyl]propanesulfonic acid ("HEPPS"), tricine, glycine, bicine, or 3-[[1,3-dihydroxy-2-(hydroxymethyl) propan-2-yl]amino]propane-1-sulfonic acid ("TAPS").

(559) In some embodiments, ammonium can act as a buffer. One or more organic solvents can also be added to the reaction.

(560) The buffer preferably contains metal salts, more preferably metal salts such as halogenides of monovalent or bivalent metals (e.g. CoCl.sub.2, MnCl.sub.2).

(561) The concentration of these metal salts in the reaction medium is preferably in the range from 1 μ M to 1 M, more preferably 1 mM to 100 mM, even more preferably 1 to 10 mM.

(562) Preferably, step (a) of the method according to the first aspect of the invention is carried out in a phosphate buffer.

(563) The pH of the reaction medium in step (a) of the method according to the first aspect of the invention is preferably in the range of from 2 to 10, more preferably in the range of from 5 to 8, more preferably 7.2 to 7.5, most preferably 7.5.

(564) Preferably, step (a) of the method according to the first aspect of the invention is carried out at a temperature in the range of from 20° C. to 70° C., more preferably in the range of from 30° C. to 55° C., most preferably 50° C.

(565) In this preferred embodiment, in which step (a) is catalyzed by a hydantoin racemase E.sub.3, the preferred reaction conditions in step (a) are the same as described for steps (b) and (c), confer items 4.5.7 and 4.5.13, respectively. It is even more preferred to carry out step (a) concomitantly with steps (b) and (c).

(566) Preferably, the total concentration of all hydantoin racemases E.sub.3 in the reaction solution in step (a) is in the range of from 1 μ M to 10 mM, preferably 10 μ M to 1 mM, more preferably 0.1 mM to 0.5 mM, most preferably 0.4 mM.

(567) In alternative preferred embodiments, the total concentration of all hydantoin racemases E.sub.3 in the reaction solution in step (a) is in the range of from 1 μ g/l to 10 g/l, preferably 0.1 mg/l to 5 g/l, more preferably 1 mg/l to 1 g/l, more preferably 5 mg/l to 500 mg/l.

(568) Preferably, step (a) is carried out in the same reaction medium in which steps (c) and (b) are

carried out. The advantage is that this allows for a one-pot synthesis in which all the steps (a), (b), and (c) are carried out.

(569) Preferably, the initial concentration of all the compounds according to formula D-(III) in the reaction medium in step (a) is in the range of from 1 μ M to 1 M, preferably of from 10 μ M to 0.5 M, more preferably of from 0.1 mM to 0.1 M, more preferably of from 1 mM to 10 mM, most preferably 1.25 mM.

(570) If compounds according to formula L-(III) are present in the reaction medium in step (a), the initial concentration of all the compounds according to formula L-(III) in the reaction medium is preferably from 1 to 100 times the concentration of all the compounds according to formula D-(III), more preferably 1 to 10 times the concentration, even more preferably 1 to 2 times even more preferably the same as the concentration of all the compounds according to formula D-(III).

(571) "Initial concentration of all the compounds according to formula L-(III)/D-(III)" refers to the concentration of the respective compound L-(III) or D-(III), respectively, in the reaction medium when the respective compounds are employed in step (a).

4.6 Second Aspect: Method for Production of an L-Glufosinate P-Ester

(572) In a second aspect, the present invention relates to a further method for the production of an L-glufosinate P-ester according to formula L-(I):

(573) ##STR00041##

(574) from a mixture M.sub.IIIA comprising at least one compound L-(III) and at least one compound D-(III):

(575) ##STR00042##

(576) wherein R^{sup.1L} is a radical selected from the above-mentioned formulae (VIII), (IX), (X), preferably selected from the above-mentioned formulae (IX), (X), more preferably R^{sup.1L} has formula (X).

(577) When R^{sup.1L} is a radical of the formula (VIII), the radical R^{sup.1L} may be selected from the above-mentioned formulae L-(VIII) and D-(VIII), preferably L-(VIII).

(578) When R^{sup.1L} is a radical of the formula (IX), the radical R^{sup.1L} may be selected from the above-mentioned formulae L-(IX) and D-(IX), preferably L-(IX).

(579) When R^{sup.1L} is a radical of the formula (X), the radical R^{sup.1L} may be selected from the above-mentioned formulae L-(X) and D-(X), preferably L-(X).

(580) R^{sup.3L} is a radical selected from the above-mentioned formulae (VIII), (IX), (X), preferably selected from the formulae (VIII), (IX), more preferably R^{sup.3L} has formula (VIII).

(581) When R^{sup.3L} is a radical of the formula (VIII), the radical R^{sup.3L} may be selected from the above-mentioned formulae L-(VIII) and D-(VIII), preferably L-(VIII).

(582) When R^{sup.3L} is a radical of the formula (IX), the radical R^{sup.3L} may be selected from the above-mentioned formulae L-(IX) and D-(IX), preferably L-(IX).

(583) When R^{sup.3L} is a radical of the formula (X), the radical R^{sup.3L} may be selected from the above-mentioned formulae L-(X) and D-(X), preferably L-(X).

(584) R^{sup.3D} is a radical selected from the above-mentioned formulae (VIII), (IX), (X), preferably selected from the formulae (VIII), (IX), more preferably R^{sup.3D} has formula (VIII).

(585) When R^{sup.3D} is a radical of the formula (VIII), the radical R^{sup.3D} may be selected from the above-mentioned formulae L-(VIII) and D-(VIII), preferably L-(VIII).

(586) When R^{sup.3D} is a radical of the formula (IX), the radical R^{sup.3D} may be selected from the above-mentioned formulae L-(IX) and D-(IX), preferably L-(IX).

(587) When R^{sup.3D} is a radical of the formula (X), the radical R^{sup.3D} may be selected from the above-mentioned formulae L-(X) and D-(X), preferably L-(X).

(588) In this method according to the second aspect of the invention, a mixture M.sub.IIIA comprising at least one compound L-(III) and at least one compound D-(III), wherein preferably R^{sup.3L}=R^{sup.3D}, is provided, and subjected to at least two enzymatic reactions (ii) and (iii), giving a composition M.sub.I comprising at least one compound L-(I) and optionally D-(I).

Therefore, the reaction according to the preferred second aspect of the invention is enzymatic.

(589) The method for the production of an L-glufosinate P-ester according to formula L-(I), according to the second aspect of the invention is L-enantioselective, in particular L-enantiospecific, because M.sub.I either comprises compounds according to formula L-(I), but not compounds according to formula D-(I) (in which case it is "L-enantiospecific") or, when M.sub.I comprises compounds according to formulae L-(I) and D-(I), the molar ratio of all compounds according to formula L-(I) in M.sub.I to all compounds according to formula D-(I) in M.sub.I is greater than the molar ratio of all compounds according to formula L-(III) in M.sub.IIIA to all compounds of formula D-(III) in M.sub.IIIA (in which case it is "L-enantioselective, but not L-enantiospecific").

(590) In a particular embodiment of the method according to the second aspect of the invention, M.sub.I comprises at least one compound according to formula L-(I) and at least one compound according to formula D-(I) and the molar ratio of all compounds according to formula L-(I) in M.sub.I to all compounds according to formula D-(I) in M.sub.I is greater than the molar ratio of all compounds according to formula L-(III) in M.sub.IIIA to all compounds of formula D-(III) in M.sub.IIIA (in which case the method according to the second aspect of the invention is "L-enantioselective, but not L-enantiospecific").

(591) In a preferred embodiment of the method according to the second aspect of the invention, a mixture M.sub.IIIA comprising at least one compound L-(III) and at least one compound D-(III), wherein $R_{sup.3L} = R_{sup.3D}$, is provided in step (a).

(592) In the method according to the second aspect of the invention, it is possible to enrich the amount of compounds according to formula L-(I) over compounds according to formula D-(I) in the product composition M.sub.I, which means that the molar ratio of all compounds according to formula L-(I) in M.sub.I to all compounds according to formula D-(I) in M.sub.I is greater than the molar ratio of all compounds according to formula L-(III) in M.sub.IIIA to all compounds of formula D-(III) in M.sub.IIIA provided in step (i-A).

(593) In particular, the molar amount of compounds of formula L-(I) per compound of formula D-(I) in M.sub.I is greater than the molar amount of compounds of formula L-(III) per compound of formula D-(III) in M.sub.IIIA, in particular by a factor of XL, wherein XL is at least 1.1, preferably at least 1.5, more preferably at least 2, more preferably at least 5, even more preferably at least 9, even more preferably at least 99. In another preferred embodiment XL is in the range of 1.1 to 99, preferably in the range of 1.5 to 9, even more preferably in the range of 2 to 5.

4.6.1 Step (i-A)

(594) In a first step of the method according to the second aspect of the invention, a mixture M.sub.IIIA comprising at least one compound L-(III) and at least one compound D-(III) is provided.

(595) In a preferred embodiment, the molar ratio of all compounds according to formula L-(III) in the mixture M.sub.IIIA to all compounds according to formula D-(III) in the mixture M_{il} is essentially 1:1.

(596) Alternatively, compounds according to formula D-(III) are comprised in an excess to compounds according to formula L-(III) in mixture M.sub.IIIA, meaning that, while L-(III) is present in mixture M.sub.IIIA, the molar ratio of all compounds according to formula L-(III) to all compounds according to formula D-(III) in mixture M.sub.IIIA is <1:1, preferably <0.9:1, more preferably <0.75:1, more preferably <0.5:1, more preferably <0.2:1, more preferably <0.1:1, more preferably <0.01:1.

(597) In other preferred embodiments, the molar ratio of all compounds according to formula L-(III) in the mixture M.sub.IIIA to all compounds according to formula D-(III) in the mixture M.sub.IIIA is in the range of from 3:2 to 1:99, more preferably in the range of from 1.01:1 to 1:99, more preferably in the range of from 1:1 to 1:99, more preferably in the range of from 1:1.01 to 1:99, more preferably in the range of from 1:1.01 to 1:9, more preferably in the range of from

1:1.01 to 1:8, more preferably in the range of from 1:1.01 to 1:3.

(598) Alternatively, D-(III) is comprised in an excess to L-(III) in mixture M.sub.IIIA, meaning that, while at least one compound according to formula L-(III) is in the mixture M.sub.IIIA, the molar ratio of all compounds according to formula L-(III) is in the mixture M.sub.IIIA to all compounds according to formula D-(III) in the mixture M.sub.IIIA is <1:1, preferably <0.9:1, more preferably <0.75:1, more preferably <0.5:1, more preferably <0.2:1, more preferably <0.1:1, more preferably <0.01:1.

(599) A mixture of these compounds can be obtained by the skilled person, for example by organic synthesis as set forth in DE 31 42 036 A1 and by Slomka et al.

(600) E. Ware, Chem. Rev. 1950, 46, 403-470 and C. Avendaño & J. C. Menendez, "Hydantoin and its derivatives" in Kirk-Othmer Encyclopedia of Chemical Technology 2000 give a general overview over hydantoin chemistry.

(601) Preferably, the mixture M.sub.IIIA is provided in step (i-A) in a reaction medium.

(602) Namely, the reaction medium in step (i-A) is preferably aqueous, more preferably an aqueous buffer.

(603) Exemplary buffers commonly used in biotransformation reactions and advantageously used in step (i-A) include Tris, phosphate, or any of Good's buffers, such as 2-(N-morpholino)ethanesulfonic acid ("MES"), N-(2-acetamido)iminodiacetic acid ("ADA"), piperazine-N,N'-bis(2-ethanesulfonic acid) ("PIPES"), N-(2-acetamido)-2-aminoethanesulfonic acid ("ACES"), P-hydroxy-4-morpholinepropanesulfonic acid ("MOPSO"), cholamine chloride, 3-(N-morpholino)propanesulfonic acid ("MOPS"), N,N-Bis(2-hydroxyethyl)-2-aminoethanesulfonic acid ("BES"), 2-[[1,3-dihydroxy-2-(hydroxymethyl)propan-2-yl]amino]ethanesulfonic acid ("TES"), 4-(2-hydroxyethyl)-1-piperazineethanesulfonic acid ("HEPES"), 3-(Bis(2-hydroxyethyl)amino)-2-hydroxypropane-1-sulfonic acid ("DIPSO"), acetamidoglycine, 3-(N-Tris(hydroxymethyl)methylamino(-2-hydroxypropane) sulfonic acid ("TAPSO"), piperazine-N,N'-bis(2-hydroxypropanesulfonic acid) ("POPSO"), 4-(2-Hydroxyethyl) piperazine-1-(2-hydroxypropanesulfonic acid) ("HEPPSO"), 3-[4-(2-Hydroxyethyl)-1-piperazinyl]propanesulfonic acid ("HEPPS"), tricine, glycineamide, bicine, or 3-[[1,3-dihydroxy-2-(hydroxymethyl)propan-2-yl]amino]propane-1-sulfonic acid ("TAPS").

(604) In some embodiments, ammonium can act as a buffer. One or more organic solvents can also be added to the reaction.

(605) The buffer preferably contains metal salts, more preferably metal salts such as halogenides of monovalent or bivalent metals (e.g. CoCl.sub.2, MnCl.sub.2).

(606) The concentration of these metal salts in the reaction medium in step (i-A) is preferably in the range from 1 μ M to 1 M, more preferably 1 mM to 100 mM, even more preferably 1 to 10 mM.

(607) Preferably, step (i-A) of the method according to the second aspect of the invention is carried out in a phosphate buffer.

(608) The pH of the reaction medium in step (i-A) of the method according to the first aspect of the invention is preferably in the range of from 2 to 10, more preferably in the range of from 5 to 8, more preferably 7.2 to 7.5, most preferably 7.5.

(609) More preferably, the concentration of all the compounds according to formula D-(III) in the reaction medium provided in step (i-A) is in the range of from 1 μ M to 1 M, preferably of from 10 μ M to 0.5 M, more preferably of from 0.1 mM to 0.1 M, more preferably of from 1 mM to 10 mM, most preferably 1.25 mM.

(610) Concerning the compounds according to formula L-(III) that are present in the reaction medium preferably provided in step (i-A), the concentration of all the compounds according to formula L-(III) in the reaction medium is preferably from 1 to a 100 times the concentration of all the compounds according to formula D-(III), more preferably 1 to 10 times the concentration, even more preferably 1 to 2 times even more preferably the same as the concentration of all the compounds according to formula D-(III).

4.6.2 Optional Step (i-B)

(611) In an optional step (i-B), at least a part of the compounds D-(III) comprised by the mixture M.sub.IIIA are reacted into compounds according to the formula L-(III), giving a composition M.sub.IIIB comprising compounds according to formula L-(III) and optionally compounds according to formula D-(III). It goes without saying that the molar ratio of all compounds according to formula L-(III) to all compounds in formula D-(III) in mixture M.sub.IIIB is greater than the molar ratio of all compounds according to formula L-(III) to all compounds in formula D-(III) in mixture M.sub.IIIA.

(612) In case step (i-B) is carried out, it is preferably carried out as described for step (a) under items 4.5.14 and 4.5.15.

(613) In particular, the reaction according to step (i-B) is catalyzed by a hydantoin racemase E.sub.3 as described with respect to step (a).

(614) Preferably, the hydantoin racemase E.sub.3 used in step (i-B) is categorized in the EC class 5.1.99.5.

(615) In another embodiment, the polypeptide sequence of the hydantoin racemase E.sub.3 used in step (i-B) is selected from the group consisting of SEQ ID NO: 17 and variants thereof, SEQ ID NO: 18 and variants thereof, SEQ ID NO: 19 and variants thereof, SEQ ID NO: 20 and variants thereof, SEQ ID NO: 21 and variants thereof, SEQ ID NO: 22 and variants thereof, SEQ ID NO: 23 and variants thereof, SEQ ID NO: 24 and variants thereof, SEQ ID NO: 25 and variants thereof, SEQ ID NO: 26 and variants thereof, most preferably selected from SEQ ID NO: 17 and variants thereof.

(616) Preferably, the hydantoin racemase E.sub.3 used in step (i-B) is selected from the group consisting of E.sub.317S and E.sub.317V (both described under item 4.5.14.7.1), E.sub.318S and E.sub.318V (both described under item 4.5.14.7.2), E.sub.319S and E.sub.319V (both described under item 4.5.14.7.3), E.sub.320S and E.sub.320V (both described under item 4.5.14.7.4), E.sub.321S and E.sub.321V (both described under item 4.5.14.7.5), E.sub.322S and E.sub.322V (both described under item 4.5.14.7.6), E.sub.323S and E.sub.323V (both described under item 4.5.14.7.7), E.sub.324S and E.sub.324V (both described under item 4.5.14.7.8), E.sub.325S and E.sub.325V (both described under item 4.5.14.7.9), E.sub.326S and E.sub.326V (both described under item 4.5.14.7.10), more preferably from E.sub.317S and E.sub.317V, even more preferably from E.sub.317S.

(617) The reaction conditions in step (i-B) are preferably as described with respect to step (a) under item 4.5.15.

(618) Namely, the reaction medium in step (i-B) is preferably aqueous, more preferably an aqueous buffer.

(619) Exemplary buffers commonly used in biotransformation reactions and advantageously used in step (i-B) include Tris, phosphate, or any of Good's buffers, such as 2-(N-morpholino)ethanesulfonic acid ("MES"), N-(2-acetamido)iminodiacetic acid ("ADA"), piperazine-N,N'-bis(2-ethanesulfonic acid) ("PIPES"), N-(2-acetamido)-2-aminoethanesulfonic acid ("ACES"), P-hydroxy-4-morpholinepropanesulfonic acid ("MOPSO"), choline chloride, 3-(N-morpholino)propanesulfonic acid ("MOPS"), N,N-Bis(2-hydroxyethyl)-2-aminoethanesulfonic acid ("BES"), 2-[[1,3-dihydroxy-2-(hydroxymethyl)propan-2-yl]amino]ethanesulfonic acid ("TES"), 4-(2-hydroxyethyl)-1-piperazineethanesulfonic acid ("HEPES"), 3-(Bis(2-hydroxyethyl)amino)-2-hydroxypropane-1-sulfonic acid ("DIPSO"), acetamidoglycine, 3-(N-Tris(hydroxymethyl)methylamino(-2-hydroxypropane) sulfonic acid ("TAPSO"), piperazine-N,N'-bis(2-hydroxypropanesulfonic acid) ("POPSO"), 4-(2-Hydroxyethyl) piperazine-1-(2-hydroxypropanesulfonic acid) ("HEPPSO"), 3-[4-(2-Hydroxyethyl)-1-piperazinyl]propanesulfonic acid ("HEPPS"), tricine, glycineamide, bicine, or 3-[[1,3-dihydroxy-2-(hydroxymethyl)propan-2-yl]amino]propane-1-sulfonic acid ("TAPS").

(620) In some embodiments, ammonium can act as a buffer. One or more organic solvents can also

be added to the reaction.

(621) The buffer preferably contains metal salts, more preferably metal salts such as halogenides of monovalent or bivalent metals (e.g. CoCl.sub.2, MnCl.sub.2).

(622) The concentration of these metal salts in the reaction medium in step (i-B) is preferably in the range from 1 μ M to 1 M, more preferably 1 mM to 100 mM, even more preferably 1 to 10 mM.

(623) Preferably, step (i-B) of the method according to the second aspect of the invention is carried out in a phosphate buffer.

(624) The pH of the reaction medium in step (i-B) of the method according to the first aspect of the invention is preferably in the range of from 2 to 10, more preferably in the range of from 5 to 8, more preferably 7.2 to 7.5, most preferably 7.5.

(625) Preferably, step (i-B) of the method according to the first aspect of the invention is carried out at a temperature in the range of from 20° C. to 70° C., more preferably in the range of from 30° C. to 55° C., most preferably 50° C.

(626) Preferably, the total concentration of all hydantoin racemases E.sub.3 in the reaction solution in step (i-B) is in the range of from 1 μ M to 10 mM, preferably 10 μ M to 1 mM, more preferably 0.1 mM to 0.5 mM, most preferably 0.4 mM.

(627) In alternative preferred embodiments, the total concentration of all hydantoin racemases E.sub.3 in the reaction solution in step (i-B) is in the range of from 1 μ g/l to 10 g/l, preferably 0.1 mg/l to 5 g/l, more preferably 1 mg/l to 1 g/l, more preferably 5 mg/l to 500 mg/l.

(628) Preferably, step (i-B) is carried out in the same reaction medium in which steps (ii) and (iii) are carried out. The advantage is that this allows for a one-pot synthesis in which all the steps (i-B), (ii), and (iii) are carried out.

(629) At the end of optional step (i-B), a composition M.sub.IIIB comprising compounds according to formula L-(III) and optionally compounds according to formula D-(III) is obtained, preferably a composition M.sub.IIIB comprising compounds according to formula L-(III) and compounds according to formula D-(III) is obtained.

4.6.3 Step (ii)

(630) In step (ii) of the method according to the second aspect of the invention, the mixture M.sub.IIIA or, in case step (i-B) is carried out, composition M.sub.IIIB, is subjected to step (b) as described above under item 4.5.8-4.5.13.

(631) In other words: in those embodiments, in which step (i-B) is not carried out, M.sub.IIIA is subjected to step (b) as described above under item 4.5.8-4.5.13.

(632) In case step (i-B) is carried out, M.sub.IIIB obtained after step (i-B) is subjected to step (b) as described above under item 4.5.8-4.5.13.

(633) The reaction according to step (ii) is catalyzed by a hydantoinase E.sub.2 as described with respect to step (b), preferably by an L-hydantoinase E.sub.2 as described with respect to step (b).

(634) More preferably, the hydantoinase E.sub.2, even more preferably the L-hydantoinase E.sub.2 used in step (ii) is categorized in the EC class 3.5.2.2.

(635) In another preferred embodiment, the polypeptide sequence of the hydantoinase E.sub.2, even more preferably the L-hydantoinase E.sub.2 used in step (ii) is selected from the group consisting of SEQ ID NO: 10 and variants thereof, SEQ ID NO: 11 and variants thereof, SEQ ID NO: 12 and variants thereof, SEQ ID NO: 13 and variants thereof, SEQ ID NO: 14 and variants thereof, SEQ ID NO: 15 and variants thereof, SEQ ID NO: 16 and variants thereof, even more preferably from SEQ ID NO: 10 and variants thereof.

(636) Preferably, the hydantoinase E.sub.2, even more preferably the L-hydantoinase E.sub.2 used in step (ii) is selected from the group consisting of E.sub.210S and E.sub.210V (both described under item 4.5.12.1), E.sub.211S and E.sub.211V (both described under item 4.5.12.2), E.sub.212S and E.sub.212V (both described under item 4.5.12.3), E.sub.213S and E.sub.213V (both described under item 4.5.12.4), E.sub.214S and E.sub.214V (both described under item 4.5.12.5), E.sub.215S and E.sub.215V (both described under item 4.5.12.6), E.sub.216S and E.sub.216V (both described

under item 4.5.12.7), more preferably from E.sub.210S and E.sub.210V, even more preferably from E.sub.210S.

(637) The reaction conditions in step (ii) are preferably as described with respect to step (b) under item 4.5.13.

(638) Namely, the reaction medium in step (ii) is preferably aqueous, more preferably an aqueous buffer.

(639) Exemplary buffers commonly used in biotransformation reactions and advantageously used in step (ii) include Tris, phosphate, or any of Good's buffers, such as 2-(N-morpholino)ethanesulfonic acid ("MES"), N-(2-acetamido)iminodiacetic acid ("ADA"), piperazine-N,N'-bis(2-ethanesulfonic acid) ("PIPES"), N-(2-acetamido)-2-aminoethanesulfonic acid ("ACES"), P-hydroxy-4-morpholinepropanesulfonic acid ("MOPSO"), choline chloride, 3-(N-morpholino)propanesulfonic acid ("MOPS"), N,N-Bis(2-hydroxyethyl)-2-aminoethanesulfonic acid ("BES"), 2-[[1,3-dihydroxy-2-(hydroxymethyl)propan-2-yl]amino]ethanesulfonic acid ("TES"), 4-(2-hydroxyethyl)-1-piperazineethanesulfonic acid ("HEPES"), 3-(Bis(2-hydroxyethyl)amino)-2-hydroxypropane-1-sulfonic acid ("DIPSO"), acetamidoglycine, 3-(N-Tris(hydroxymethyl)methylamino(-2-hydroxypropane) sulfonic acid ("TAPSO"), piperazine-N, N-bis(2-hydroxypropanesulfonic acid) ("POPSO"), 4-(2-Hydroxyethyl) piperazine-1-(2-hydroxypropanesulfonic acid) ("HEPPSO"), 3-[4-(2-Hydroxyethyl)-1-piperazinyl]propanesulfonic acid ("HEPPS"), tricine, glycineamide, bicine, or 3-[[1,3-dihydroxy-2-(hydroxymethyl)propan-2-yl]amino]propane-1-sulfonic acid ("TAPS").

(640) In some embodiments, ammonium can act as a buffer. One or more organic solvents can also be added to the reaction.

(641) The buffer preferably contains metal salts, more preferably metal salts such as halogenides of monovalent or bivalent metals (e.g. CoCl.sub.2, MnCl.sub.2).

(642) The concentration of these metal salts in the reaction medium is preferably in the range from 1 μ M to 1 M, more preferably 1 mM to 100 mM, even more preferably 1 to 10 mM.

(643) Preferably, step (ii) of the method according to the second aspect of the invention is carried out in a phosphate buffer.

(644) The pH of the reaction medium in step (ii) of the method according to the first aspect of the invention is preferably in the range of from 2 to 10, more preferably in the range of from 5 to 8, more preferably 7.2 to 7.5, most preferably 7.5.

(645) Preferably, step (ii) of the method according to the first aspect of the invention is carried out at a temperature in the range of from 20° C. to 70° C., more preferably in the range of from 30° C. to 55° C., most preferably 50° C.

(646) Preferably, the total concentration of all hydantoinases E.sub.2 in the reaction solution in step (ii) is in the range of from 1 μ M to 10 mM, preferably 10 μ M to 1 mM, more preferably 0.1 mM to 0.5 mM, most preferably 0.4 mM.

(647) In alternative preferred embodiments, the total concentration of all hydantoinases E.sub.2 in the reaction solution in step (ii) is in the range of from 1 μ g/l to 10 g/l, preferably 0.1 mg/l to 5 g/l, more preferably 1 mg/l to 1 g/l, more preferably 5 mg/l to 500 mg/l.

(648) Preferably, step (ii) is carried out in the same reaction medium in which step (iii) is carried out.

(649) At the end of step (ii), a composition M.sub.II comprising at least one compound according to formula L-(II) and optionally at least one compound according to formula D-(II), wherein L-(II) and D-(II) have the above-mentioned formulae, is obtained.

4.6.4 Step (iii)

(650) In step (iii) of the method according to the second aspect of the invention, M.sub.II obtained in step (ii) is subjected to step (c) as described above under items 4.5.1 to 4.5.7 wherein the reaction according to step (iii) is catalyzed by an L-carbamoylase E.sub.1.

(651) This brings about that compounds according to formula L-(I) are enriched over compounds

according to formulas D-(I) in M.sub.I, i.e., compared to the molar ratio of compounds according to formula L-(I) to compounds according to formulas D-(I) in M.sub.IIIA.

(652) More preferably, the L-carbamoylase E.sub.1 used in step (iii) is categorized in the EC class 3.5.1.87. In another preferred embodiment, the polypeptide sequence of the L-carbamoylase E.sub.1 used in step (iii) is selected from the group consisting of SEQ ID NO: 1 and variants thereof, SEQ ID NO: 2 and variants thereof, SEQ ID NO: 3 and variants thereof, SEQ ID NO: 4 and variants thereof, SEQ ID NO: 5 and variants thereof, SEQ ID NO: 6 and variants thereof, SEQ ID NO: 7 and variants thereof, SEQ ID NO: 8 and variants thereof, SEQ ID NO: 9 and variants thereof.

(653) Preferably, the L-carbamoylase E.sub.1 used in step (iii) is selected from the group consisting of E.sub.101S and E.sub.101V (both described under item 4.5.6.1), E.sub.102S and E.sub.102V (both described under item 4.5.6.2), E.sub.103S and E.sub.103V (both described under item 4.5.6.3), E.sub.104S and E.sub.104V (both described under item 4.5.6.4), E.sub.105S and E.sub.105V (both described under item 4.5.6.5), E.sub.106S and E.sub.106V (both described under item 4.5.6.6), E.sub.107S and E.sub.107V (both described under item 4.5.6.7), E.sub.108S and E.sub.108V (both described under item 4.5.6.8), E.sub.109S and E.sub.109V (both described under item 4.5.6.9), more preferably from E.sub.101S and E.sub.101V, even more preferably from E.sub.101S.

(654) The reaction conditions in step (iii) are preferably as described with respect to step (c) under item 4.5.7.

(655) Namely, the reaction medium in step (iii) is preferably aqueous, more preferably an aqueous buffer.

(656) Exemplary buffers commonly used in biotransformation reactions and advantageously used in step (iii) include Tris, phosphate, or any of Good's buffers, such as 2-(N-morpholino)ethanesulfonic acid ("MES"), N-(2-acetamido)iminodiacetic acid ("ADA"), piperazine-N,N'-bis(2-ethanesulfonic acid) ("PIPES"), N-(2-acetamido)-2-aminoethanesulfonic acid ("ACES"), P-hydroxy-4-morpholinepropanesulfonic acid ("MOPSO"), choline chloride, 3-(N-morpholino)propanesulfonic acid ("MOPS"), N,N-Bis(2-hydroxyethyl)-2-aminoethanesulfonic acid ("BES"), 2-[[1,3-dihydroxy-2-(hydroxymethyl)propan-2-yl]amino]ethanesulfonic acid ("TES"), 4-(2-hydroxyethyl)-1-piperazineethanesulfonic acid ("HEPES"), 3-(Bis(2-hydroxyethyl)amino)-2-hydroxypropane-1-sulfonic acid ("DIPSO"), acetamidoglycine, 3-(N-Tris(hydroxymethyl)methylamino(-2-hydroxypropane) sulfonic acid ("TAPSO"), piperazine-N,N'-bis(2-hydroxypropanesulfonic acid) ("POPSO"), 4-(2-Hydroxyethyl) piperazine-1-(2-hydroxypropanesulfonic acid) ("HEPPSO"), 3-[4-(2-Hydroxyethyl)-1-piperazinyl]propanesulfonic acid ("HEPPS"), tricine, glycine, or 3-[[1,3-dihydroxy-2-(hydroxymethyl)propan-2-yl]amino]propane-1-sulfonic acid ("TAPS").

(657) In some embodiments, ammonium can act as a buffer. One or more organic solvents can also be added to the reaction.

(658) The buffer preferably contains metal salts, more preferably metal salts such as halogenides of monovalent or bivalent metals (e.g. CoCl.sub.2, MnCl.sub.2).

(659) The concentration of these metal salts in the reaction medium is preferably in the range from 1 μ M to 1 M, more preferably 1 mM to 100 mM, even more preferably 1 to 10 mM.

(660) Preferably, step (iii) of the method according to the second aspect of the invention is carried out in a phosphate buffer.

(661) The pH of the reaction medium in step (iii) of the method according to the first aspect of the invention is preferably in the range of from 2 to 10, more preferably in the range of from 5 to 8, more preferably 7.2 to 7.5, most preferably 7.5.

(662) Preferably, step (iii) of the method according to the first aspect of the invention is carried out at a temperature in the range of from 20° C. to 70° C., more preferably in the range of from 30° C. to 55° C., most preferably 50° C.

(663) Preferably, the total concentration of all carbamoylases E.sub.1 in the reaction solution in step (iii) is in the range of from 1 μ M to 10 mM, preferably 10 μ M to 1 mM, more preferably 0.1 mM to 0.5 mM, most preferably 0.4 mM.

(664) In alternative preferred embodiments, the total concentration of all carbamoylases E.sub.1 in the reaction solution in step (iii) is in the range of from 1 μ g/l to 10 g/l, preferably 0.1 mg/l to 5 g/l, more preferably 1 mg/l to 1 g/l, more preferably 5 mg/l to 500 mg/l.

(665) At the end of step (i), a mixture M.sub.I comprising compounds according to formula L-(I) and optionally compounds according to formula D-(I) is obtained, preferably a mixture M.sub.I comprising compounds according to formula L-(I) and compounds according to formula D-(I) is obtained.

4.7 Saponification

(666) The L-glufosinate P ester according to formula L-(I) produced with the method according to the first or second aspect can then be saponified to produce LGA, for example in an acidic aqueous medium, preferably at pH<7, even more preferably at an pH between <6, more preferably at an pH<3, even more preferably at an pH of <1. These saponification conditions are known to the skilled person and described e.g by H. J. Zeiss, J. Org. Chem. 1991, 56, 1783-1788.

5. EXAMPLES

5.1 Example 1 Identification of Suitable Enzymes and Construction of Plasmids

(667) Genes of different origins encoding a hydantoinase (dihydropyrimidinase, EC 3.5.2.2), L-carbamoylase (N-carbamoyl-L-amino-acid hydrolase, EC 3.5.1.87) and hydantoin racemase (EC 5.1.99.5) were tested for their ability to react with different hydantoin substrates according to formula L-(III) and D-(III) to form the respective enantioselective L-glufosinate derivative according to formula L-(I).

5.1.1 Examined Enzymes

(668) Details of the strains and genes of the respective enzymes that were used in the examples are summarized in table 4.

(669) TABLE-US-00004 TABLE 4 SEQ ID NO: of SEQ ID NO: of Gene the poly- the poly-
Plasmid Enzyme Donor Strain name nucleotides peptides pOM22c L-carbamoylase *Arthrobacter*
hyuC SEQ ID NO: 27 SEQ ID NO: 1 sp. DSM 3747 pOM22c hydantoinase *Arthrobacter* hyuH
SEQ ID NO: 28 SEQ ID NO: 10 sp. DSM 9771 pOM21c hydantoin *Arthrobacter* hyuR SEQ ID
NO: 29 SEQ ID NO: 17 racemase sp. DSM 3745

5.1.2 Cloning of the Enzymes

(670) Cloning of the hydantoin racemase and generation of the plasmid pOM21c (FIG. 1) was carried out as described in example 1 of WO 2004/111227 A2. In particular, a polynucleotide was used which comprised the respective gene (SEQ ID NO: 29) and additional sequences for NdeI and PstI restriction sites.

(671) Cloning of hydantoinase and L-carbamoylase into the rhamnose expression vector pJOE4036 was carried out in a plasmid derivative of the rhamnose expression vector pJOE4036.

(672) Polynucleotides comprising the genes of the respective enzymes (SEQ ID NOs: 27, 28) were synthesized by GeneArt (ThermoFisher Scientific (Waltham, USA)). The polynucleotides carried additional sequences for EcoRI and HindIII restriction sites. Both enzymes were cloned into pJOE4036 using those restriction sites resulting in the plasmid pOM22c, under the control of a rhamnose promoter (FIG. 2).

5.2 Example 2: Production of Strains Positive for Hydantoinase, L-Carbamoylase and Hydantoin Racemase

(673) Chemically competent *E. coli* ET5 cells (as described in WO 2004/042047 A1) were transformed with 10 ng of the plasmid pOM22c generated according to Example 1.

(674) The generated strain which was positive for hydantoinase- and carbamoylase was rendered chemically competent and transformed with 10 ng of the plasmid pOM21c.

(675) An *E. coli* ET5 strain transformed with pOM21c or pOM22c was incubated under shaking

(250 U/min) at 30° C. for 18 hours in LB medium containing ampicillin (100 µg/l), chloramphenicol (50 µg/l), and 2 g/l rhamnose.

(676) The biomass was separated by centrifugation, resuspended in 50 mM phosphate buffer (pH 7.2) and applied in biotransformation tests in the following examples. The concentration of the biomass in the solution was 40-50 g/l. The solution was used as catalyst ("catalyst 1") in the following.

(677) The concentration of the respective polypeptide carbamoylase, hydantoinase and racemase in the obtained solution may be determined by SDS page and analysis of the respective bands via the software GelQuant® (BiochemLabSolutions).

5.3 Inventive Example 11 and Comparative Examples C1 to C4: Production of P-Alkyl Phosphinothricin Ester Using the Strains Produced in Examples 1 and 2

(678) In the following examples, different hydantoin substrates were tested to determine whether the respective polypeptide catalyzed the reaction of the respective substrate enantioselectively to give the corresponding L amino acid.

5.3.1 Comparative Example C1

(679) In comparative example C1, a racemic mixture Mc1 of 1.25 mmol L-(IV).sub.c1 and 1.25 mmol D-(IV).sub.c1, wherein L-(IV).sub.c1 and D-(IV).sub.c1 are hydantoins with the following formulae, was used as substrate:

(680) ##STR00043##

(681) The mixture Mc1 was dissolved in a stirring reactor with 25 ml water. 2.4 g of the catalyst 1 was added, followed by the addition of 50 µl CoCl.sub.2 solution. The suspension was set to pH 7.5 with 0.5 M NaOH. The total volume was replenished with water to 50 ml, so that the concentration of each enantiomer L-(IV).sub.c1 and D-(IV).sub.c1 was 0.025 mol/l. The pH was held between 7.0 and 7.5 by HCl-titration or NaOH-titration. The temperature was maintained at 37° C. by a thermostat during the reaction.

(682) The reaction was stopped after 120 hours by addition of 2 N HCl until pH 2.5 was reached. The biomass was separated by centrifugation or filtration.

(683) The enzyme reaction was monitored by ninhydrin test to determine the formation of amino acids. No formation of amino acids was detected by the ninhydrine test.

5.3.2 Comparative Example C2

(684) In comparative example C2, a racemic mixture Mc2 of 1.25 mmol L-(V).sub.c2 and 1.25 mmol D-(V).sub.c2, wherein L-(V).sub.c2 and D-(V).sub.c2 are carbamoylates with the following formulae, was used as substrate:

(685) ##STR00044##

(686) The mixture Mc2 was dissolved in a stirring reactor with 25 ml water. 2.4 g of the catalyst 1 was added, followed by the addition of 50 µl CoCl.sub.2 solution. The suspension was set to pH 7.5 with 0.5 M NaOH. The total volume was replenished with water to 50 ml, so that the concentration of each enantiomer L-(V).sub.c2 and D-(V).sub.c2 was 0.025 mol/l. The pH was held between 7.0 and 7.5 by HCl-titration or NaOH-titration. The temperature was maintained at 37° C. by a thermostat during the reaction.

(687) The reaction was stopped after 120 hours by addition of 2 N HCl until pH 2.5 was reached. The biomass was separated by centrifugation or filtration.

(688) The enzyme reaction was monitored by ninhydrin test to determine the formation of amino acids. No formation of amino acids was detected by the ninhydrine test.

5.3.3 Example C3 (not According to the Invention)

(689) In example C3, a racemic mixture Mc3 of 1.25 mmol L-(VI).sub.c3 and 1.25 mmol D-(VI).sub.c3, wherein L-(VI).sub.c3 and D-(VI).sub.c3 were hydantoins with the following formulae, was used as substrate:

(690) ##STR00045##

(691) The mixture M.sub.II was dissolved in a stirring reactor with 25 ml water. 2.4 g of the

catalyst 1 was added, followed by the addition of 50 µl CoCl.sub.2 solution. The suspension was set to pH 7.5 with 0.5 M NaOH. The total volume was replenished with water to 50 ml, so that the concentration of each enantiomer L-(VI).sub.c3 and D-(VI).sub.c3 was 0.025 mol/l. The pH was held between 7.0 and 7.5 by HCl-titration or NaOH-titration. The temperature was maintained at 37° C. by a thermostat during the reaction.

(692) The reaction was stopped after 120 hours by addition of 2 N HCl until pH 2.5 was reached. The biomass was separated by centrifugation or filtration.

(693) The enzyme reaction was monitored by ninhydrin test to determine the formation of amino acids. Formation of amino acids was detected by the ninhydrine test.

(694) In a final step the reaction mixture was saponified at 100° C. for 10 hours by adding 6 M HCl to obtain L-glufosinate and, if present, D-glufosinate. The final reaction mixture was analysed by LC-MS with a CR-I column as described under item 5.4 to determine the enantiomeric excess (“ee”) of either D- or L-glufosinate. An ee of LGA of 79% over the D-enantiomer was detected. The ee of the L-enantiomer (“ee.sub.L”) is determined by the following formula in the context of the invention, wherein m.sub.L and m.sub.D are the detected molar masses of L- and D-glufosinate, respectively:

$$(695) ee_L = \frac{(m_L - m_D) \times 100\%}{(m_L + m_D)}.$$

5.3.4 Example C4 (not According to the Invention)

(696) In example C4, a racemic mixture Mc4 of 1.25 mmol L-(VII).sub.c4 and 1.25 mmol D-(VII).sub.c4, wherein L-(VII).sub.c4 and D-(VII).sub.c4 were hydantoins with the following formulae, was used as substrate:

(697) ##STR00046##

(698) The mixture Mc4 was dissolved in a stirring reactor with 25 ml water. 2.4 g of the catalyst 1 was added, followed by the addition of 50 µl CoCl.sub.2 solution. The suspension was set to pH 7.5 with 0.5 M NaOH. The total volume was replenished with water to 50 ml, so that the concentration of each enantiomer L-(VII).sub.c4 and D-(VII).sub.c4 was 0.025 mol/l. The pH was held between 7.0 and 7.5 by HCl-titration or NaOH-titration. The temperature was maintained at 37° C. by a thermostat during the reaction.

(699) The reaction was stopped after 120 hours by addition of 2 N HCl until pH 2.5 was reached. The biomass was separated by centrifugation or filtration.

(700) The enzyme reaction was monitored by ninhydrin test to determine the formation of amino acids. Formation of amino acids was detected by the ninhydrine test.

(701) In a final step the reaction mixture was saponified at 100° C. for 10 hours by adding 6 M HCl to obtain L-glufosinate and, if present, D-glufosinate. The final reaction mixture was analysed by LC-MS with a CR-I column as described under item 5.4 to determine the enantiomeric excess (“ee”) of either D- or L-glufosinate. An ee of LGA of 77% over the D-enantiomer was detected.

5.3.5 Inventive Example 11

(702) In example 11, a mixture Mm of 0.625 mmol L-(VIII).sub.I1, 0.625 mmol L-(VIII).sub.I1', 0.625 mmol D-(VIII).sub.I1, and 0.625 mmol D-(VIII).sub.I1', wherein L-(VIII).sub.I1, L-(VIII).sub.I1', D-(VIII).sub.I1, and D-(VIII).sub.I1' were hydantoins with the following formulae, is used as substrate:

(703) ##STR00047##

(704) These compounds may be obtained from mixtures of L-homoserine/D-homoserine by a synthesis method as set forth in CN 111662325 A.

(705) The mixture Mm is dissolved in a stirring reactor with 25 ml water. 2.4 g of the catalyst 1 is added, followed by the addition of 50 µl CoCl.sub.2 solution. The suspension is set to pH 7.5 with 0.5 M NaOH. The total volume is replenished with water to 50 ml, so that the concentration of each isomer L-(VIII).sub.I1, L-(VIII).sub.I1', D-(VIII).sub.I1, and D-(VIII).sub.I1' is 0.0125 mol/l. The pH is held between 7.0 and 7.5 by HCl-titration or NaOH-titration. The temperature is maintained at 37° C. by a thermostat during the reaction.

(706) The reaction is stopped after 120 hours by addition of 2 N HCl until pH 2.5 is reached. The biomass is separated by centrifugation or filtration.

(707) The enzyme reaction is monitored by ninhydrin test to determine the formation of amino acids. Formation of amino acids may be detected by the ninhydrine test.

(708) In a final step the reaction mixture is saponified at 100° C. for 10 hours by adding 6 M HCl to obtain L-glufosinate and, if present, D-glufosinate. The final reaction mixture is analysed by LC-MS with a CR-I column as described under item 5.4 to determine the enantiomeric excess (“ee”) of either D- or L-glufosinate. An ee of LGA over the D-enantiomer may be detected (not quantified). The ee of the L-enantiomer (“ee.sub.L”) is determined by the following formula in the context of the invention, wherein m.sub.L and m.sub.D are the detected molar masses of L- and D-glufosinate, respectively:

$$(709) ee_L = \frac{(m_L - m_D) \times 100\%}{(m_L + m_D)}.$$

5.3.5 Results

(710) The results of the examples C1, C2, C3, C4 are summarized in table 5.

(711) TABLE-US-00005 TABLE 5 Example substrate Ninhydrine test ee.sub.L C1 L-(IV).sub.C1, D-(IV).sub.C1 Negative — C2 L-(V).sub.C2, D-(V).sub.C2 Negative — C3 L-(VI).sub.C3, D-(VI).sub.C3 Positive 79% C4 L-(VII).sub.C4, D-(VII).sub.C4 Positive 77%

5.3.6 Conclusion

(712) The results as summarized in table 5 show that certain hydantoin esters such as the alkylated hydantoin esters can be used as substrates for the enzymatic enantioselective synthesis of L-glufosinate (C3, C4). In contrast, substrates in which the phosphinic acid group is not protected are not accepted by the catalytic system (C1, C2). In this regard, C2 suggests that at least one reason for this is that the L-carbamoylase enzyme does not accept the respective substrate in which the phosphinic acid function is not protected by an ester group.

(713) Analogous results are obtained, when compound mixtures according to the scope of the invention, i.e. of compounds according to L-(III) and D-(III), wherein R.sup.3L and R.sup.3D are preferably identical, such as in inventive example 11, are used.

(714) Moreover, in contrast to the process for enantioselective production according to the prior art (CN 111662325 A), the ee excess could be maintained throughout the process, because saponification is carried out not at the hydantoin stage, but at the stage of the amino acid ester.

5.4 Analytical Methods

(715) L-glufosinate and D-glufosinate were detected by LC-MS (“Liquid Chromatography-Mass Spectrometry”) with a chiral column [Daicel CROWNPAK CR-I(-)] as follows. For hydantoins, a Daicel Chiralpak IA-U column may also be used.

(716) This detection method may also be used for detection and quantification of the the LGA P-(n-butyl) ester according to formula L-(I) custom character according to Assay A.sub.L (item 4.5.3.1); the D-glufosinate P-(n-butyl) ester according to formula D-(I) custom character according to Assay A.sub.D (item 4.5.3.2); the carbamoyl LGA P-(n-butyl) ester according to formula L-(II) custom character according to Assay D.sub.L (item 4.5.9.1); the carbamoyl D-glufosinate P-(n-butyl) ester according to formula D-(II) custom character according to Assay D.sub.D (item 4.5.9.2); the L-enantiomer of a n-butyl ester of hydantoin glufosinate of the formula L-(III) custom character according to Assay G (item 4.5.14.3). the D-enantiomer of a n-butyl ester of hydantoin glufosinate of the formula D-(III) custom character according to Assay G (item 4.5.14.3).

5.4.1 Acquisition Method Details

(717) CR-I(-), 3.0 mm I.D.×150 mm, 5 μm, Crownpak, Part. No. 54784 MS QQQ Mass Spectrometer 6420, Agilent ESI+ MS2 SIM, Glufosinat m/z 182.06 Unit Positive Centroid

5.4.2 Source Parameter

(718) Parameter Value (+); Gas Temp (° C.): 350; Gas Flow (l/min): 12; Nebulizer (psi): 25; Capillary (V): 3000; Sampler Module: G1329B; Injection Volume 1.00 μL; 5° C.

5.4.3 Binary Pump Module

(719) G1312B Flow: 0.200 mL/min; Channel A: H.sub.2O with trifluoroacetic acid (TFA) pH1.15; Channel B: ACN (acetonitrile); isocratic 80.0% A/20.0% B; stop time 7.00 min Column Comp. Module: G1316A Temperature 5.0° C.

Claims

1. A method for the production of an L-glufosinate P-ester according to formula L-(I), the method comprising: (c) reacting a compound according to formula L-(II) to give a compound according to formula L-(I) ##STR00048## wherein the reaction according to (c) is catalyzed by an L-carbamoylase E.sub.1 categorized in EC class 3.5.1.87, wherein R.sup.1L and R.sup.2L are radicals selected from the group consisting of formulae (VIII), (IX), and (X) ##STR00049## and wherein R.sup.1L=R.sup.2L.
2. The method according to claim 1, wherein R.sup.1L and R.sup.2L are radicals that are each independently selected from the group consisting of formulae (IX) and (X).
3. The method according to claim 1, wherein a polypeptide sequence of the L-carbamoylase E.sub.1 is selected from the group consisting of SEQ ID NO: 1, SEQ ID NO: 2, SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 9, and variant amino acid sequences having at least 90% sequence identity to one of SEQ ID NOS: 1-9.
4. The method according to claim 1, wherein the compound according to formula L-(II) is obtained by (b) in which a compound according to formula L-(III) is reacted to give a compound according to formula L-(II): ##STR00050## wherein the reaction according to (b) is catalyzed by a hydantoinase E.sub.2 categorized in EC class 3.5.2.2, and wherein R.sup.3L is a radical that is selected from the group consisting of formulae (VIII), (IX), and (X).
5. The method according to claim 4, wherein R.sup.3L is a radical of the formula (VIII).
6. The method according to claim 4, wherein a polypeptide sequence of the hydantoinase E.sub.2 is selected from the group consisting of SEQ ID NO: 10, SEQ ID NO: 11, SEQ ID NO: 12, SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 16, and variant amino acid sequences having at least 90% sequence identity to one of SEQ ID NOS: 10-16.
7. The method according to claim 4, wherein the hydantoinase E.sub.2 is an L-hydantoinase E.sub.2.
8. The method according to claim 4, wherein the compound according to formula L-(III) is obtained by (a) in which a compound according to formula D-(III) is reacted to give a compound according to formula L-(III): ##STR00051## and wherein R.sup.3D is a radical that is selected from the group consisting of formulae (VIII), (IX), and (X).
9. The method according to claim 8, wherein R.sup.3D is a radical of the formula (VIII).
10. The method according to claim 8, wherein the reaction according to (a) is catalyzed by a hydantoin racemase E.sub.3 categorized in EC class 5.1.99.5.
11. The method according to claim 10, wherein a polypeptide sequence of the hydantoin racemase E.sub.3 is selected from the group consisting of SEQ ID NO: 17, SEQ ID NO: 18, SEQ ID NO: 19, SEQ ID NO: 20, SEQ ID NO: 21, SEQ ID NO: 22, SEQ ID NO: 23, SEQ ID NO: 24, SEQ ID NO: 25, SEQ ID NO: 26, and variant amino acid sequences having at least 90% sequence identity to one of SEQ ID NOS: 18-26.
12. The method according to claim 1, wherein the compound according to formula L-(I) is saponified to give L-glufosinate.
13. A method for the production of an L-glufosinate P-ester according to formula L-(I): ##STR00052## the method comprising: (i-A) providing a mixture MIIA comprising at least one compound L-(III) and at least one compound D-(III), wherein L-(III) and D-(III) have the following formulae: ##STR00053## (i-B) (a) reacting at least a part of the compounds according to formula D-(III) comprised by the mixture M.sub.IIIA into the compound according to formula L-

(III) to form a composition M.sub.IIIB comprising compounds according to formula L-(III) and compounds according to formula D-(III), and (ii) subjecting composition M.sub.IIIB to a reaction according to (b), giving a composition Mu comprising at least one compound according to formula L-(II) and at least one compound according to formula D-(II), wherein the reaction according to (b) is catalyzed by a hydantoinase E.sub.2 categorized in EC class 3.5.2.2, wherein L-(II) and D-(II) have the following formulae: ##STR00054## and (iii) subjecting composition Mu to a reaction according to (c), wherein the reaction according to (c) is catalyzed by an L-carbamoylase E.sub.1, giving a composition M.sub.I comprising at least one compound according to formula L-(I) and at least one compound according to formula D-(I), wherein a molar ratio of all compounds according to formula L-(I) in composition M.sub.1 to all compounds according to formula D-(I) in composition M.sub.I is greater than a molar ratio of all compounds according to formula L-(III) in M.sub.IIIA to all compounds of formula D-(III) in M.sub.IIIA, wherein D-(I) has the following formula: ##STR00055## and wherein R.sup.1L, R.sup.1D, R.sup.2L, R.sup.2D, R.sup.3L and R.sup.3D are each the same radical selected from the group consisting of formulae (VIII), (IX), and (X) having the following formulae: ##STR00056##
